Celebrating AutoPkg with a new Runner

Hi everyone. Good morning!

For those who don't know me, I'm Scott...

Celebrating AutoPkg with a new Runner



Scott Blake

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@MScottBlake

I've been in IT for over 20 years, and I have been focussing on Apple deployments since 2011. Back then, we walked uphill both ways, in the snow, to deploy computers.

This came up at dinner the other night... who remembers the MacEnterprise Listserv? We used this before Slack and before JamfNation even existed. Did you know that listserv was hosted by Penn State? Did you know they maintain an archive with messages dating all the way back to 2004?

Now I feel old. Anyway...
Nowadays, you can find me on Slack, LinkedIn, Github... I use @MScottBlake for just about everything professionally.



https://github.com/MScottBlake/Presentations

I uploaded a copy of these slides to my Github repo. If you want to follow along, please try not to crash the wifi.

You can take notes if you want... but don't stress over it.

This slide will appear again at the end, so don't worry if you don't get it in time.

Agenda

- AutoPkg History
- Cloud AutoPkg Runner
- Demos
- Takeaways
- What's next?

As this session title implies, we're going to go through the history of AutoPkg and celebrate some big milestones and noteworthy tools.

Then we'll talk about a new tool I wrote and why I think it's awesome. We'll get to see it in action and then talk about what's on the horizon.

AutoPkg History

Since there are 3 or 4 other AutoPkg-related sessions this week, and I feel like I'm kicking off a mini conference-track, I wanted to take some time and celebrate some history.

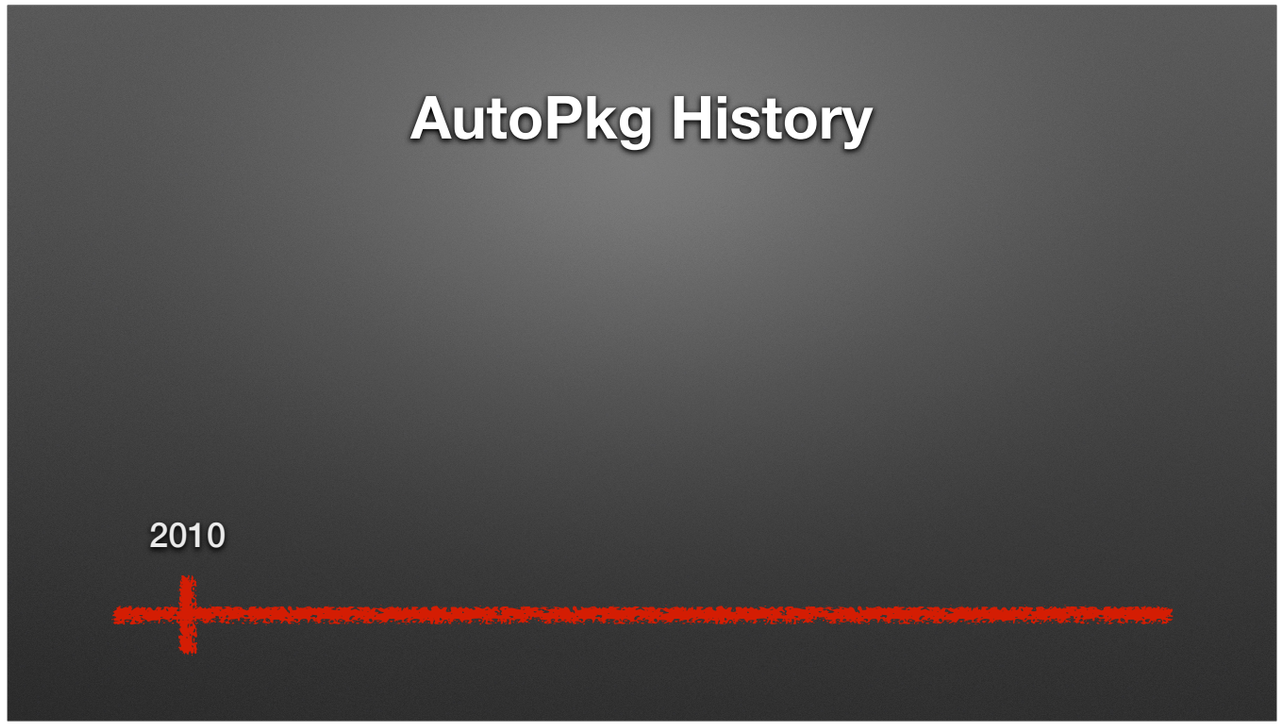
As the session title suggests, I want this to be a celebration of efforts, so if I say your name and you're sitting here, please raise your hand to be recognized. I wouldn't be opposed to standing and taking a bow or something either. I might also have something for you if you do that.

AutoPkg History

As a grey-beard, I know most of the people that I call out in this presentation, but even I don't know everyone. Plus, there are also people in here that are new to the scene and don't know who any of you are.

Without further ado, let's take a look at how we got here.

AutoPkg History



Our first stop down memory lane is in 2010. A whole 15 years ago!

AutoPkg History

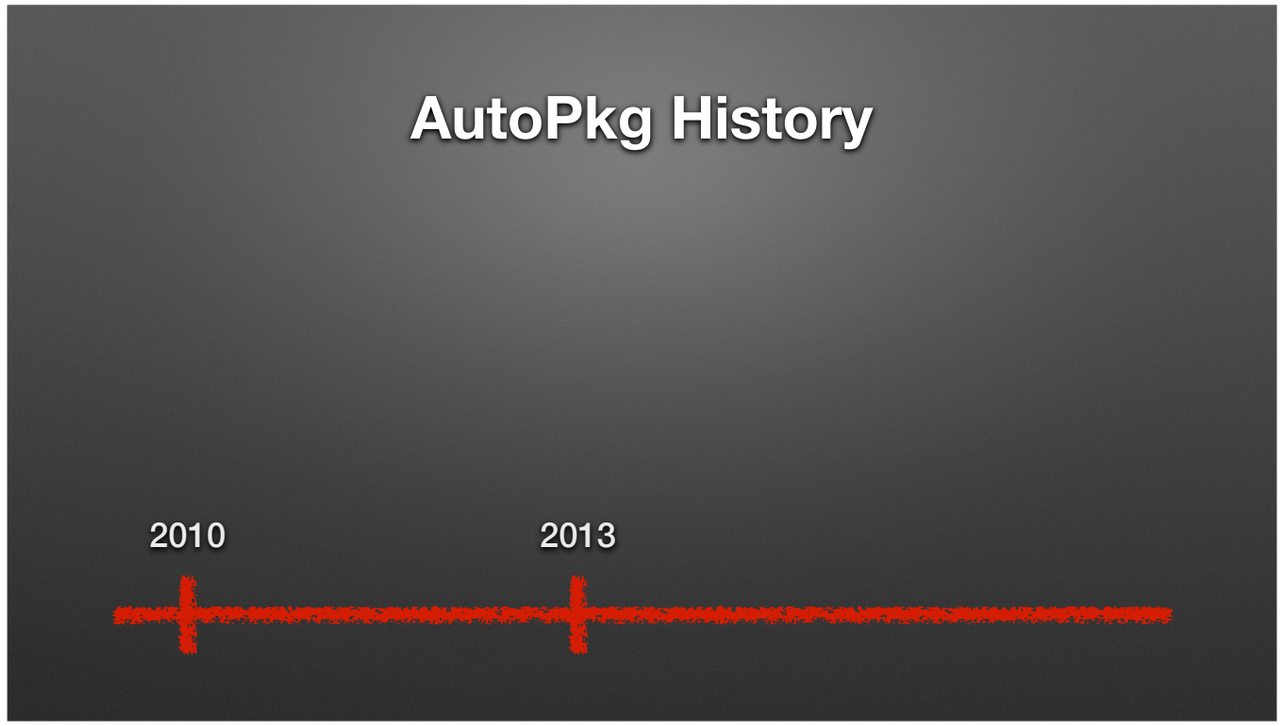
```
#!/usr/local/autopkg/python
#
# Copyright 2010 Per Olofsson
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
```

2010



Here we find the first code for what would eventually be AutoPkg, written by Per Olofsson. At that time, the project was named MacAutoPkg.

You can still find that code on the Google Code archive. It's very possible that parts of the core code are still unchanged.



Next we skip to 2013.



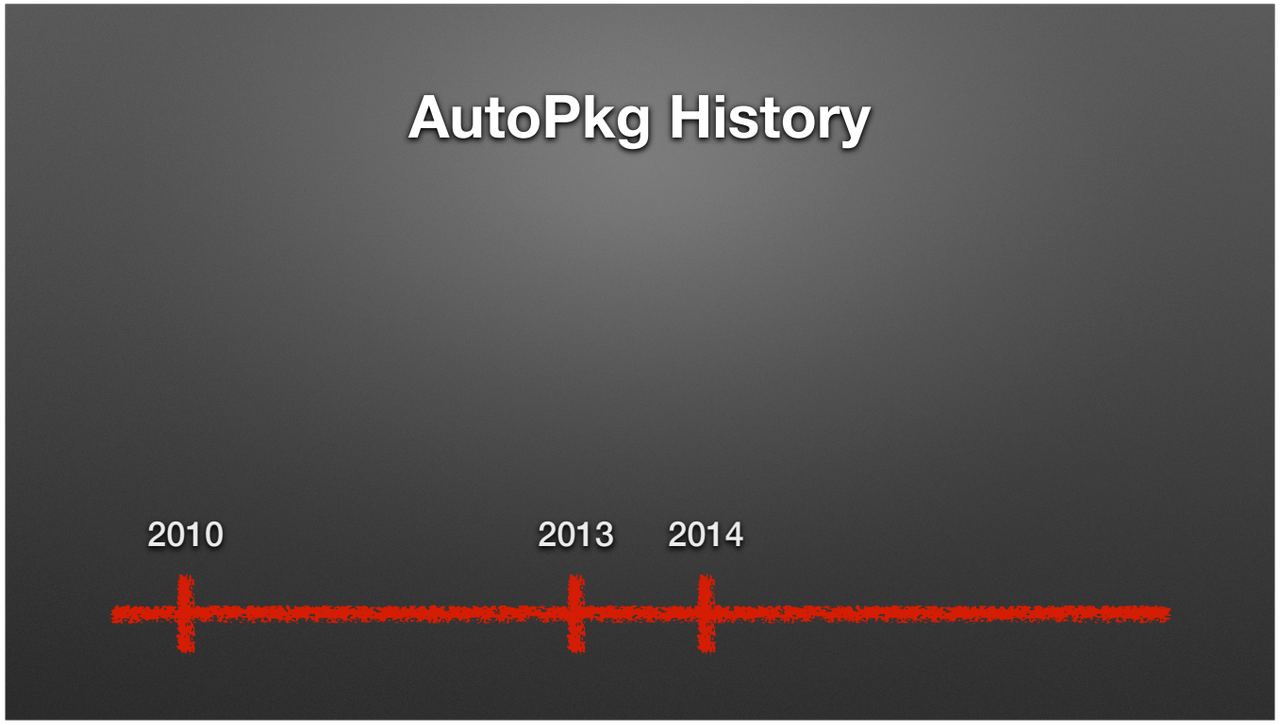
Next up is the first official release of AutoPkg in August 2013.

At this point, the project was starting to get some attention. It was a little rough around the edges and it was missing a lot of today's features, but the concept was the same as it is today and that got a lot of people excited.



I started playing with it myself shortly after this. It was probably October/November, after the "new semester" rush (I worked in a Higher Ed Library at the time).

I remember thinking it was great, but I didn't really have a huge need for it yet. I was managing about 70 lab machines that only changed in the summer.



I was preparing to use AutoPkg for the next image, but I ended up getting a new job before then. I'll come back to that in a minute.

In July 2014, The Linde Group released AutoPkgr (with an R).



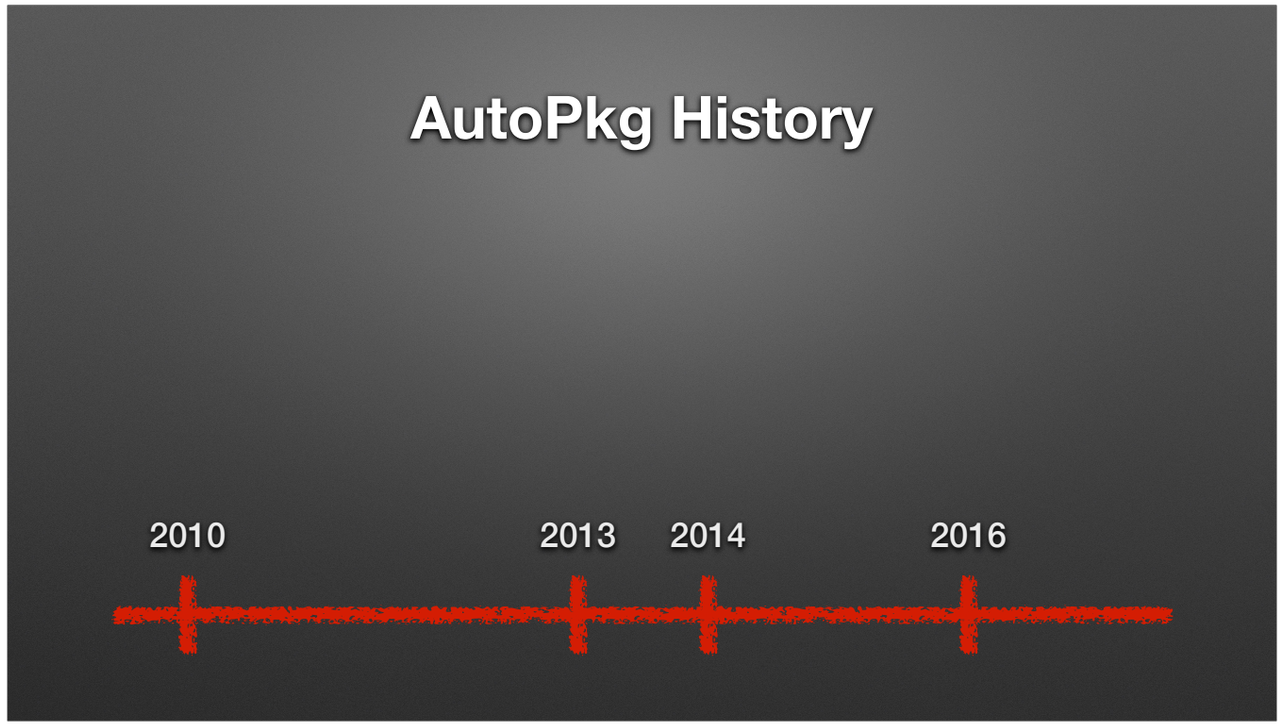
I feel like this is where the AutoPkg project really gained a lot of steam. AutoPkgr provided a GUI that helped make the project more accessible to non-technical people and it reduced the learning curve to get started. It also provided an easy interface to schedule recipe runs, which further simplified adoption.

AutoPkgr opened the door for AutoPkg in my workplace. I was able to train my new coworkers on the GUI interface, while I used the CLI for most things myself.



This new role was in central IT and I was being asked to stand up a new multi-tenant MDM that could be used by the entire university. AutoPkg enabled me to very quickly build out *and maintain* a large software catalog.

Today, we all know that's possible, but back then, it was still a very novel concept.



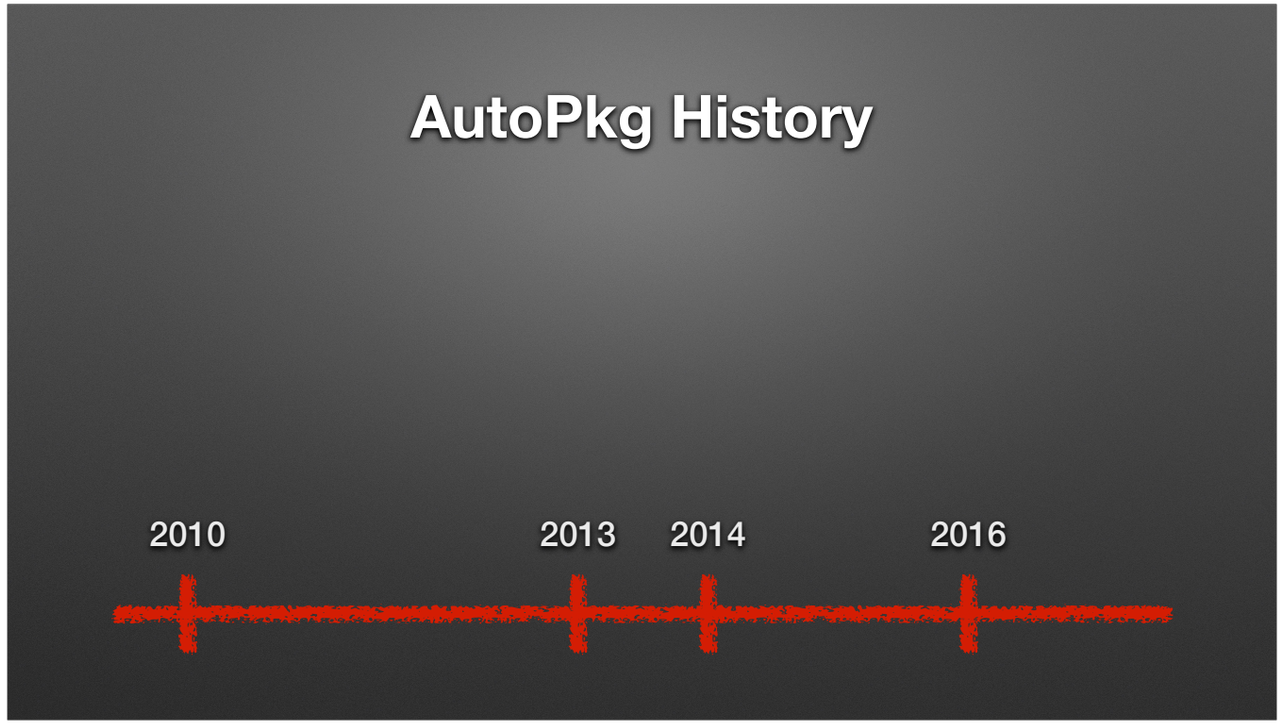
We continued to get steady updates, each with new features, but it's not until 2016 that we get 1.0.



By this point, most of the functionality we know and love has been added. We have `autopkg run`, `search`, `repo-add`, `repo-update`, `verify-trust-info`, `update-trust-info`...

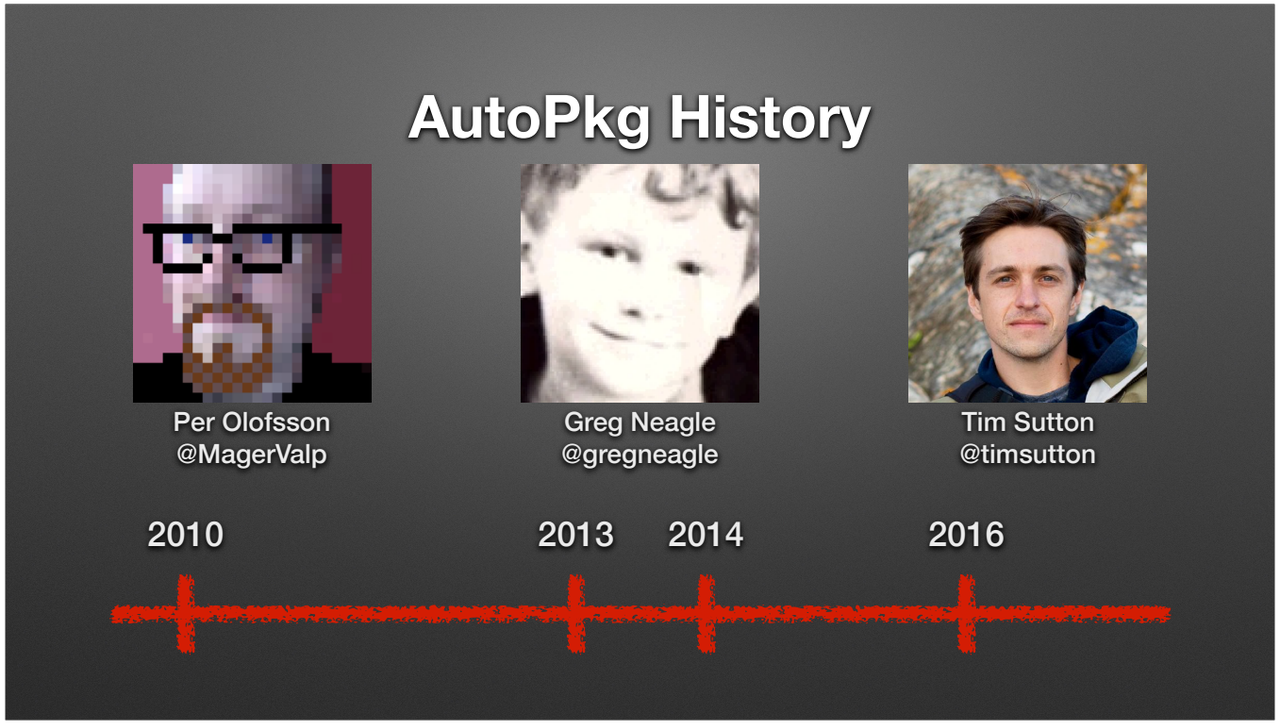
At this point, AutoPkg news was everywhere. If you went to a conference, there was a guarantee that there was a session about AutoPkg. For instance, there were 4 sessions right here at Penn State in 2016 and then 4 more in 2017.

It was inescapable.



I want to pause here for a minute.

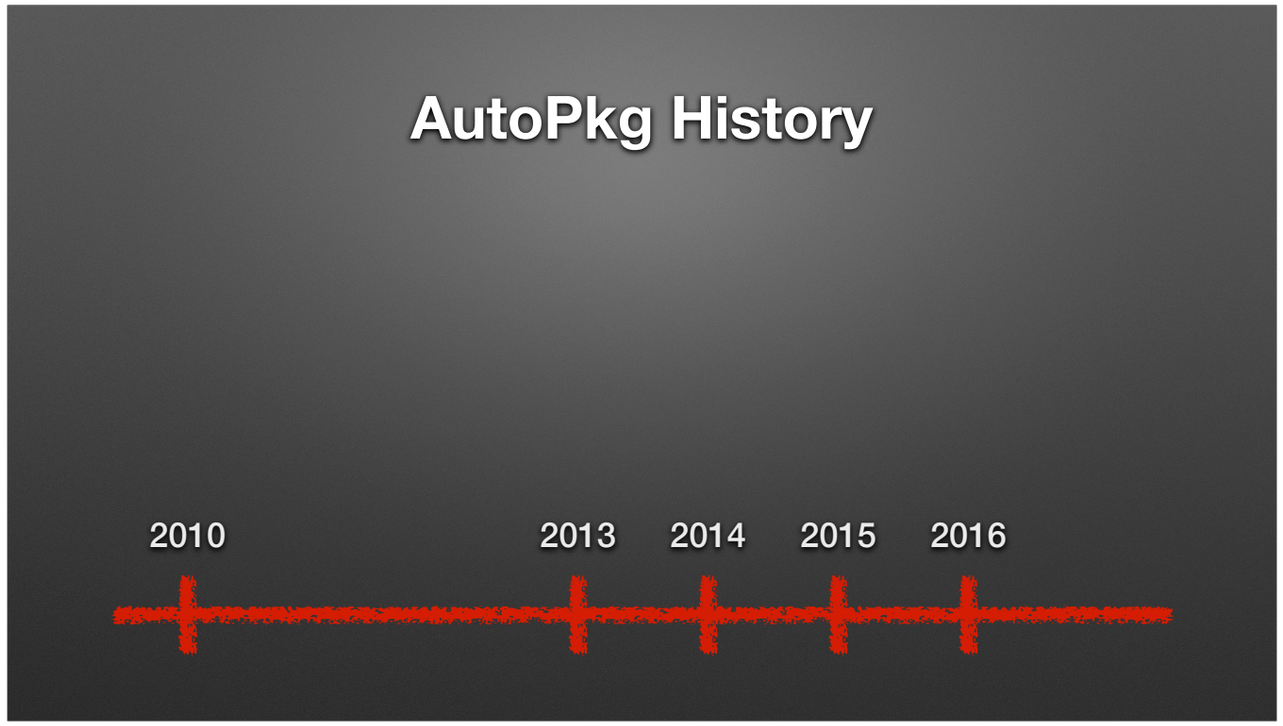
Up to this point, most of the development work was done by Per Olofsson, Greg Neagle, and Tim Sutton.



Now, I think these three deserve a huge round of applause. What do you think?

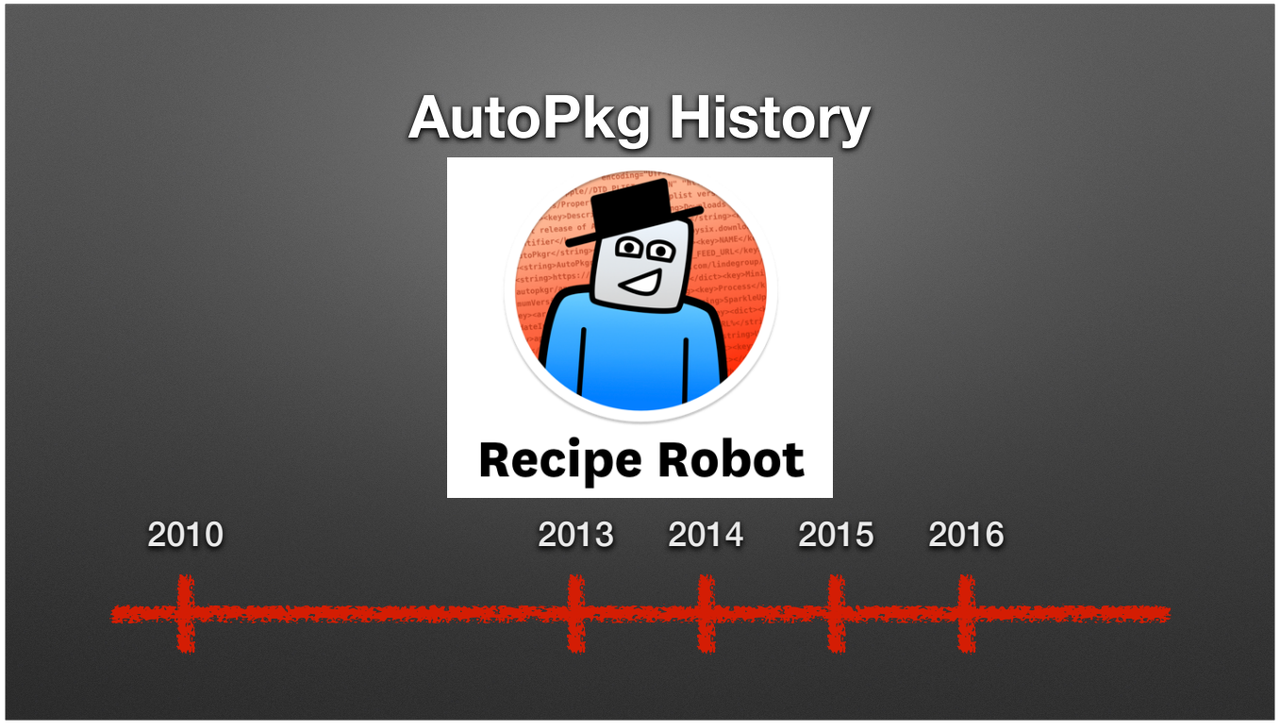
These guys laid the groundwork for everything that followed and that is no easy task. They built an expandable, low-code solution that solved real problems. Now here we are, going on 15 years and counting.

I think that's proof enough that this deserves a celebration.



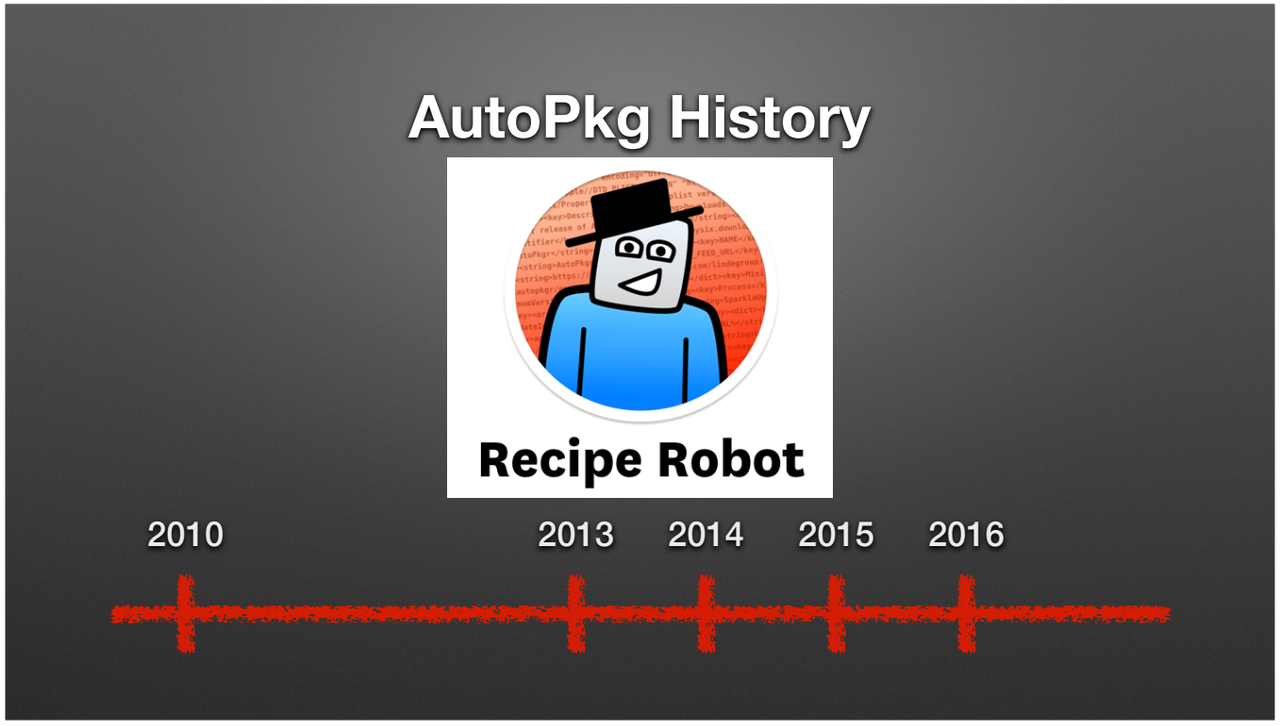
Going back in time just a little bit, we got a new GUI tool in late 2015.

I just realized that I didn't define GUI. I think most of us know it, but for those of you who don't, it means Graphical User Interface. Any app that you click on has a GUI.



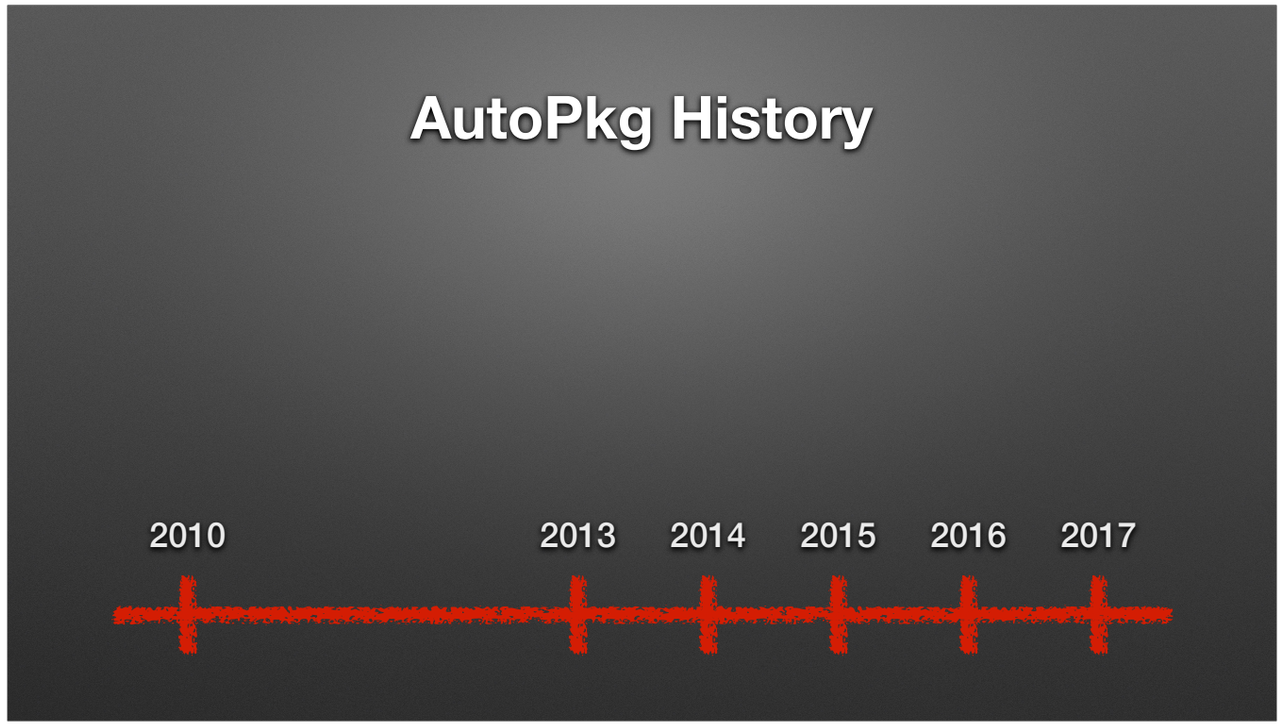
Recipe Robot was developed by Elliot Jordan to make it easier to create AutoPkg Recipes. More so, it also serves as a learning tool to show new users what a well-crafted recipe should look like.

Just like AutoPkgr did for tool adoption, Recipe Robot opened the flood gates for new recipes. As you might imagine, the number of recipes grew astronomically.



No longer was AutoPkg just for popular software. More and more niche software titles had ready-to-use recipes, and if you found one that was missing, Recipe Robot could help you create one.

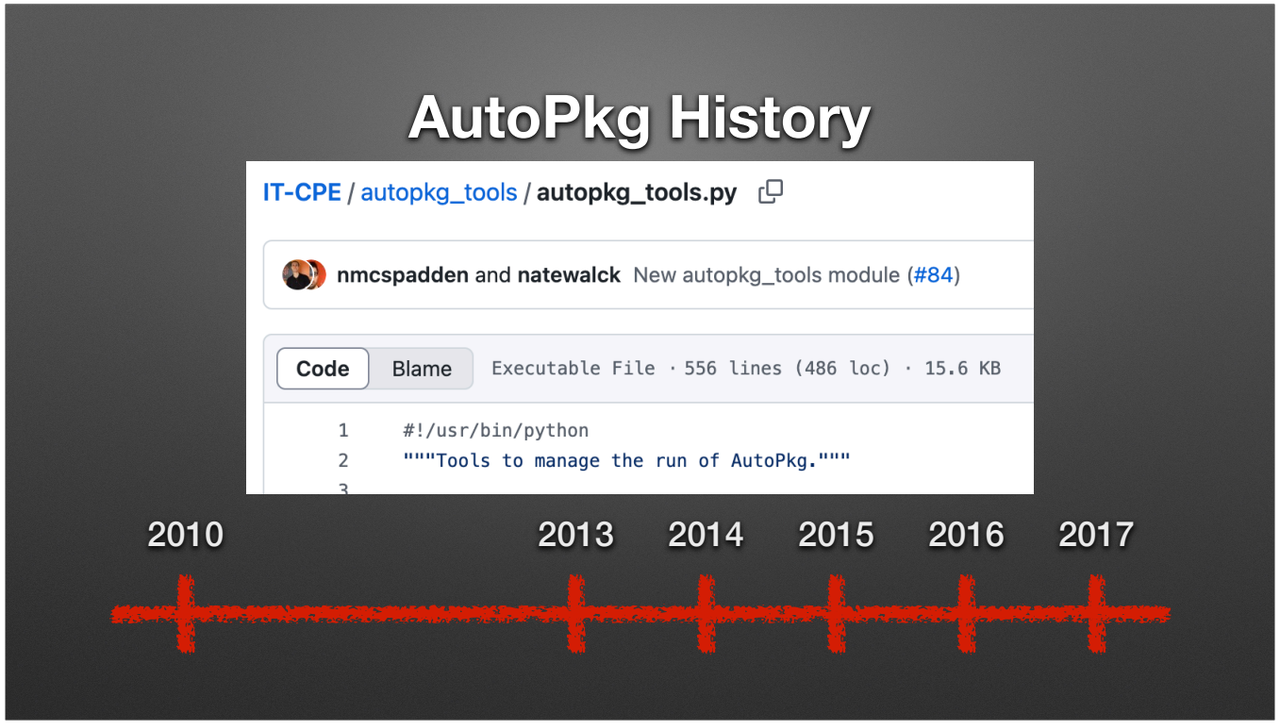
Elliot states that as of today, over 20% of the recipes in the AutoPkg organization were created with the help of Recipe Robot.



The landscape started changing in 2017.

People were looking to automate the automation tool.

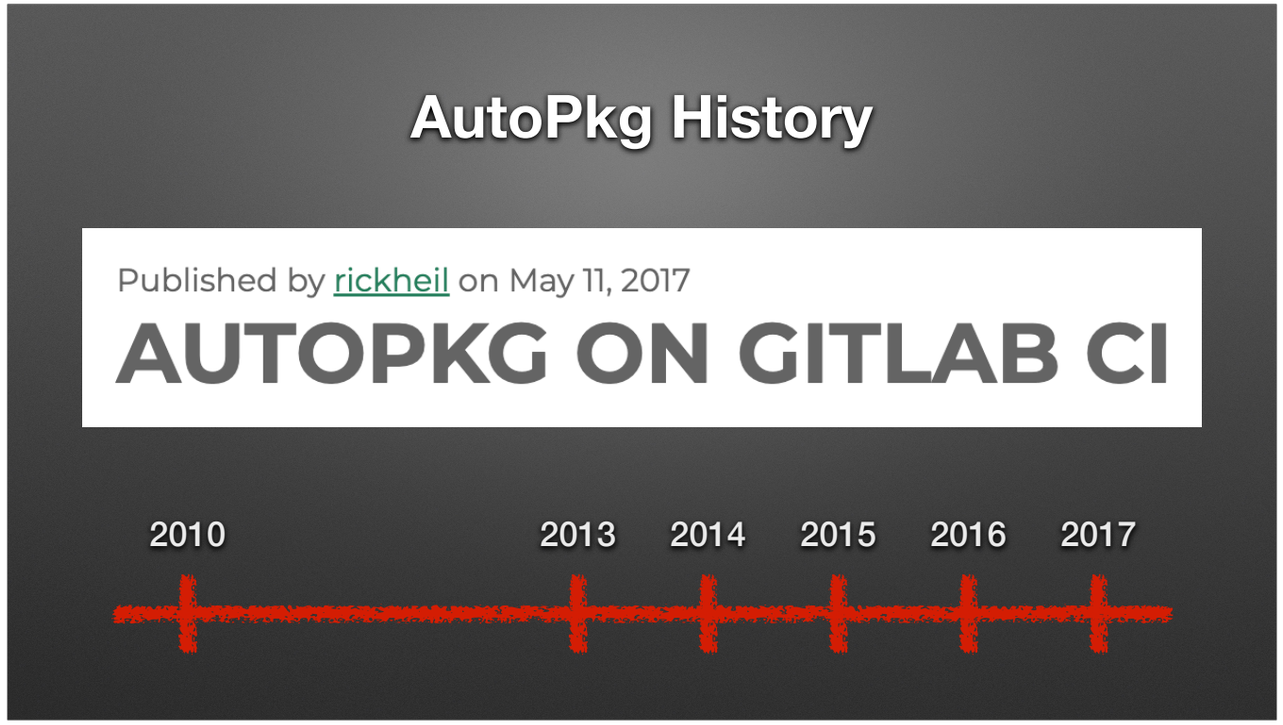
By this time, AutoPkg has gotten pretty mature, so I think this is a pretty normal progression.



A couple Facebook employees (Nick McSpadden and Nate Walck for sure) released a wrapper script that they named "autopkg_tools.py". It created a separate git branch and created a commit for each item that was imported into a Munki repo during an AutoPkg run.

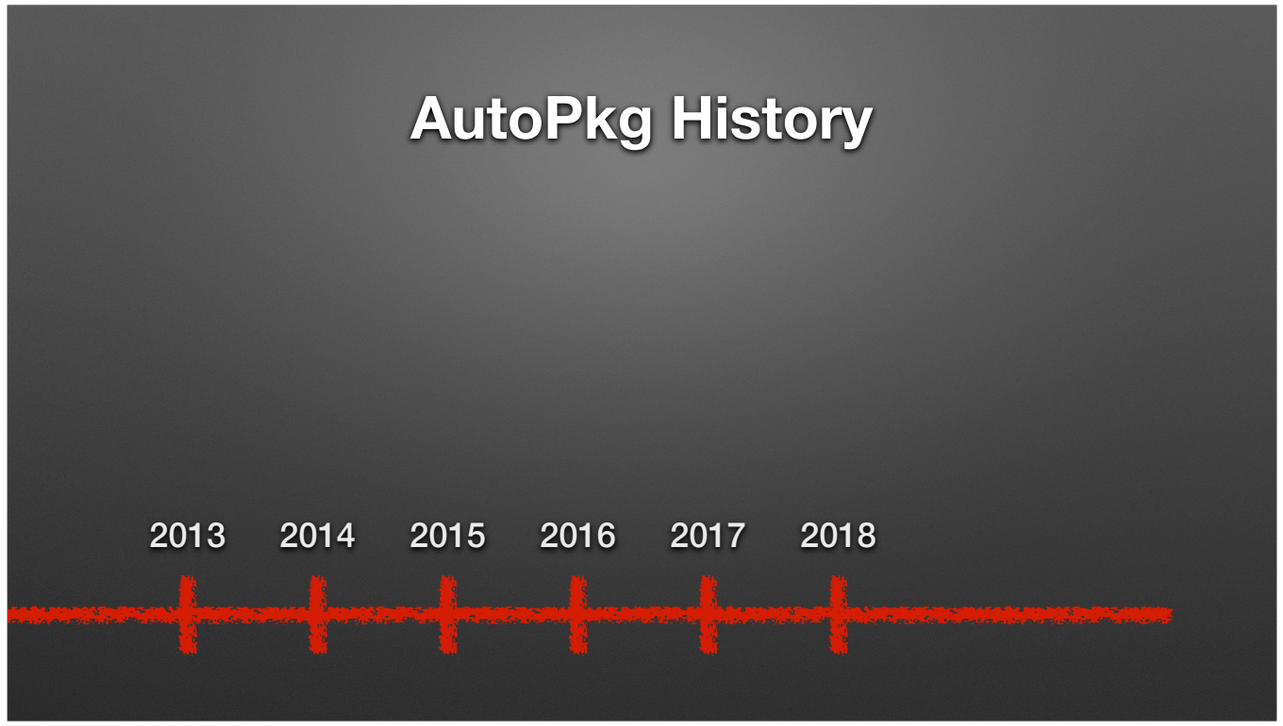
In the last 8 years, this script has been forked, modified, and forked again countless times. I'll talk about this in a minute.

AutoPkg History

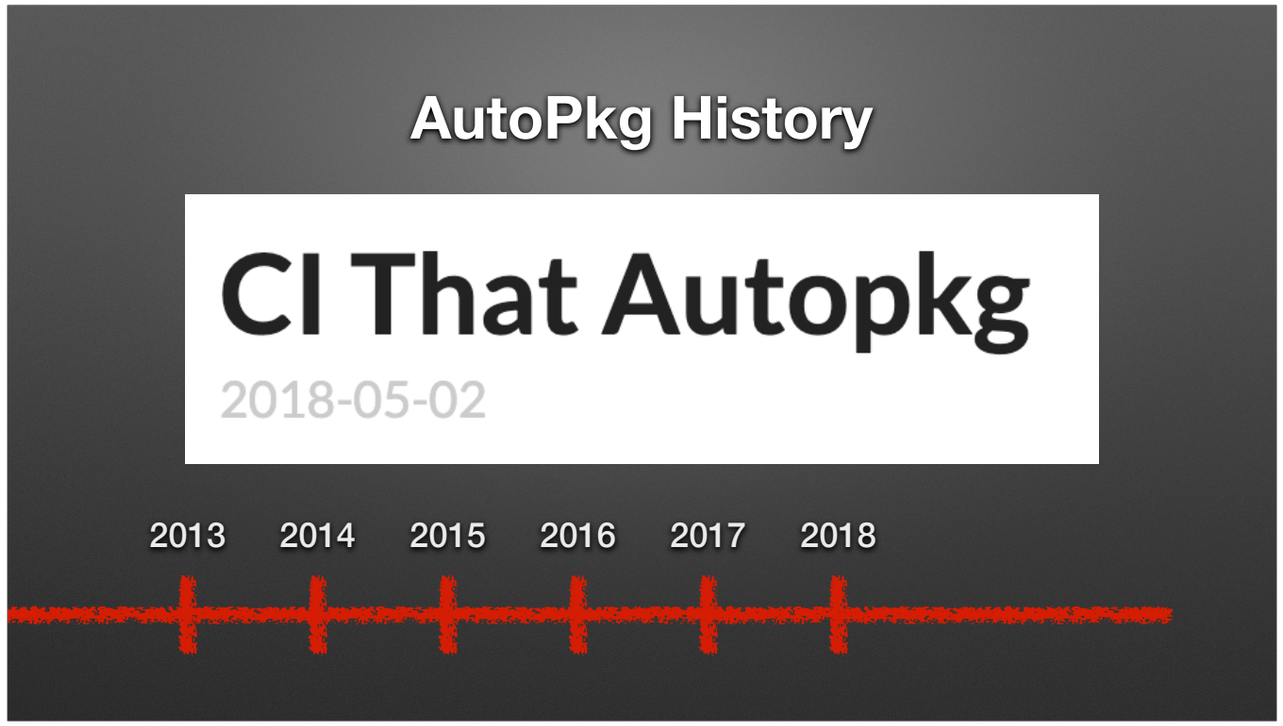


Around this time, some other enterprising MacAdmins started working to run AutoPkg as part of CI/CD pipelines.

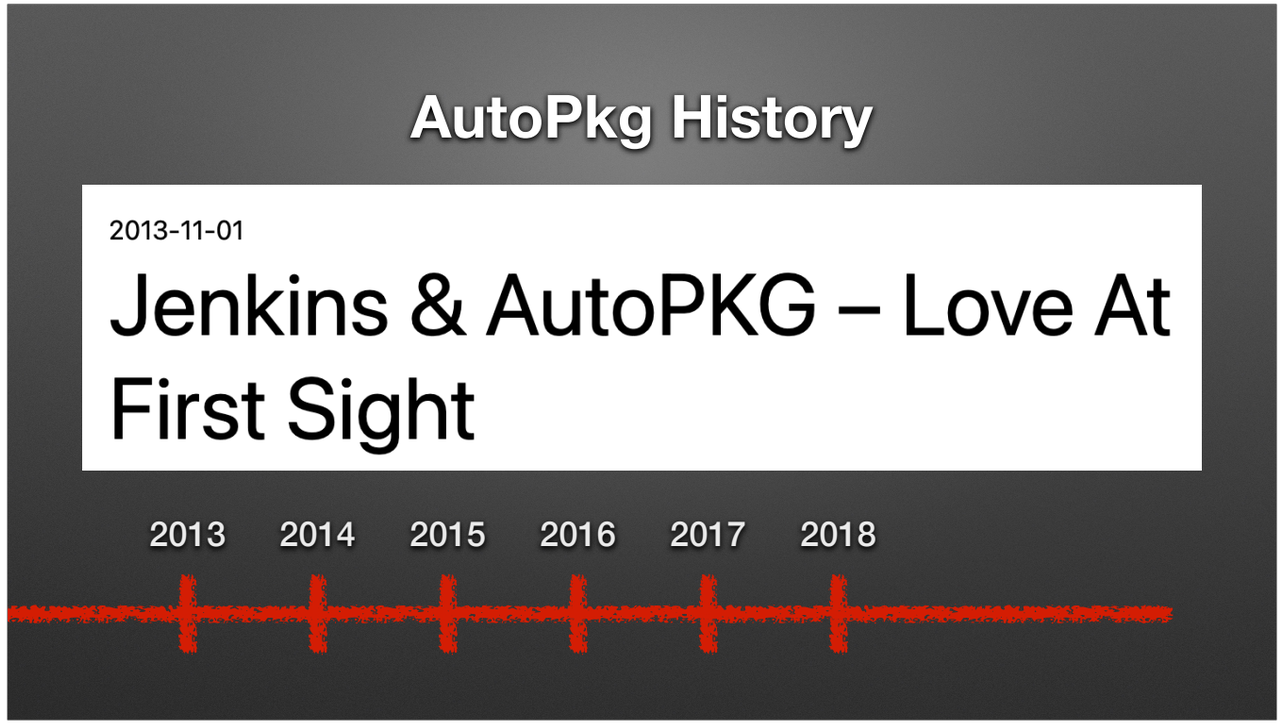
This blog post from Rick Heil is not the first, but it seemed to re-ignite the initiative. This process used autopkg_tools.py under the hood and run as part of a GitLab CI.



Moving into 2018, we got a blog post from Tim Grebien (Gre-BEEN), aka @tig, which also centered around using GitLab CI.



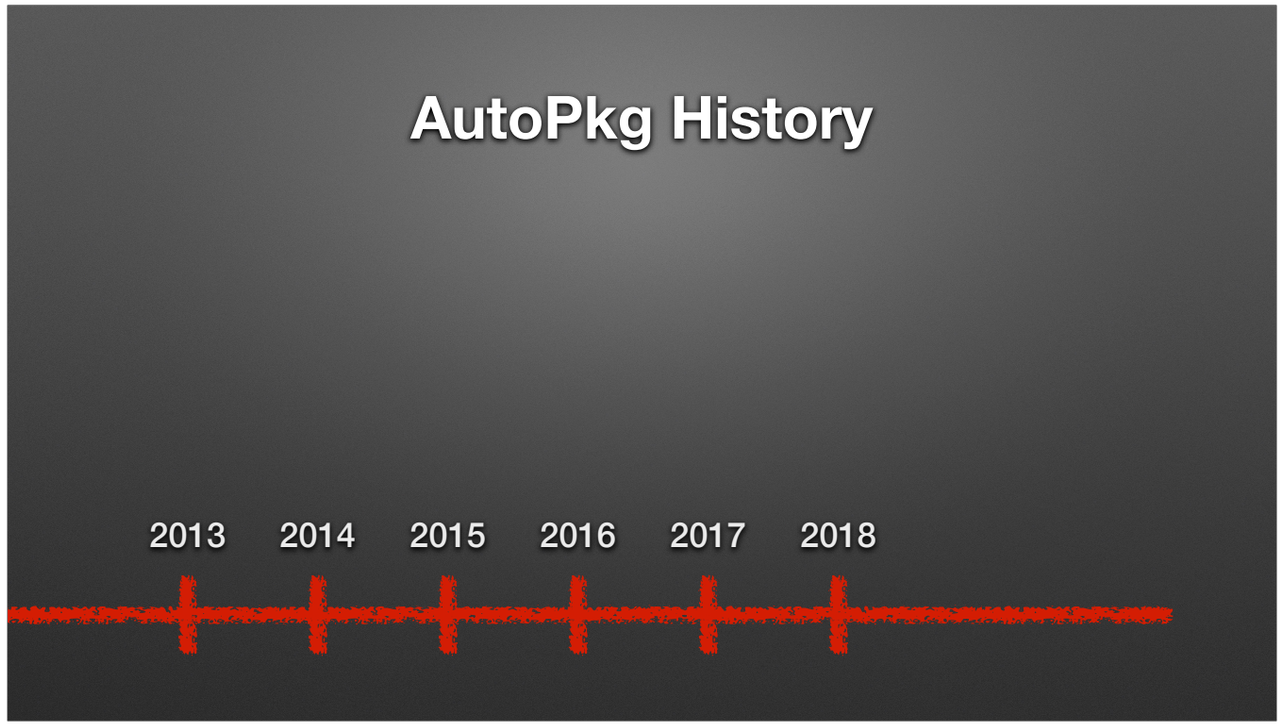
Moving into 2018, we got a blog post from Tim Grebien (Gre-BEEN), aka @tig, which also centered around using GitLab CI.



Neither of those posts were the first though.

That award goes Steve Wood all the way back in 2013.

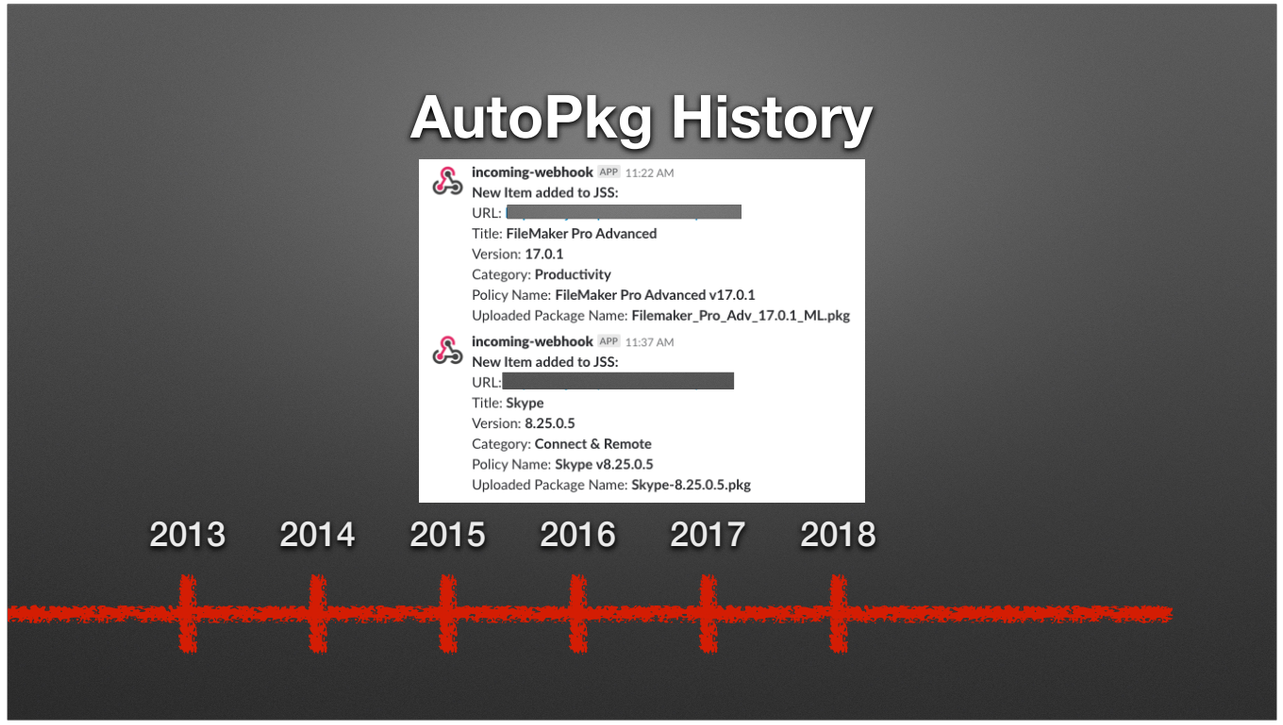
Though I don't think Jenkins ever really made it mainstream.



Due to a lack of updates, many community members were looking to move past AutoPkgr in 2018.

It didn't support the latest OS, while changes to AutoPkg as a result of SSL issues were pushing people in that direction.

It had some features that people wanted to take with them though.

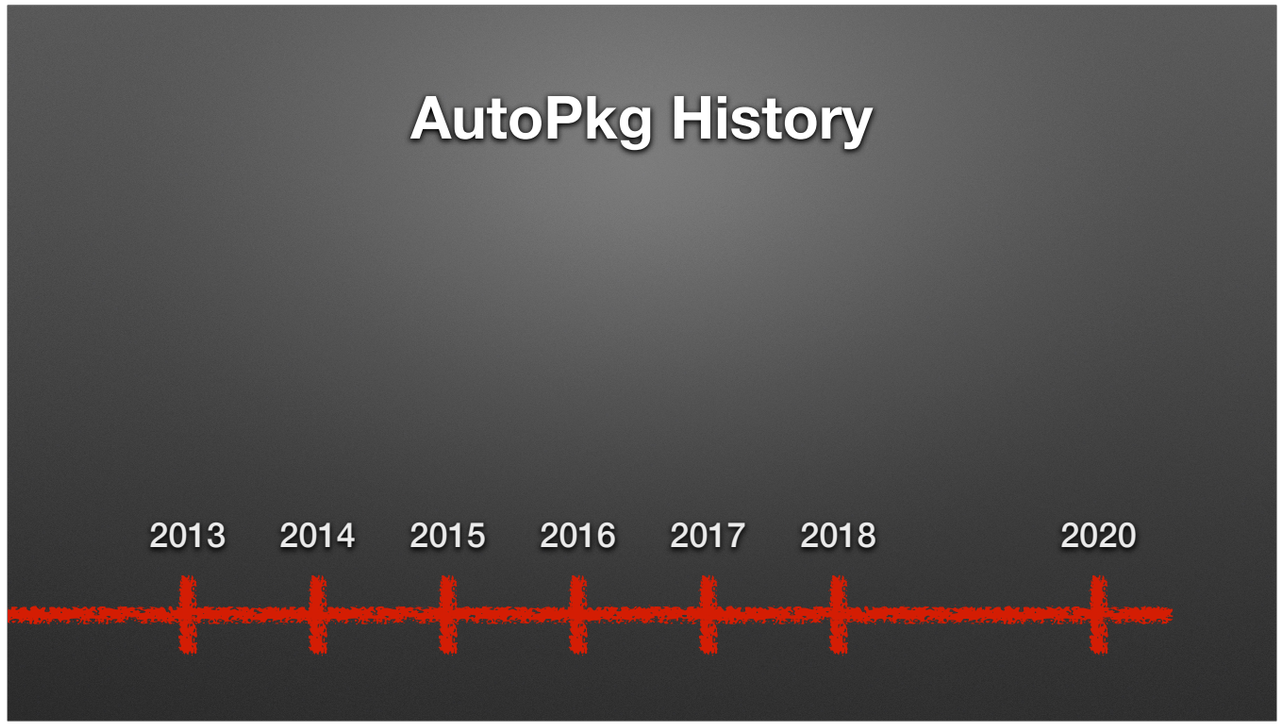


Graham Pugh released the "Slacker" post-processor, which allowed admins to easily send Slack notifications as part of an AutoPkg run, without needing AutoPkgr, or a full python wrapper like autopkg_tools.py.



Soon after, Rich Trouton released "autopkg-conductor". This tool is essentially a bash script with sample launch daemons. It made it easier for admins to schedule AutoPkg runs even when no one was signed into the computer and included the Slacker post-processor for increased visibility and later added Teams support as well.

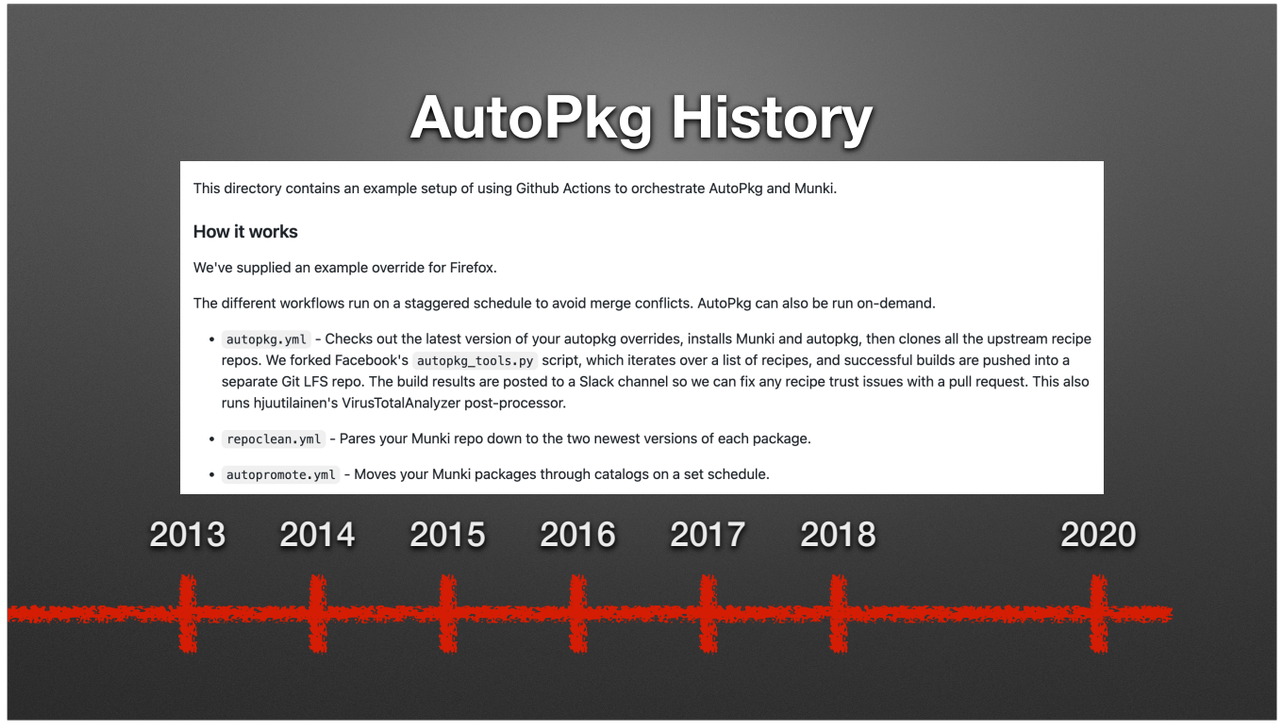
The novel thing here was that you could get a lot of the features from AutoPkgr or autopkg_tools.py with a command line script.



Skipping ahead to 2020, we were faced with the end of Python 2.

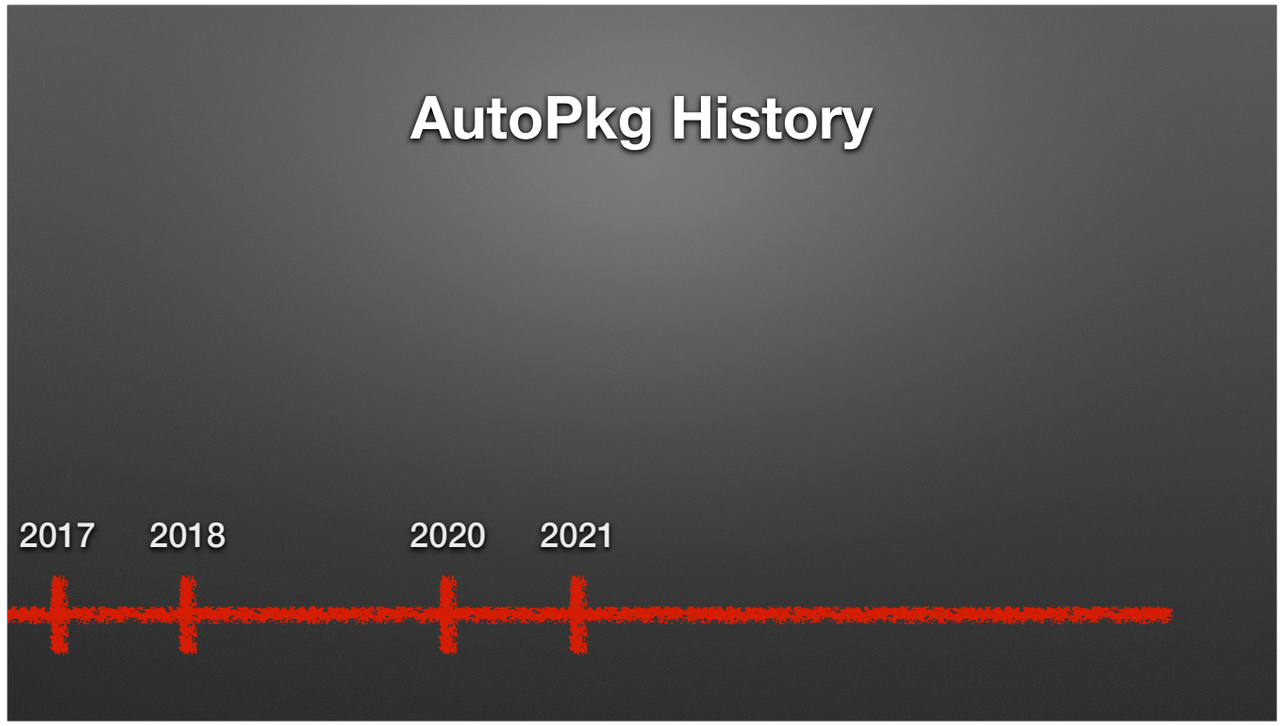


A huge initiative to upgrade the AutoPkg codebase was spearheaded by Nick McSpadden, and eventually, AutoPkg 2.0 was released as Python 3-only, dropping support for Python 2.



Additionally, William Theaker (THEE-ker) and Harry Seeber from Gusto, made a new fork of autopkg_tools and included GitHub Actions examples.

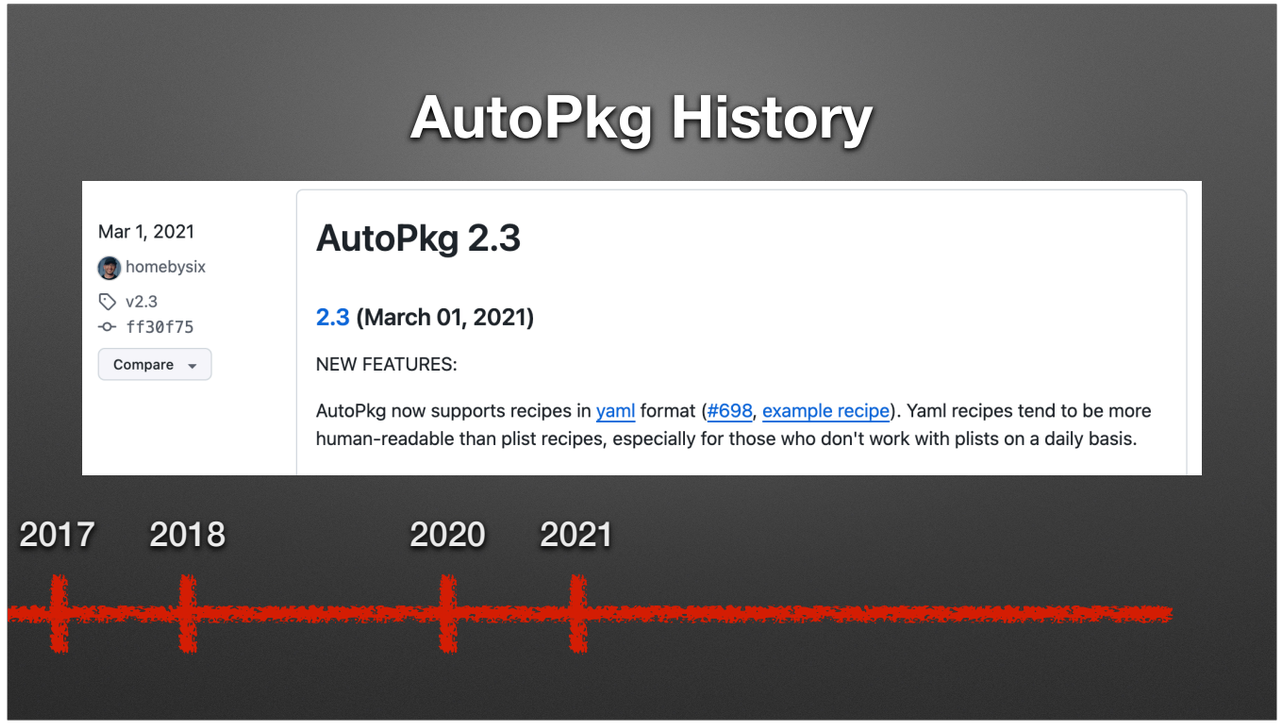
I believe this project set a "new standard" and I think it is probably the most prevalent in CI/CD pipelines today. I have found quite a few cases of people and organizations posting about their workflows and when I look at them, they all seem to be variants of this Gusto version.



In 2021, AutoPkg 2.3 was released, bringing YAML recipe functionality.

I am personally hoping for yaml to become the default at some point. In the meantime, I'll be keeping my fingers crossed.

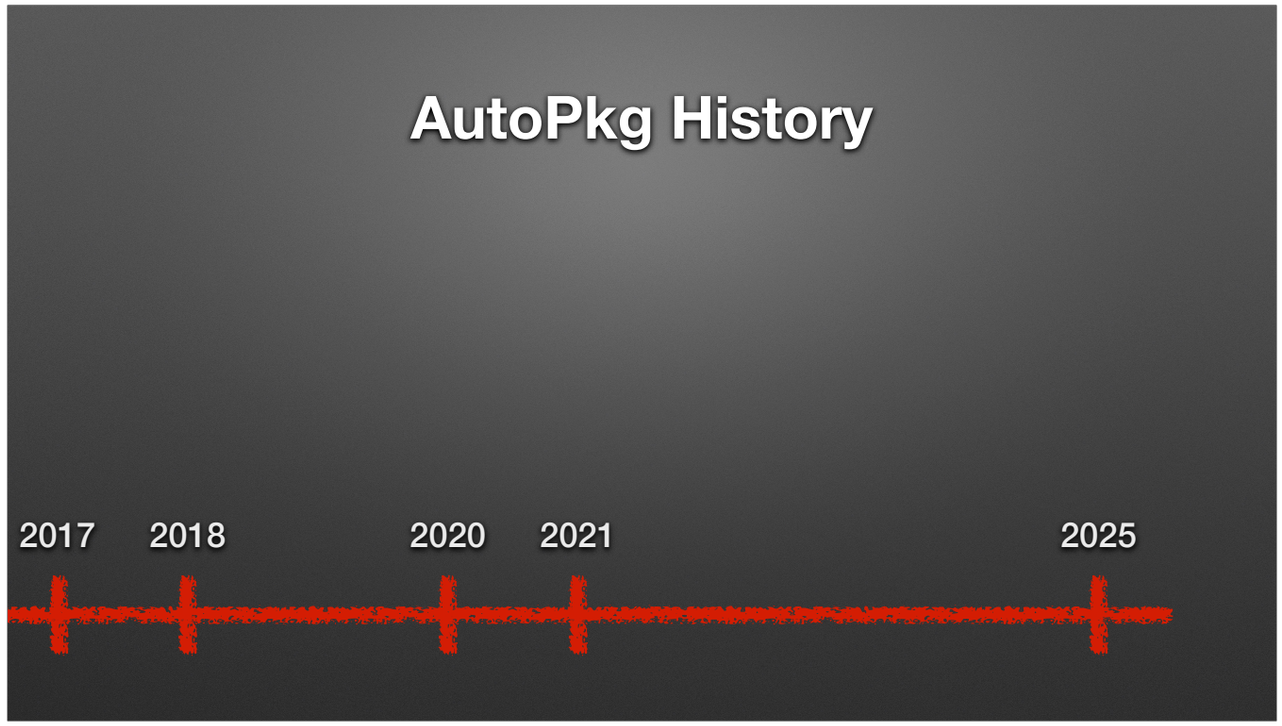
We have only gotten minor changes and bug fixes that have made it to full a release since then.



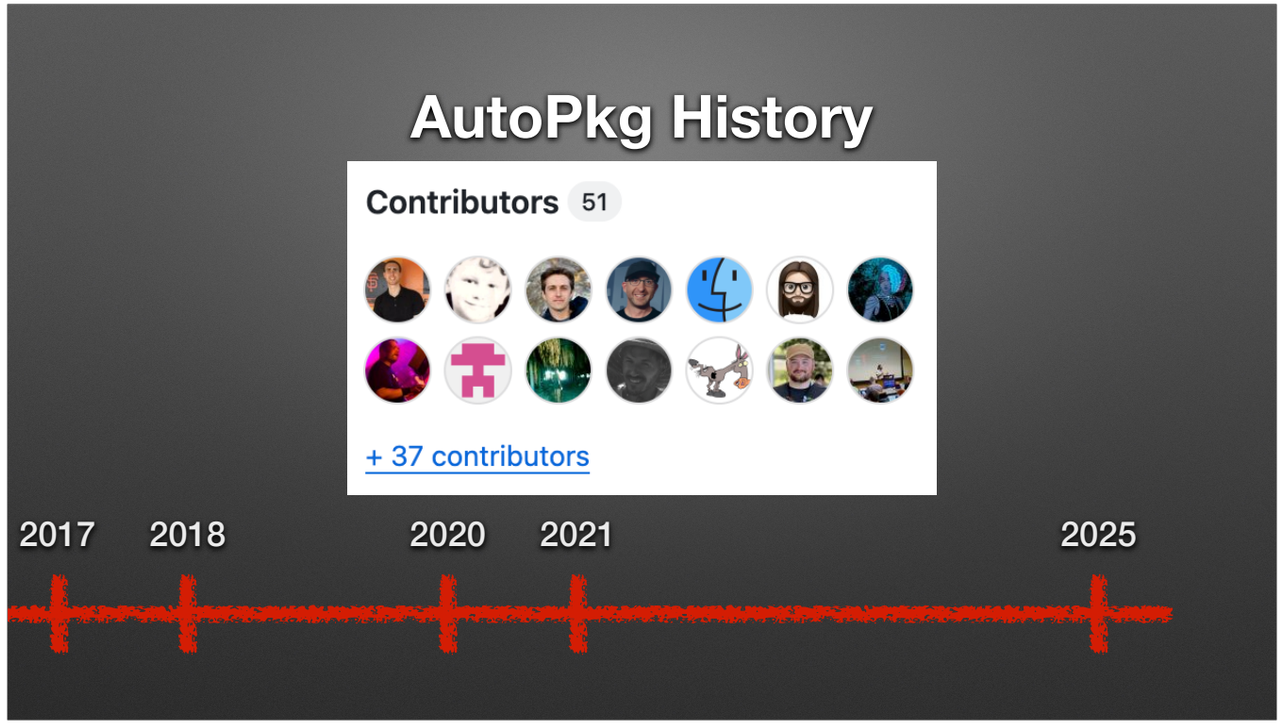
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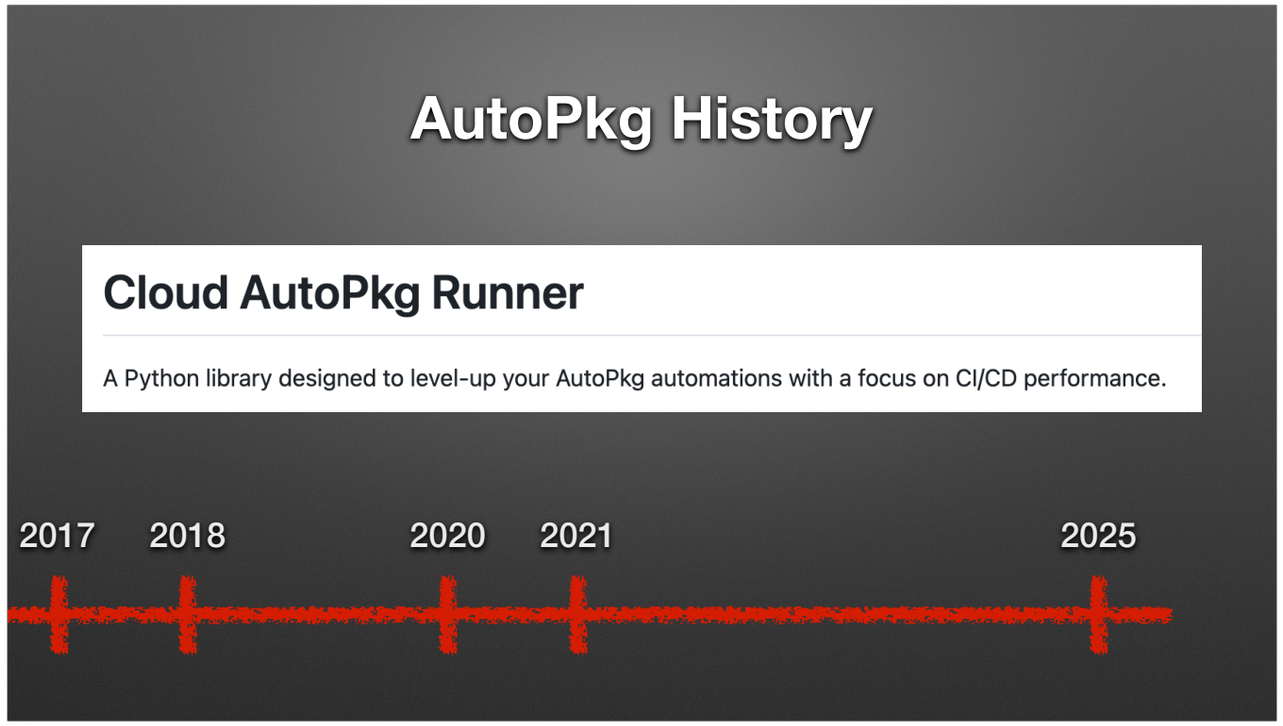


That brings us up to present day. AutoPkg is a tool that has saved me hundreds, maybe thousands, of hours. A lot of care and feeding has gone into the project to keep it running for so long.



If you are one of the 51 contributors, you have my thanks. Please take 2 pats on the back from the petty cash fund for your efforts. Seriously though, thank you.

Same goes to anyone that has built something AutoPkg-related, even if it didn't get named earlier, thank you. The eco-system thrives as more people contribute to it.



There have been *many* projects related to AutoPkg that didn't make this list, and time will tell if this one should... but it's mine... and this is my presentation... so I get to add it.

Cloud AutoPkg Runner

Long before I ever started on this project, I have been trying to figure out a way to handle situations where the AutoPkg Cache does not persist between runs. In this type of environment, all files are downloaded every time it runs.

[Story about POC]

Basically, I decided that I wanted to write a tool that could bring this concept to the masses, not just my own org. So I set some goals for myself.

Project Goals

- Store download metadata
- Easy to use
- Fast
- Strict typing

The main goal of this project was to make it easier for CI/CD workflows by storing the metadata for each file that AutoPkg downloads. I also wanted to keep storage costs low and use this data to reconstitute the cache on subsequent runs.

Project Goals

- Store download metadata
  - Don't store the entire download
  - Smaller storage requirements
  - Create fake files from previous run
- Easy to use
- Fast
- Strict typing

The main goal of this project was to make it easier for CI/CD workflows by storing the metadata for each file that AutoPkg downloads. I also wanted to keep storage costs low and use this data to reconstitute the cache on subsequent runs.

Project Goals

- Store download metadata
- Easy to use
  - Use existing preferences wherever possible
  - Sane defaults
- Fast
- Strict typing

I also wanted to make it easy to use.

I didn't want to *have* to redefine AutoPkg preferences, and I wanted to keep the learning curve as low as possible. Ideally, most people won't have to define a whole bunch of command line arguments because the defaults are good enough for most environments.

Project Goals

- Store download metadata
- Easy to use
- Fast
  - Enable concurrency
  - Create threads for blocking tasks
- Strict typing

I wanted it to be fast.

AutoPkg does basically everything sequentially and I felt there was room for improvement just by adding concurrency and multithreading.

Project Goals

- Store download metadata
- Easy to use
- Fast
- Strict typing
  - Improves code readability and maintainability
  - Early error detection

Lastly, this is for the developers in the room. I planned to use Python and wanted strict typing across the board.

Type annotations make for more reliable, readable, and maintainable code. They don't do anything for performance, but if definitely makes it easier to for others to learn the codebase. I wanted to make it easier for you to contribute back to the project or to import it into your own code and extend its capabilities.

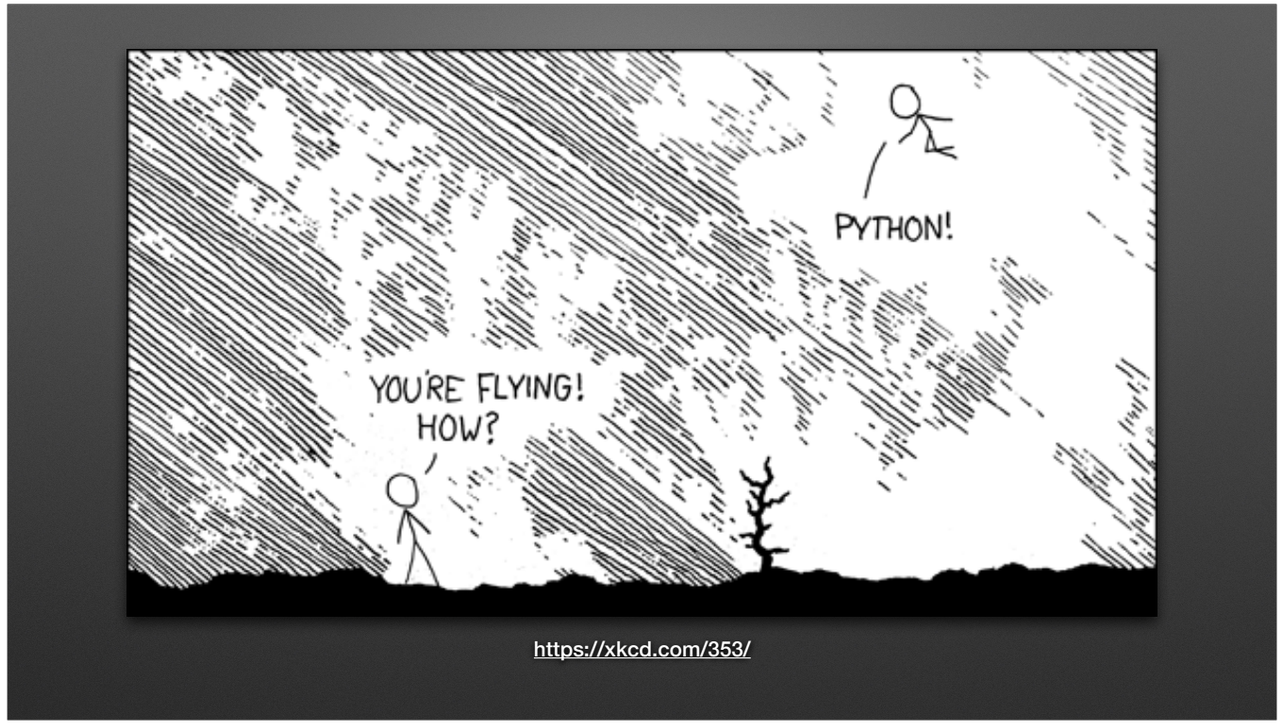
Features

- Python library that can run as a CLI binary
- Metadata caching avoids redundant downloads
- Concurrent recipe processing
- Optimized for CI/CD, but works everywhere

So I wrote the tool, and this is what I came up with.

I decided that in order to make it easy to use, I would set it up so that it could be run as a Python import where you can interact with the classes and functions I wrote, but it can also be run as a command line binary.

However, the most compelling feature is the metadata caching. It works brilliantly.



https://xkcd.com/353/

I wrote it in Python, so I believe that I am now contractually obligated to show this XKCD.

Please correct me if I’m wrong wrong.

Sorry. Moving on...

Features

- Python library that can run as a CLI binary
- Metadata caching avoids redundant downloads
- Concurrent recipe processing
- Optimized for CI/CD, but works everywhere

I was successful in making it run recipes concurrently. The faster your internet connection, the bigger the difference in runtime vs native AutoPkg.

Lastly, even though the name implies that it can only be run in the cloud, that's not a requirement. You will see speed improvements regardless.

Installation

```
pip install cloud-autopkg-runner
```

Since it's a Python Library, you can install it with pip like any other Python dependency. That said... don't use pip. Use UV instead. It is superior in every way. Go convert your existing projects. You have my blessing to do so.

Technically, you could also install it with poetry, but seriously, use UV.

If you've never heard of UV, make sure you look it up tonight. If you want me to come back next year with a UV for MacAdmins talk, let me know on the feedback form for this session.

Anyway... let's get back on track.

Installation

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Installation

```
poetry add cloud-autopkg-runner
```

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Installation

```
uv add cloud-autopkg-runner
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv add cloud-autopkg-runner[json]
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv add cloud-autopkg-runner[sqlite]
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv add cloud-autopkg-runner[azure]
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv add cloud-autopkg-runner[gcs]
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv add cloud-autopkg-runner[s3]
```

The default metadata cache storage is a JSON file and all dependencies are included by default. If you want to be explicit about it, you could add JSON as an optional dependency.

Other options are SQLite, an Azure Container, Google Cloud Storage, and S3. Adding this bit to the end ensures that the dependencies you need for your environment are installed. I'll come back to why that's necessary in a minute.

Installation

```
uv tool install cloud-autopkg-runner
```

Now, if you want to run it as a command line tool, use the "uv tool install" command to install it. This will install it as a legit binary on your system.

It's also super easy to upgrade as changes are released.

UV handles all the dependencies for both install and upgrade. You do still need to specify any optional dependencies though.

Installation

```
uv tool upgrade cloud-autopkg-runner
```

Now, if you want to run it as a command line tool, use the "uv tool install" command to install it. This will install it as a legit binary on your system.

It's also super easy to upgrade as changes are released.

UV handles all the dependencies for both install and upgrade. You do still need to specify any optional dependencies though.

Usage

Any questions so far? We still have 45 minutes.

Let's switch gears from installation to usage.

Command Line Usage

uv run cloud-autopkg-runner --recipe Firefox.pkg.recipe

uv run cloud-autopkg-runner --recipe-list path/to/list.json

If you used "uv add", then you need to "uv run" then your command.

Command Line Usage

cloud-autopkg-runner --recipe Firefox.pkg.recipe

cloud-autopkg-runner --recipe-list path/to/list.json

However, if you installed it with "uv tool install" calling it is as easy as any other binary on your machine.

That's why I think "uv tool install" is the best approach.

Command Line Usage

-r, --recipe

--pre-processor

--post-processor

-v, --verbose

There are a handful of command line arguments that you can use. I'm not going to list them all here. Check the project wiki to get them all.

These 4 arguments can be listed multiple times. For instance, you can list multiple pre or post processors and they will be run in order after each recipe.

All parameters are optional, but if you don't specify at least one recipe, nothing's going to happen.

Command Line Usage

--cache-plugin

--cache-file

--cloud-container-name

https://github.com/MScottBlake/cloud-autopkg-runner/wiki/Command-Line-Arguments

Here's a few more that I think will be more common. Again, the rest are documented in the Wiki.

The cache plugin argument is where you specify how you want the cache stored. Currently, the options are JSON, SQLite, S3, Azure, and Google Cloud Storage. The default is JSON.

The cache-file argument defines the path to the file where the cache will be stored. If it is in a cloud container, this will serve as the path within the bucket or container.

Similarly, the cloud container name is the name of the bucket or container (if you are using cloud storage). The cache file will be automatically downloaded from the bucket, modified, and then re-uploaded.

Python Import

```
import cloud_autopkg_runner

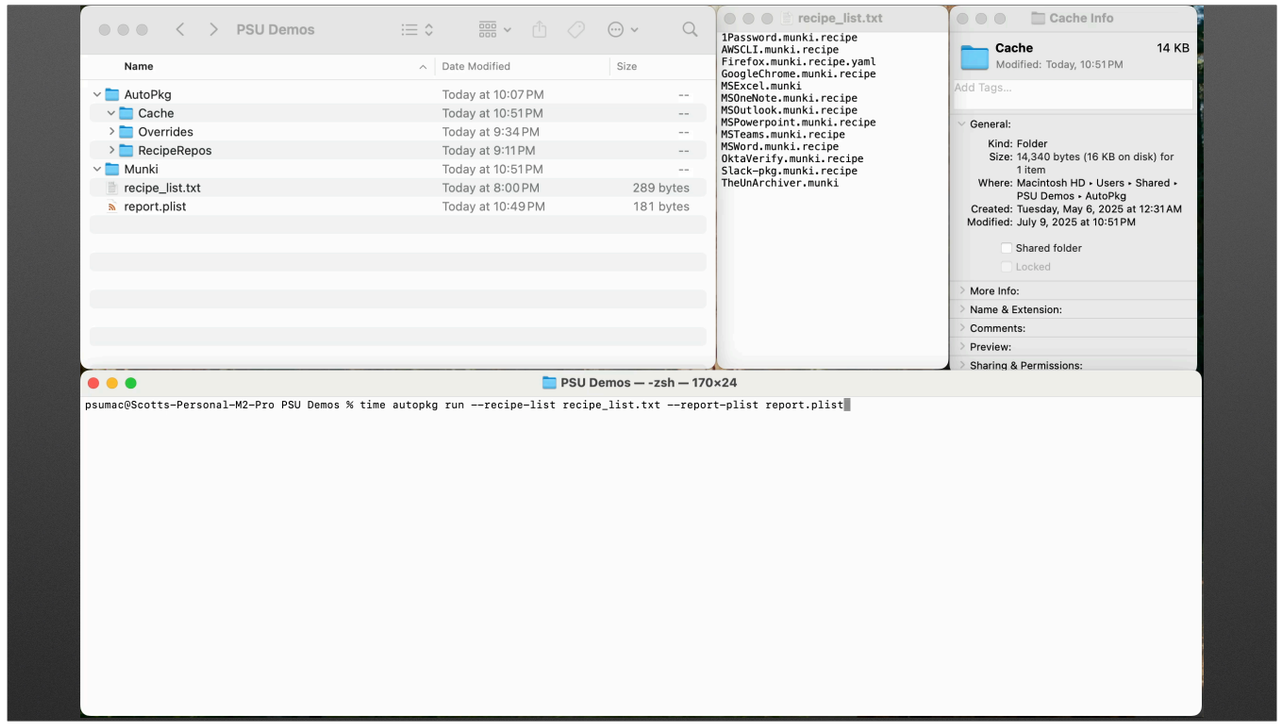
from cloud_autopkg_runner import Recipe, Settings
```

You can import it exactly like any other Python library.

My goal is that the CLI usage can handle the majority of use cases. *But*, if you don't fit that mold, the same tool can be used for your more customized or intricate workflows. I tried to document the code well enough that anyone can use it this way, but it does require a bit of Python knowledge.

I won't go into too much detail here, but I will say that there is an example in the Wiki, and I am happy to talk through it with you after the session while we're here or later in Slack. Find me after the session and we can talk.

Demo

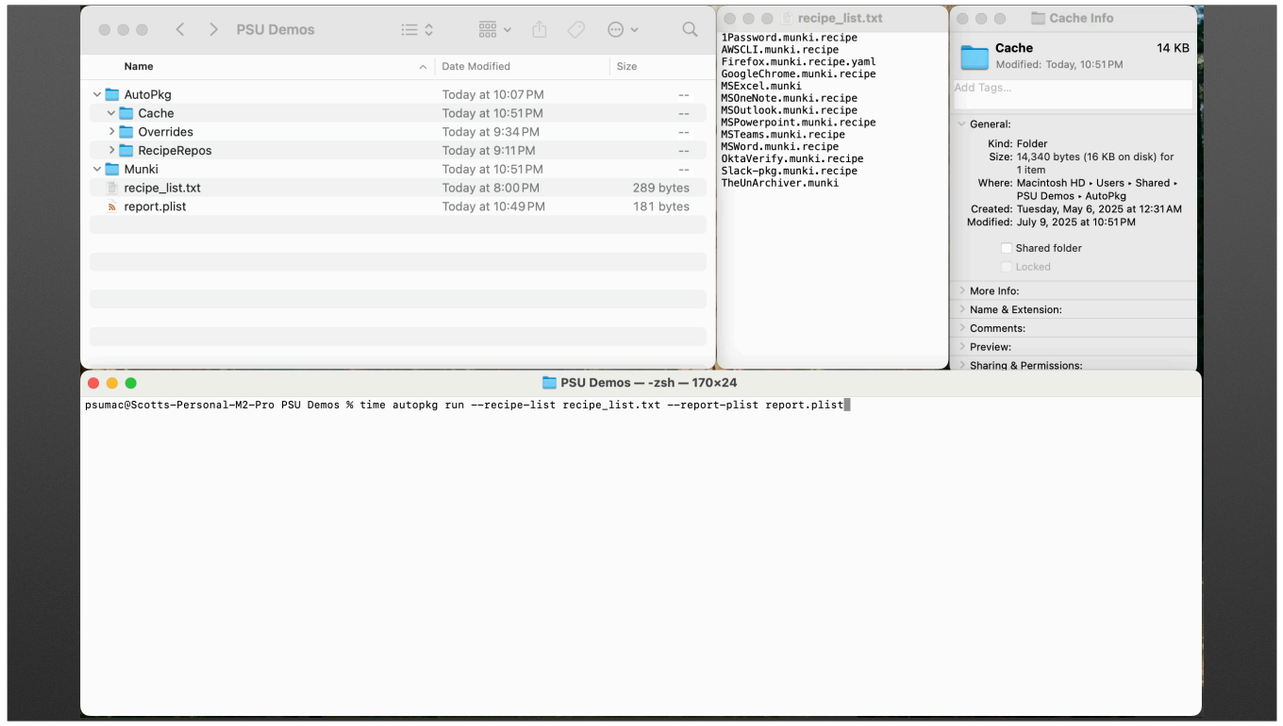


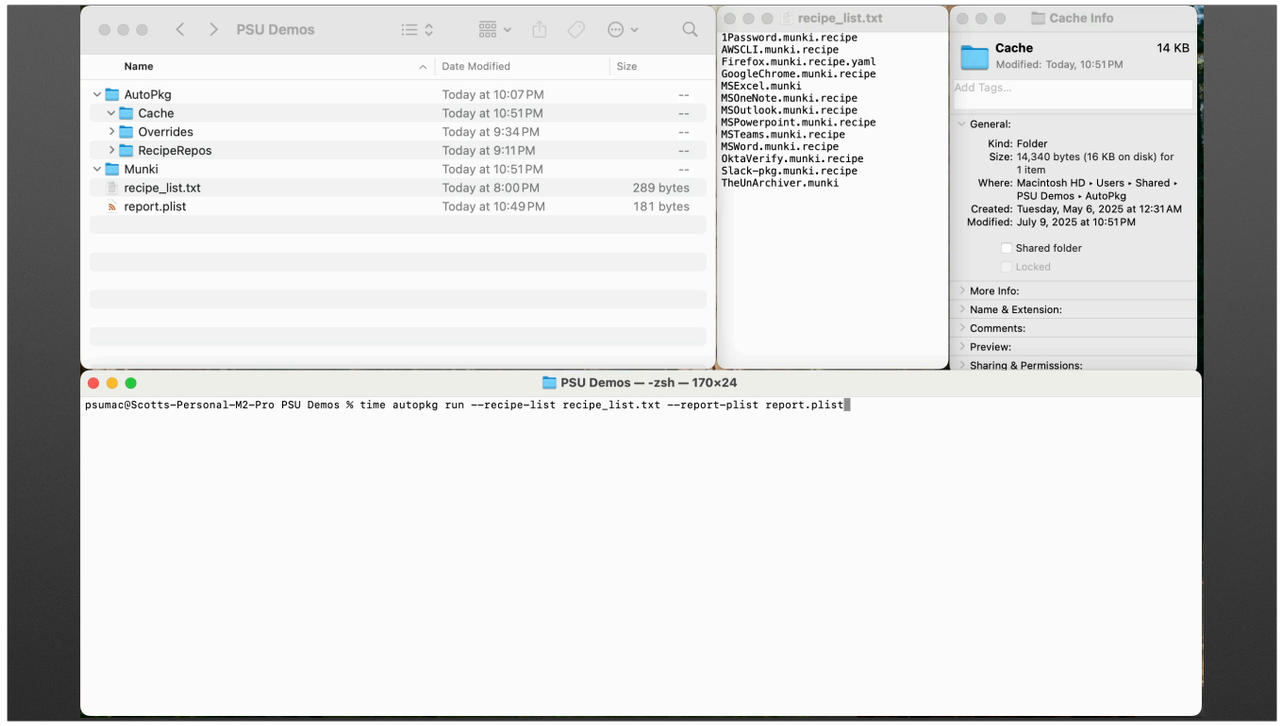
Okay, I lied a little. I wanted to start with a normal AutoPkg run with the recipe-list parameter.

I'm running this on a MacBook Pro. This recording was taken after everyone else in the house had gone to bed. I have a 1 gig fiber connection, but I was on wifi.

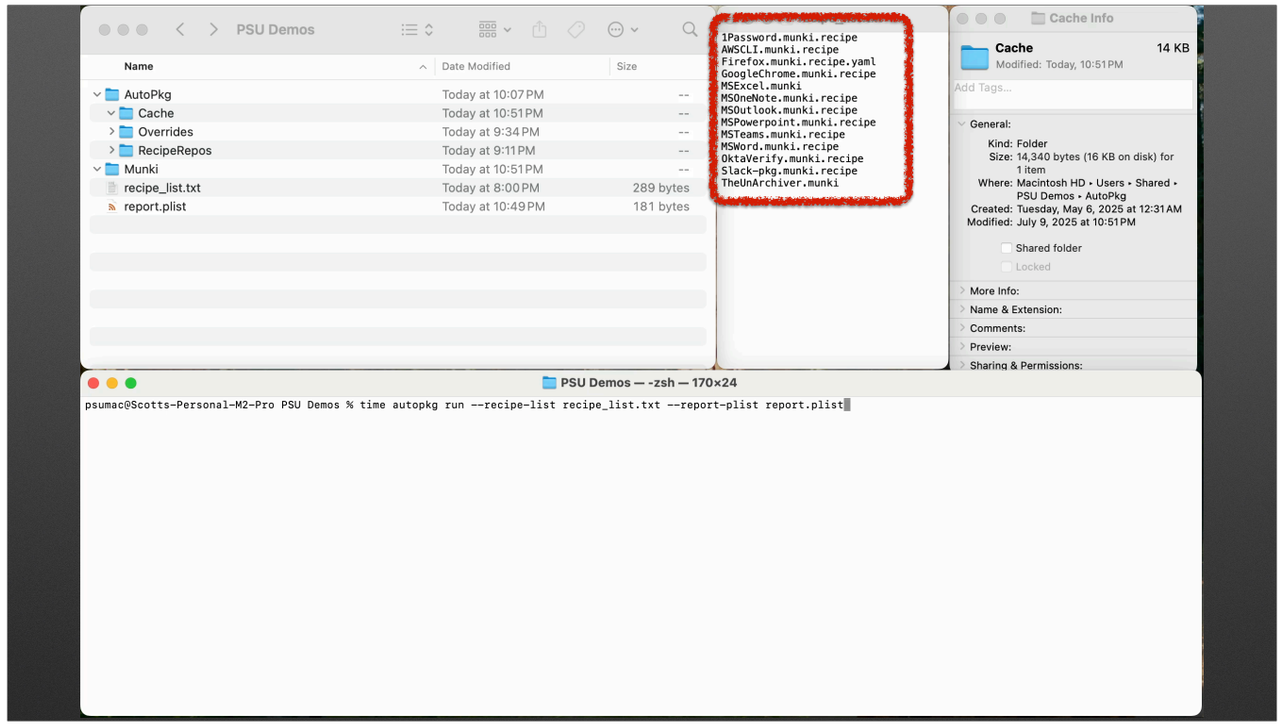
I mention all this so you can try to gauge how this compares to your environments.

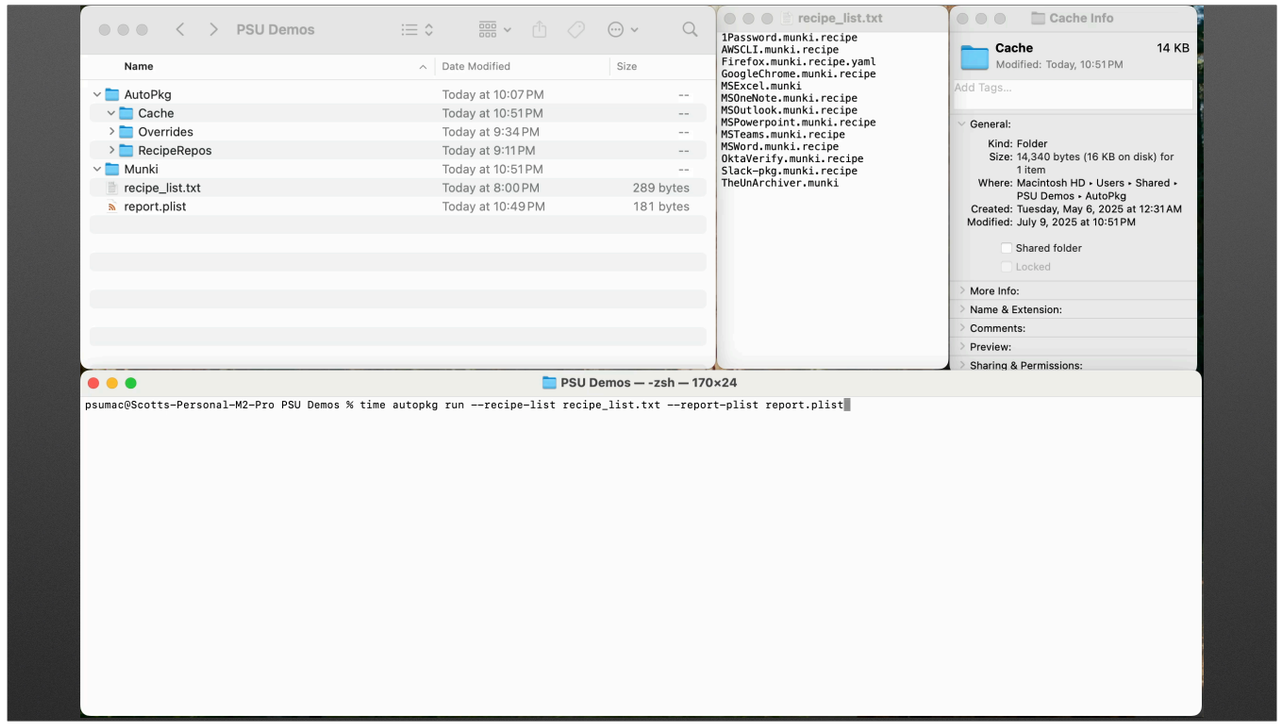
This is going to take a bit, so let's talk about what's on the screen.

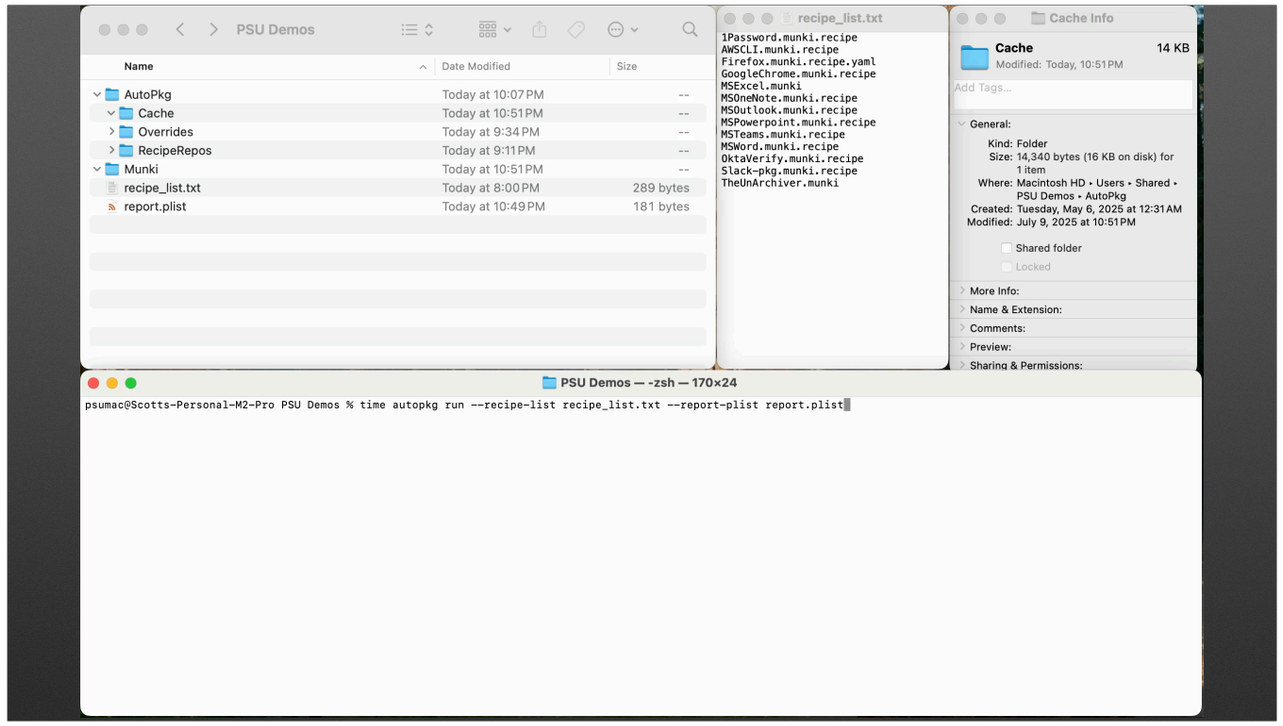

PSU Demos
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recipe_list.txt
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289 bytes
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MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
14 KB
Modified: Today, 10:51 PM
Add Tags...
General:
Kind: Folder
Size: 14,340 bytes (16 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:51 PM
Shared folder
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More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time autopkg run --recipe-list recipe_list.txt --report-plist report.plist



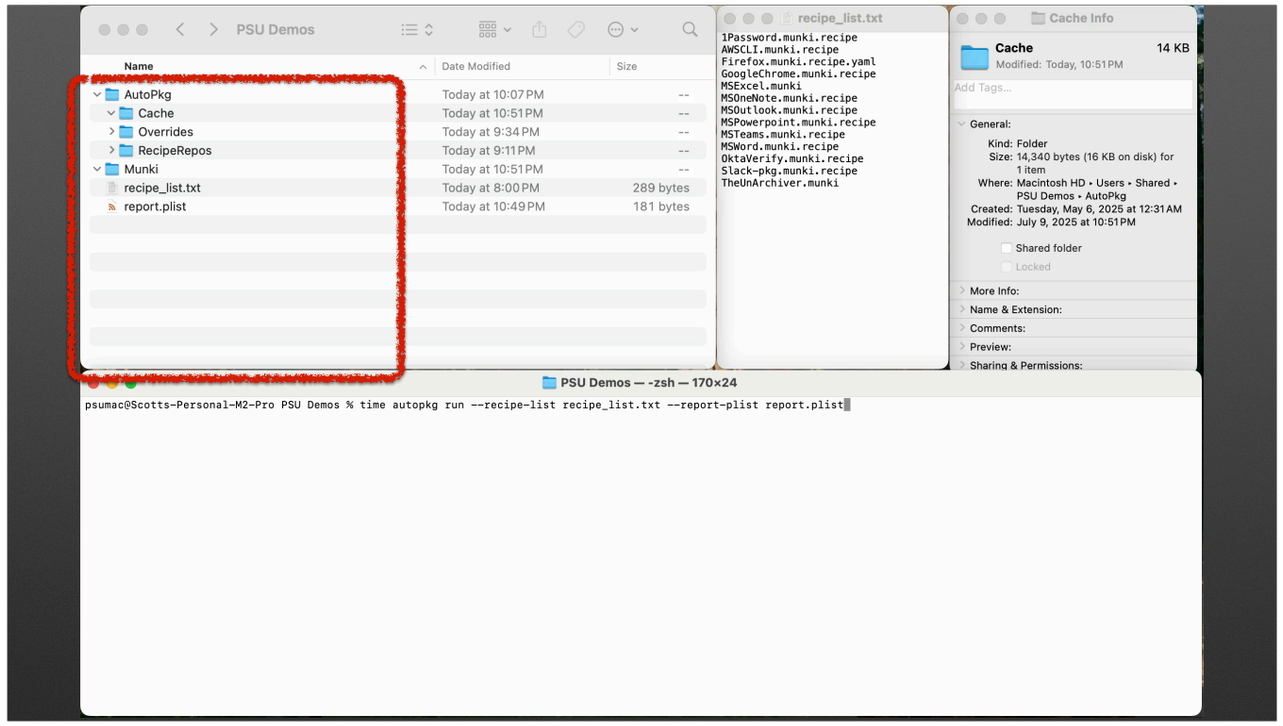
This is the contents of the recipe_list.txt file. It's a total of 13 recipes. We have 1Password, a couple browsers (Chrome and Firefox), Microsoft Office, Slack, Okta Verify, etc.

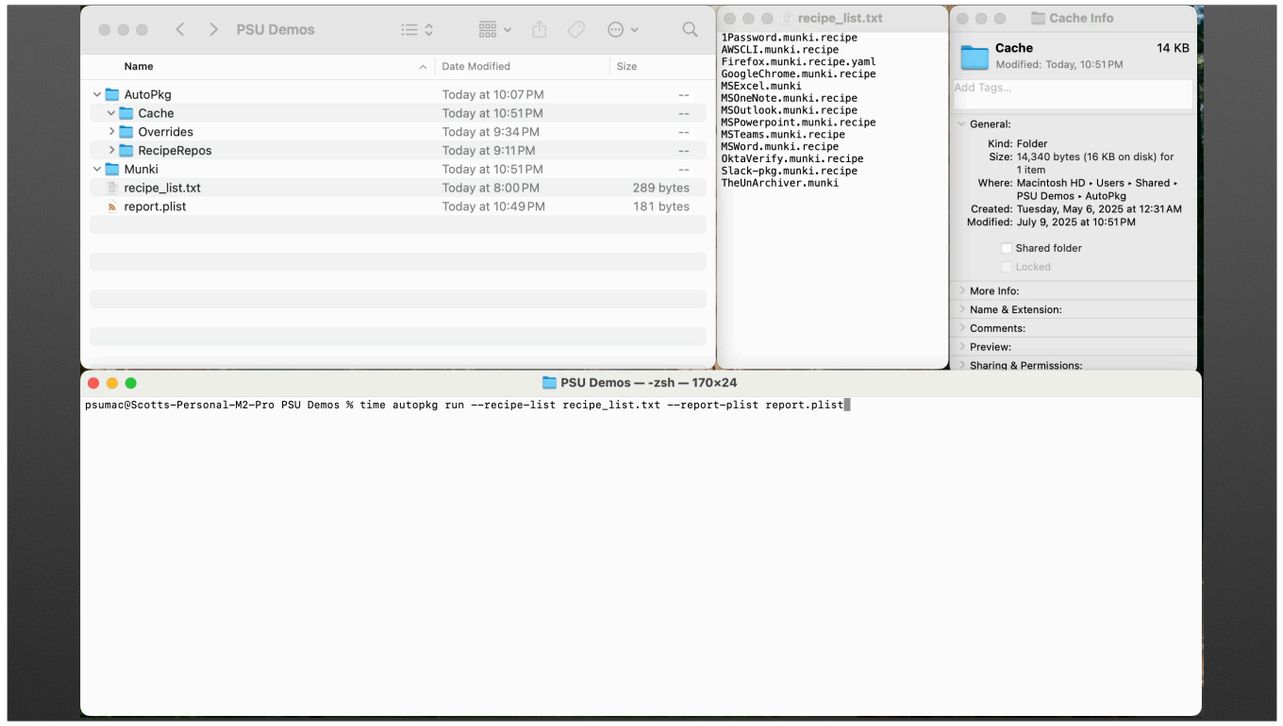

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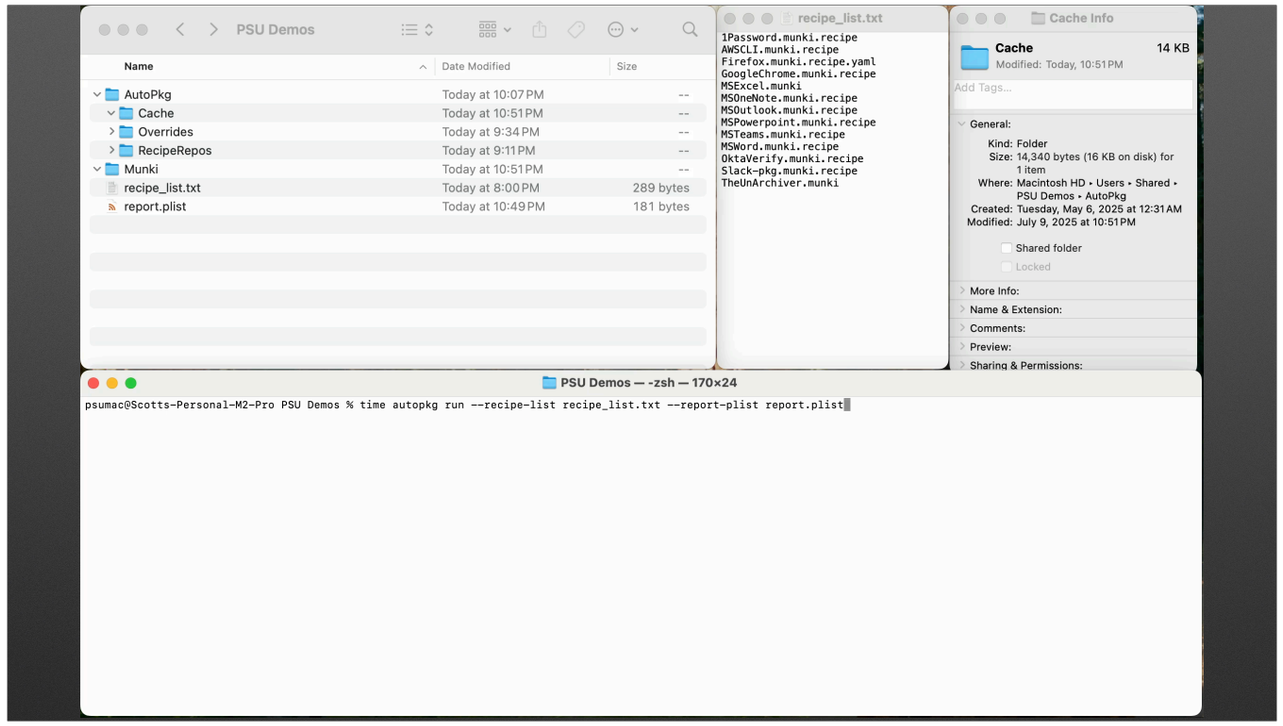

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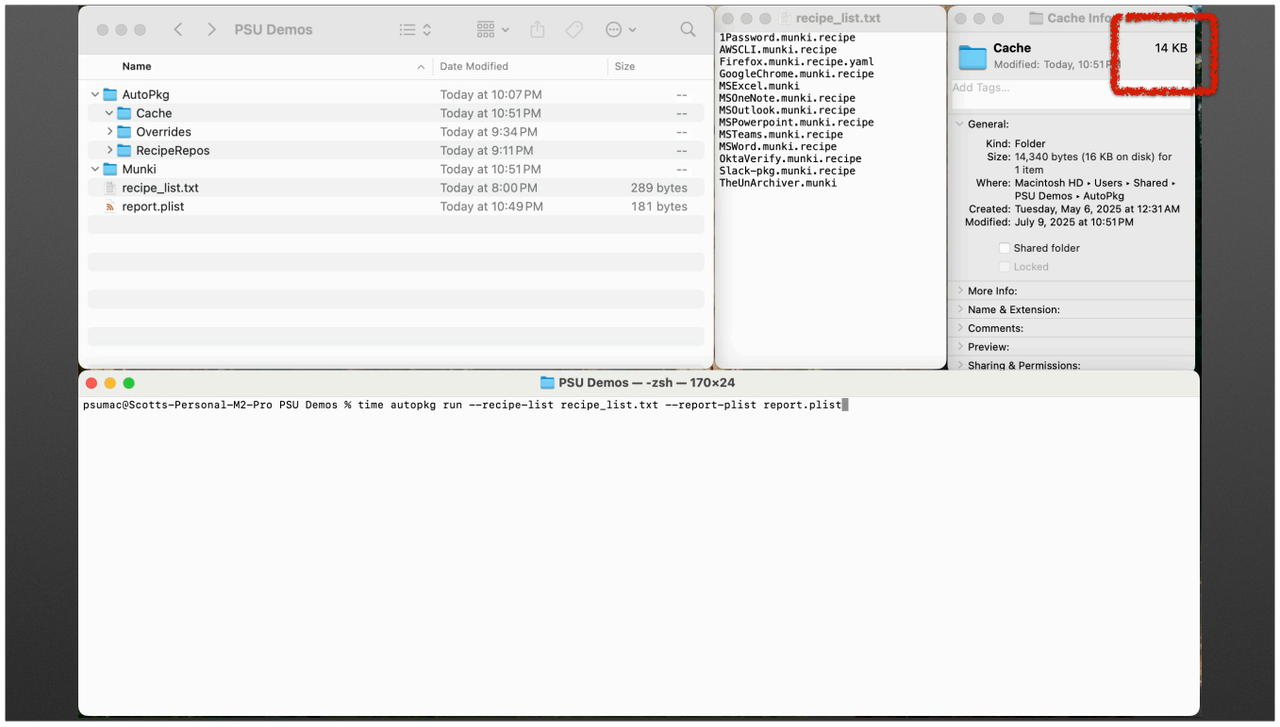
They are all Munki recipes, so you can see the output into the Munki repo over here. The AutoPkg cache is also populating as each recipe is processed and files are downloaded.

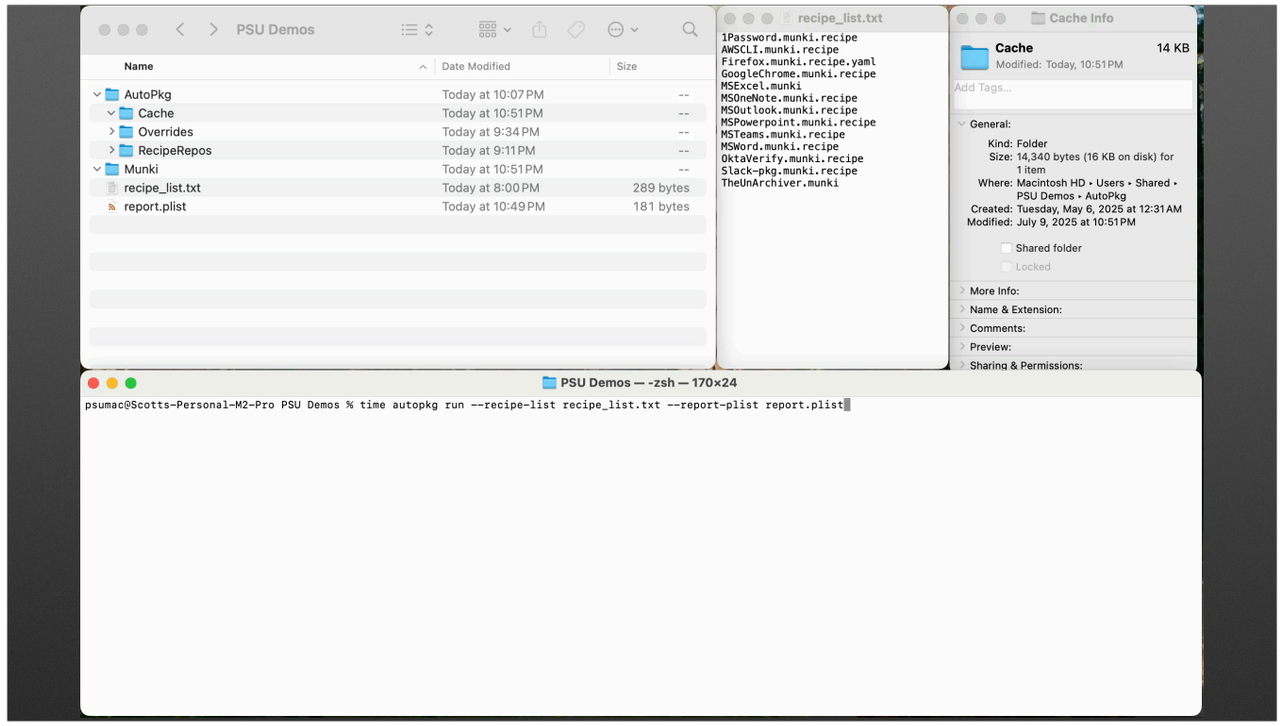

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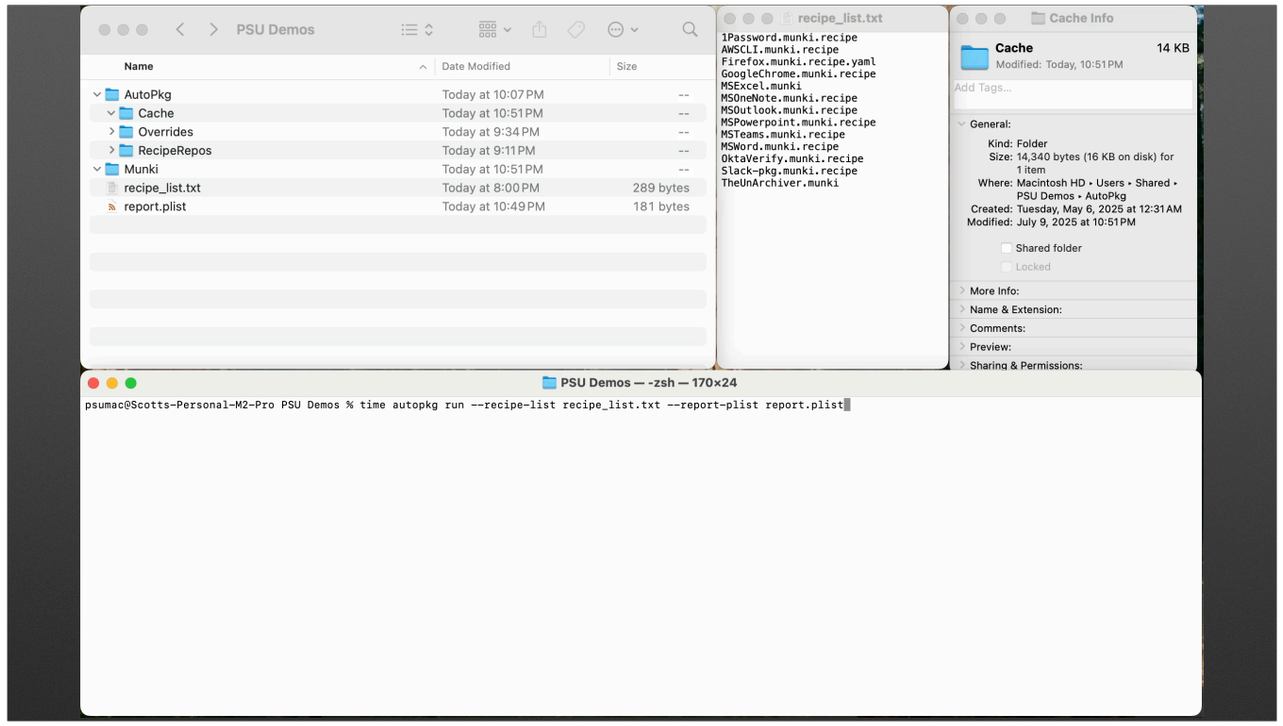

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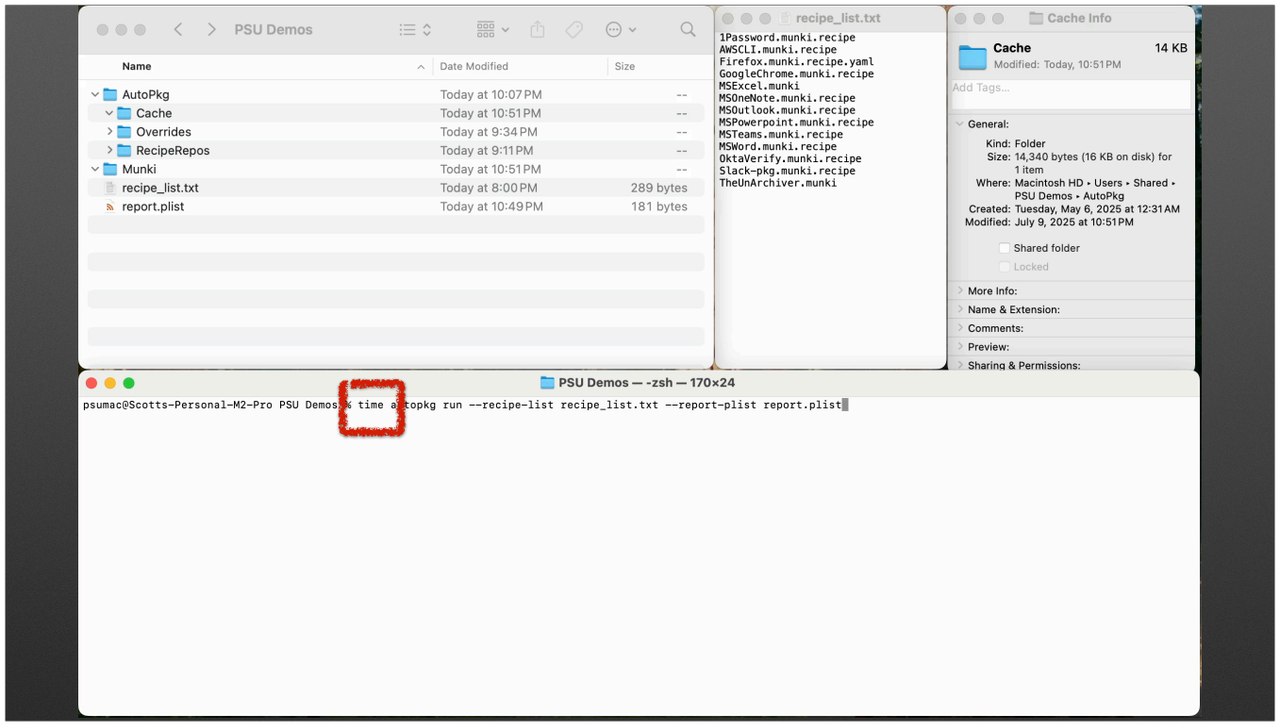
You can see the cache climbing here. The total download size from these recipes is a little over 6.5 GB, so we have some more time.

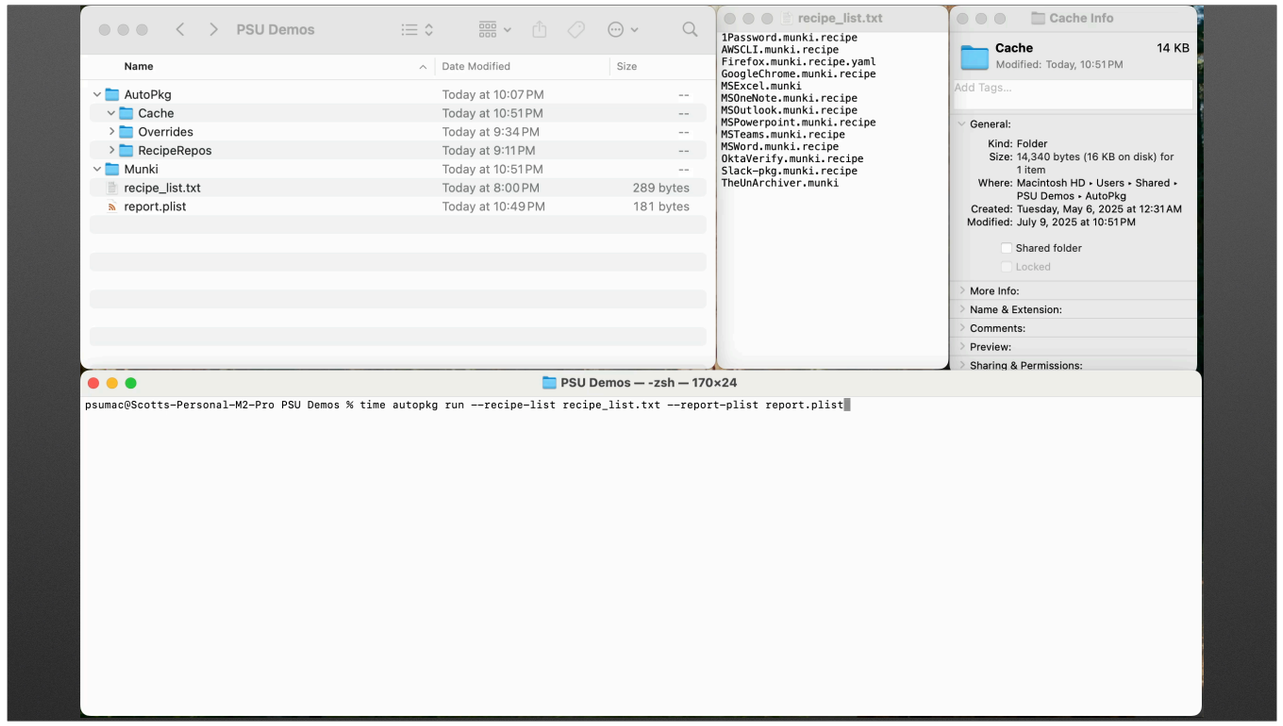

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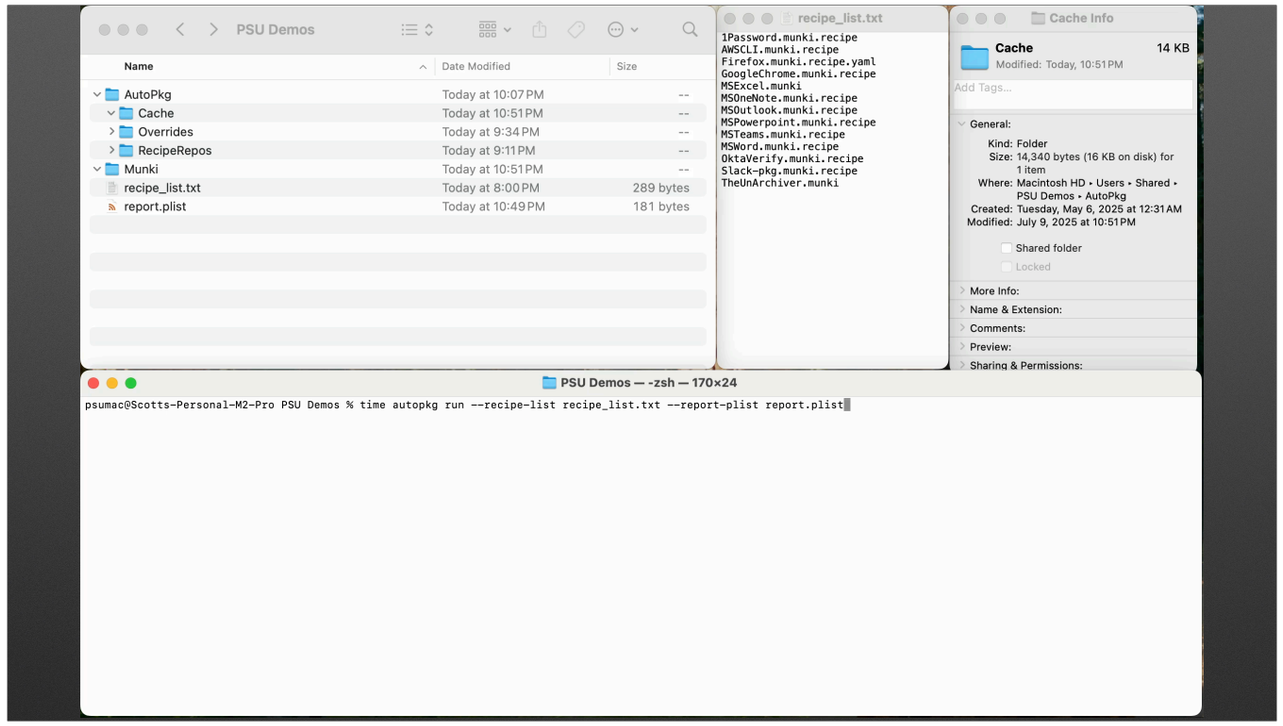

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psumac@Scotts-Personal-M2-Pro PSU Demos % time autopkg run --recipe-list recipe_list.txt --report-plist report.plist



Speaking of time, I made sure to run this with the time binary so that I would get an accurate run length when it completes. This recording is real-time.

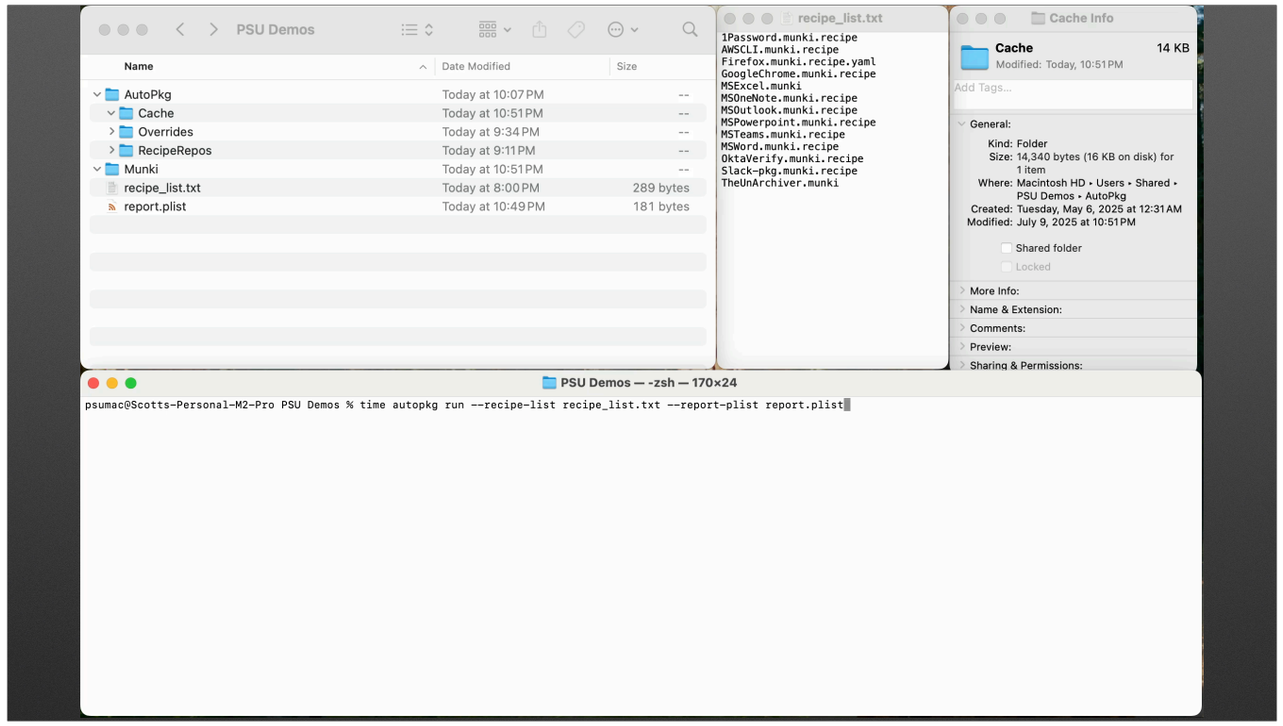

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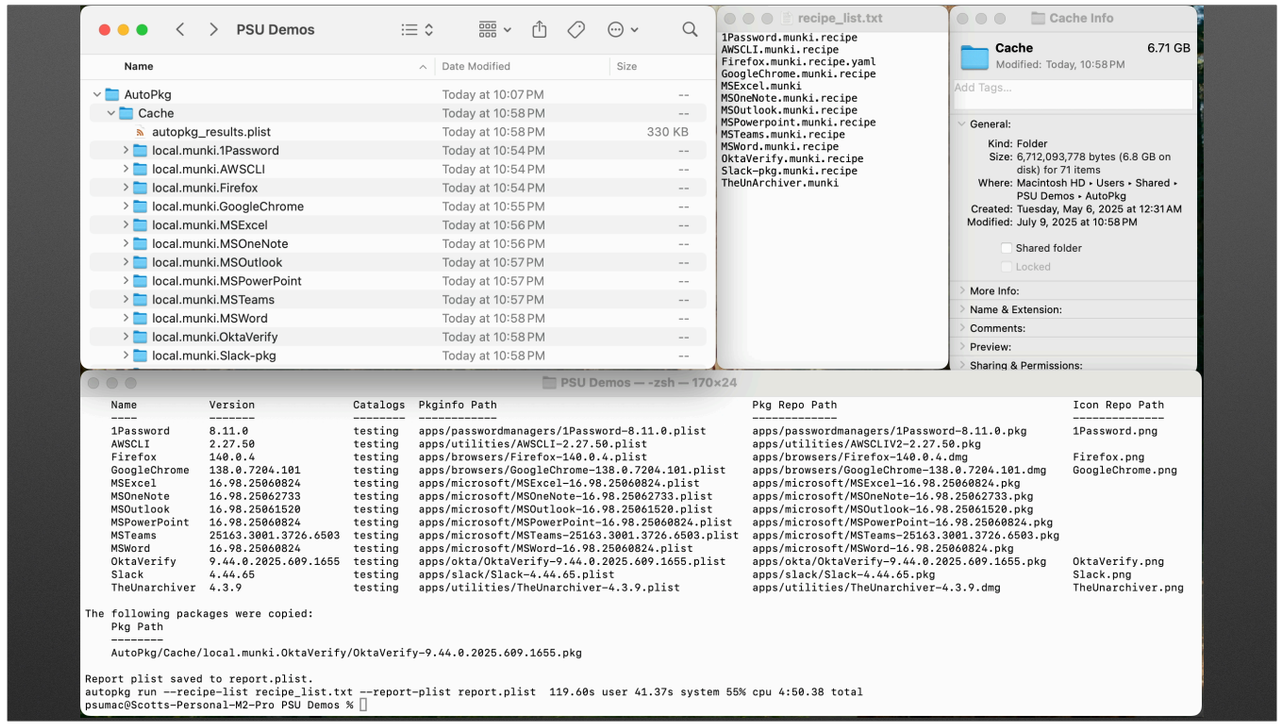

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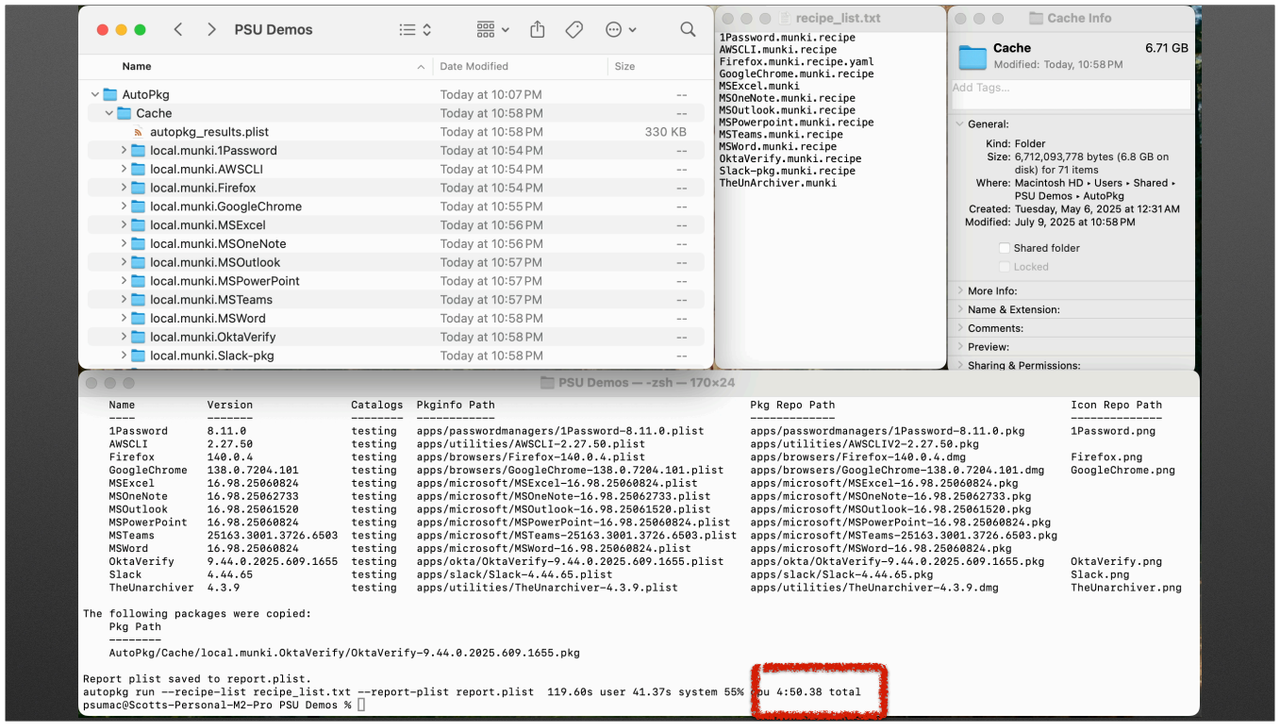
Okay, it seems we have some time to kill while this continues to run...
How many recipes do you run?
How often do you run them?
Would you run them more often if it was faster?
Is anyone in here NOT using AutoPkg?

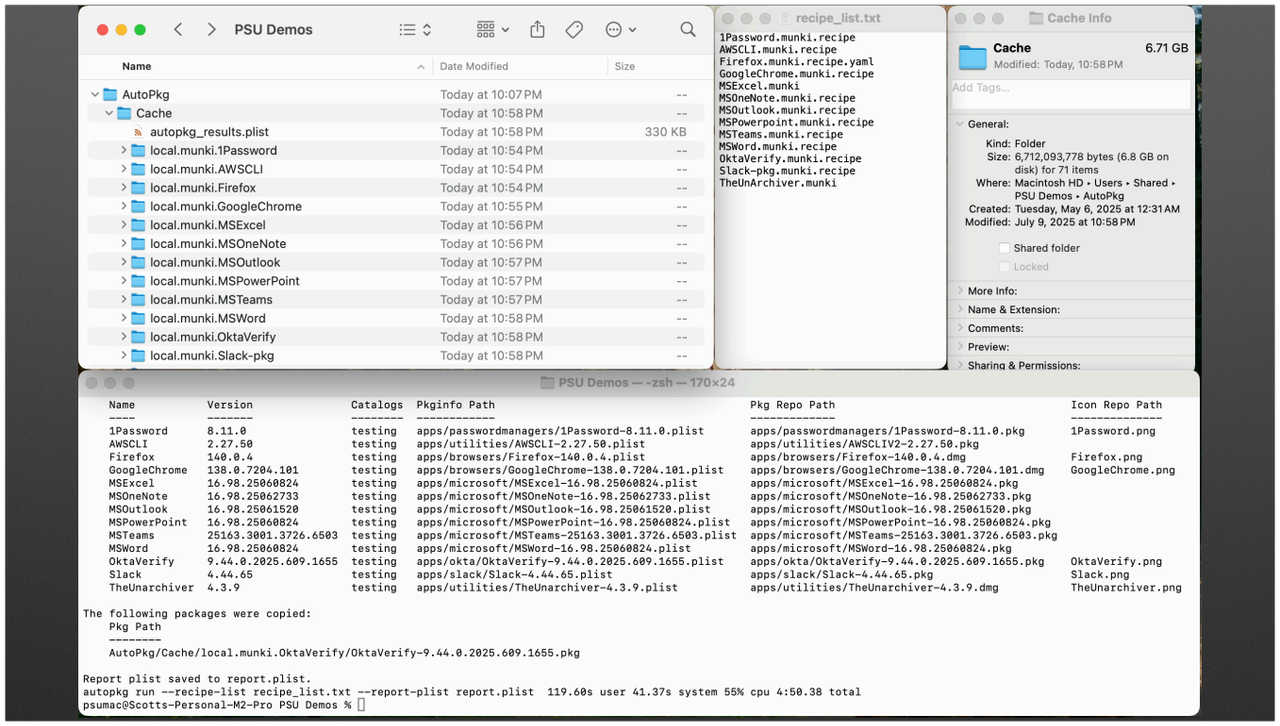
[Do not continue until video finishes.]

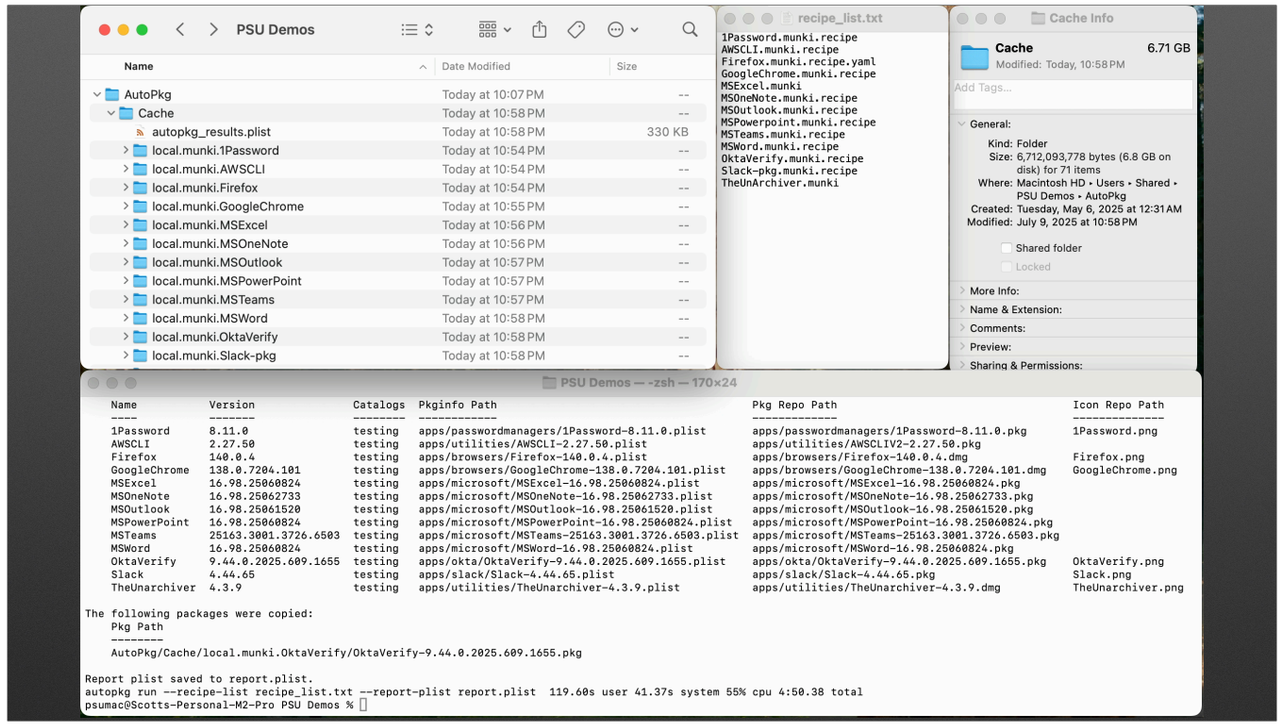

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 10:51 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 10:51 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
report.plist
Today at 10:49 PM
181 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
14 KB
Modified: Today, 10:51 PM
Add Tags...
General:
Kind: Folder
Size: 14,340 bytes (16 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:51 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time autopkg run --recipe-list recipe_list.txt --report-plist report.plist



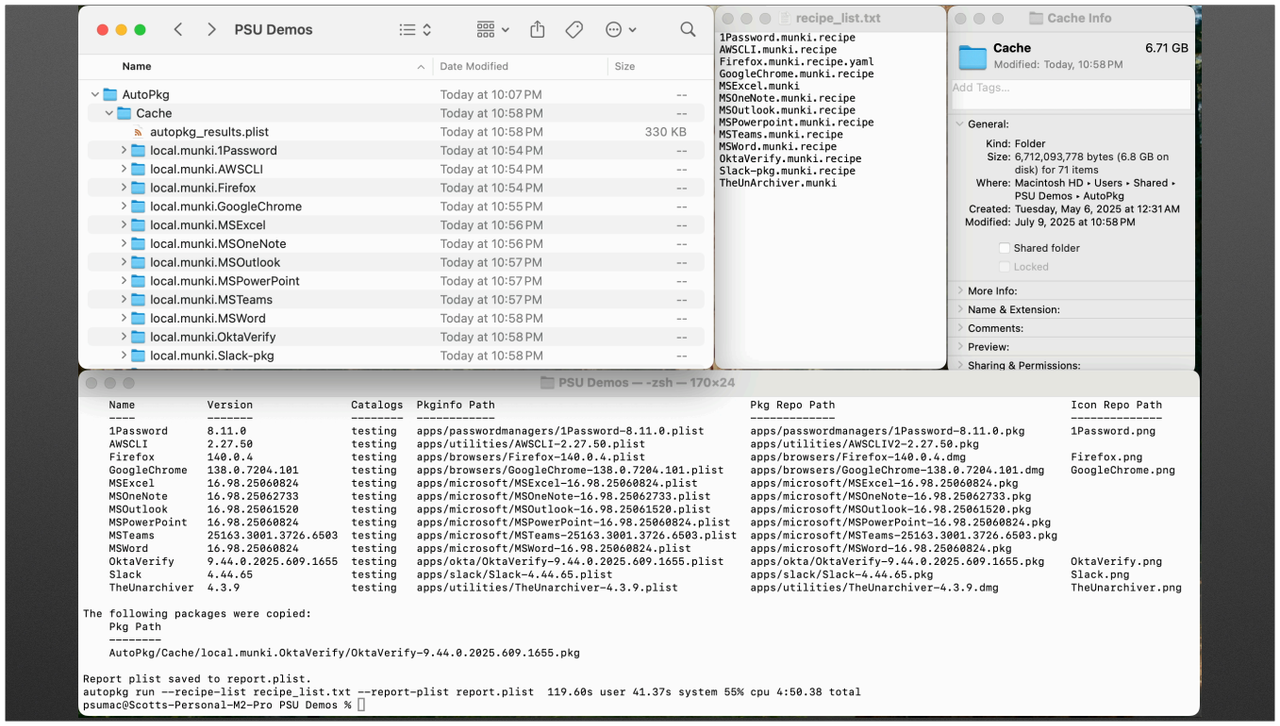
All 13 recipes successfully imported into Munki and it took 4 minutes and 50 seconds. Almost 5 minutes.

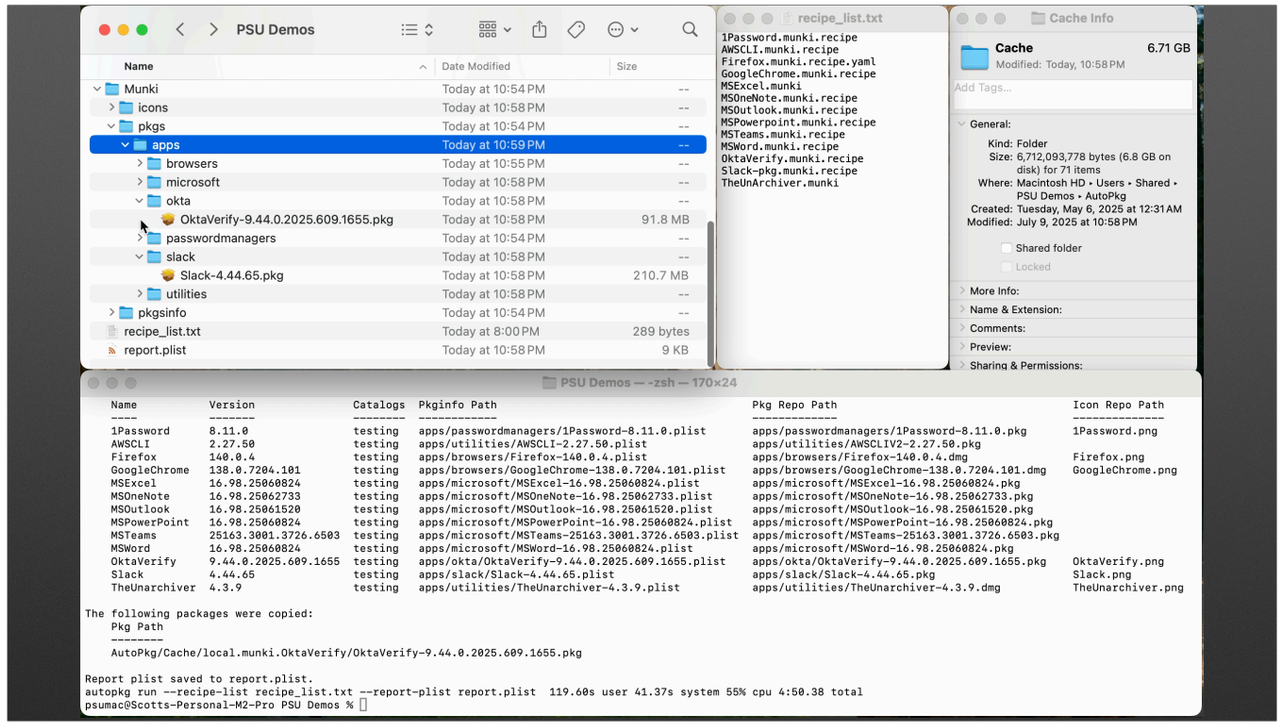

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 10:58 PM
autopkg_results.plist
Today at 10:58 PM
330 KB
local.munki.1Password
local.munki.AWSCLI
local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
local.munki.MSWord
local.munki.OktaVerify
local.munki.Slack-pkg
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Name Version Catalogs Pkginfo Path Pkg Repo Path Icon Repo Path
1Password 8.11.0 testing apps/passwordmanagers/1Password-8.11.0.plist apps/passwordmanagers/1Password-8.11.0.pkg 1Password.png
AWSCLI 2.27.50 testing apps/utilities/AWSCLI-2.27.50.plist apps/utilities/AWSCLIV2-2.27.50.pkg
Firefox 140.0.4 testing apps/browsers/Firefox-140.0.4.plist apps/browsers/Firefox-140.0.4.dmg Firefox.png
GoogleChrome 138.0.7204.101 testing apps/browsers/GoogleChrome-138.0.7204.101.plist apps/browsers/GoogleChrome-138.0.7204.101.dmg GoogleChrome.png
MSExcel 16.98.25060824 testing apps/microsoft/MSExcel-16.98.25060824.plist apps/microsoft/MSExcel-16.98.25060824.pkg
MSOneNote 16.98.25062733 testing apps/microsoft/MSOneNote-16.98.25062733.plist apps/microsoft/MSOneNote-16.98.25062733.pkg
MSOutlook 16.98.25061520 testing apps/microsoft/MSOutlook-16.98.25061520.plist apps/microsoft/MSOutlook-16.98.25061520.pkg
MSPowerPoint 16.98.25060824 testing apps/microsoft/MSPowerPoint-16.98.25060824.plist apps/microsoft/MSPowerPoint-16.98.25060824.pkg
MSTeams 25163.3001.3726.6503 testing apps/microsoft/MSTeams-25163.3001.3726.6503.plist apps/microsoft/MSTeams-25163.3001.3726.6503.pkg
MSWord 16.98.25060824 testing apps/microsoft/MSWord-16.98.25060824.plist apps/microsoft/MSWord-16.98.25060824.pkg
OktaVerify 9.44.0.2025.609.1655 testing apps/okta/OktaVerify-9.44.0.2025.609.1655.plist apps/okta/OktaVerify-9.44.0.2025.609.1655.pkg OktaVerify.png
Slack 4.44.65 testing apps/slack/Slack-4.44.65.plist apps/slack/Slack-4.44.65.pkg Slack.png
TheUnarchiver 4.3.9 testing apps/utilities/TheUnarchiver-4.3.9.plist apps/utilities/TheUnarchiver-4.3.9.dmg TheUnarchiver.png
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 119.60s user 41.37s system 55% cpu 4:50.38 total
psumac@Scotts-Personal-M2-Pro PSU Demos %


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Today at 10:07 PM
Cache
Today at 10:58 PM
autopkg_results.plist
Today at 10:58 PM
330 KB
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local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
local.munki.MSWord
local.munki.OktaVerify
local.munki.Slack-pkg
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Name Version Catalogs Pkginfo Path Pkg Repo Path Icon Repo Path
1Password 8.11.0 testing apps/passwordmanagers/1Password-8.11.0.plist apps/passwordmanagers/1Password-8.11.0.pkg 1Password.png
AWSCLI 2.27.50 testing apps/utilities/AWSCLI-2.27.50.plist apps/utilities/AWSCLIV2-2.27.50.pkg
Firefox 140.0.4 testing apps/browsers/Firefox-140.0.4.plist apps/browsers/Firefox-140.0.4.dmg Firefox.png
GoogleChrome 138.0.7204.101 testing apps/browsers/GoogleChrome-138.0.7204.101.plist apps/browsers/GoogleChrome-138.0.7204.101.dmg GoogleChrome.png
MSExcel 16.98.25060824 testing apps/microsoft/MSExcel-16.98.25060824.plist apps/microsoft/MSExcel-16.98.25060824.pkg
MSOneNote 16.98.25062733 testing apps/microsoft/MSOneNote-16.98.25062733.plist apps/microsoft/MSOneNote-16.98.25062733.pkg
MSOutlook 16.98.25061520 testing apps/microsoft/MSOutlook-16.98.25061520.plist apps/microsoft/MSOutlook-16.98.25061520.pkg
MSPowerPoint 16.98.25060824 testing apps/microsoft/MSPowerPoint-16.98.25060824.plist apps/microsoft/MSPowerPoint-16.98.25060824.pkg
MSTeams 25163.3001.3726.6503 testing apps/microsoft/MSTeams-25163.3001.3726.6503.plist apps/microsoft/MSTeams-25163.3001.3726.6503.pkg
MSWord 16.98.25060824 testing apps/microsoft/MSWord-16.98.25060824.plist apps/microsoft/MSWord-16.98.25060824.pkg
OktaVerify 9.44.0.2025.609.1655 testing apps/okta/OktaVerify-9.44.0.2025.609.1655.plist apps/okta/OktaVerify-9.44.0.2025.609.1655.pkg OktaVerify.png
Slack 4.44.65 testing apps/slack/Slack-4.44.65.plist apps/slack/Slack-4.44.65.pkg Slack.png
TheUnarchiver 4.3.9 testing apps/utilities/TheUnarchiver-4.3.9.plist apps/utilities/TheUnarchiver-4.3.9.dmg TheUnarchiver.png
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 119.60s user 41.37s system 55% cpu 4:50.38 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

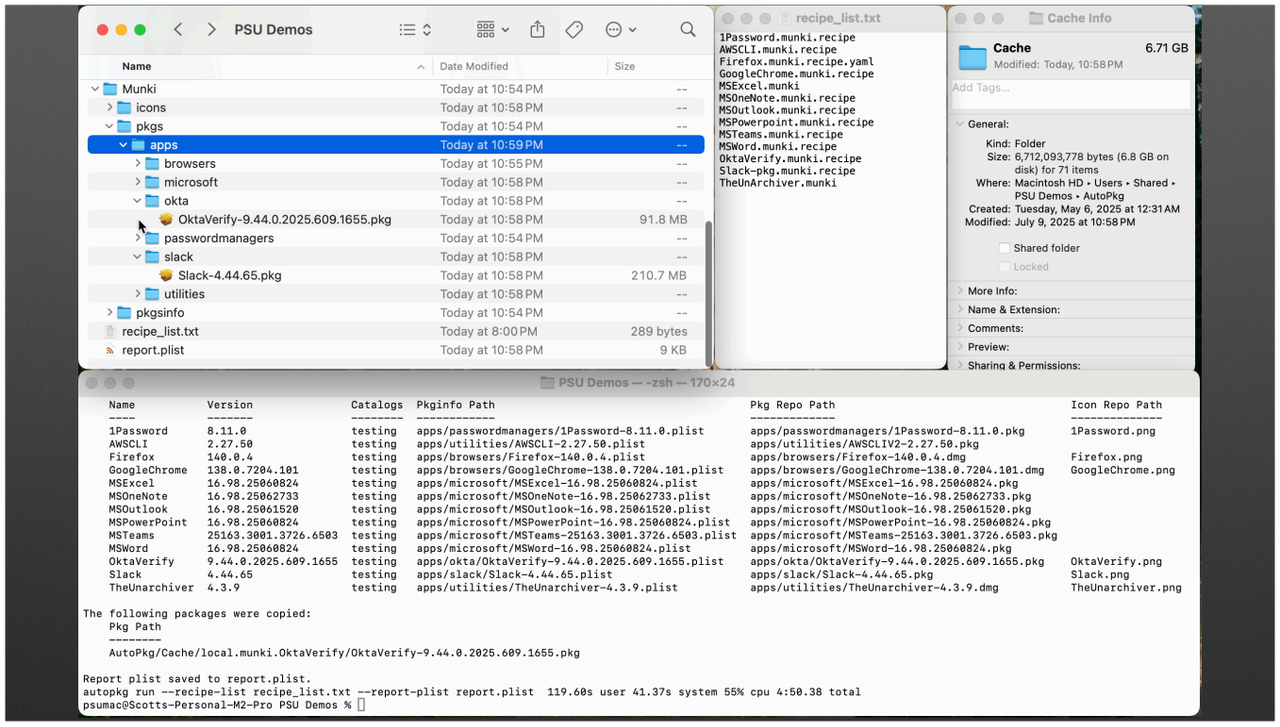


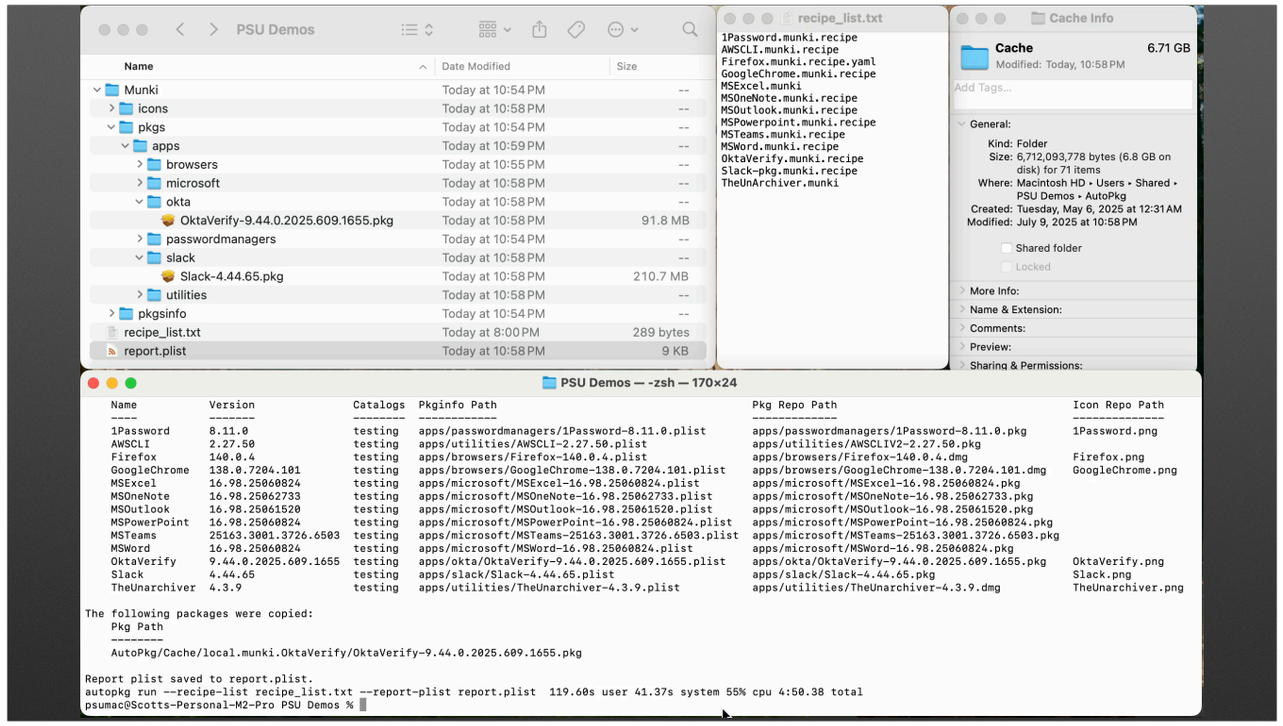
Let's take a look at the output. If we scroll down, we can see the installers were all stored in the munki directory as expected.


PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 10:58 PM
autopkg_results.plist
Today at 10:58 PM
330 KB
local.munki.1Password
local.munki.AWSCLI
local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
local.munki.MSWord
local.munki.OktaVerify
local.munki.Slack-pkg
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Name Version Catalogs Pkginfo Path Pkg Repo Path Icon Repo Path
1Password 8.11.0 testing apps/passwordmanagers/1Password-8.11.0.plist apps/passwordmanagers/1Password-8.11.0.pkg 1Password.png
AWSCLI 2.27.50 testing apps/utilities/AWSCLI-2.27.50.plist apps/utilities/AWSCLIV2-2.27.50.pkg
Firefox 140.0.4 testing apps/browsers/Firefox-140.0.4.plist apps/browsers/Firefox-140.0.4.dmg Firefox.png
GoogleChrome 138.0.7204.101 testing apps/browsers/GoogleChrome-138.0.7204.101.plist apps/browsers/GoogleChrome-138.0.7204.101.dmg GoogleChrome.png
MSExcel 16.98.25060824 testing apps/microsoft/MSExcel-16.98.25060824.plist apps/microsoft/MSExcel-16.98.25060824.pkg
MSOneNote 16.98.25062733 testing apps/microsoft/MSOneNote-16.98.25062733.plist apps/microsoft/MSOneNote-16.98.25062733.pkg
MSOutlook 16.98.25061520 testing apps/microsoft/MSOutlook-16.98.25061520.plist apps/microsoft/MSOutlook-16.98.25061520.pkg
MSPowerPoint 16.98.25060824 testing apps/microsoft/MSPowerPoint-16.98.25060824.plist apps/microsoft/MSPowerPoint-16.98.25060824.pkg
MSTeams 25163.3001.3726.6503 testing apps/microsoft/MSTeams-25163.3001.3726.6503.plist apps/microsoft/MSTeams-25163.3001.3726.6503.pkg
MSWord 16.98.25060824 testing apps/microsoft/MSWord-16.98.25060824.plist apps/microsoft/MSWord-16.98.25060824.pkg
OktaVerify 9.44.0.2025.609.1655 testing apps/okta/OktaVerify-9.44.0.2025.609.1655.plist apps/okta/OktaVerify-9.44.0.2025.609.1655.pkg OktaVerify.png
Slack 4.44.65 testing apps/slack/Slack-4.44.65.plist apps/slack/Slack-4.44.65.pkg Slack.png
TheUnarchiver 4.3.9 testing apps/utilities/TheUnarchiver-4.3.9.plist apps/utilities/TheUnarchiver-4.3.9.dmg TheUnarchiver.png
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 119.60s user 41.37s system 55% cpu 4:50.38 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

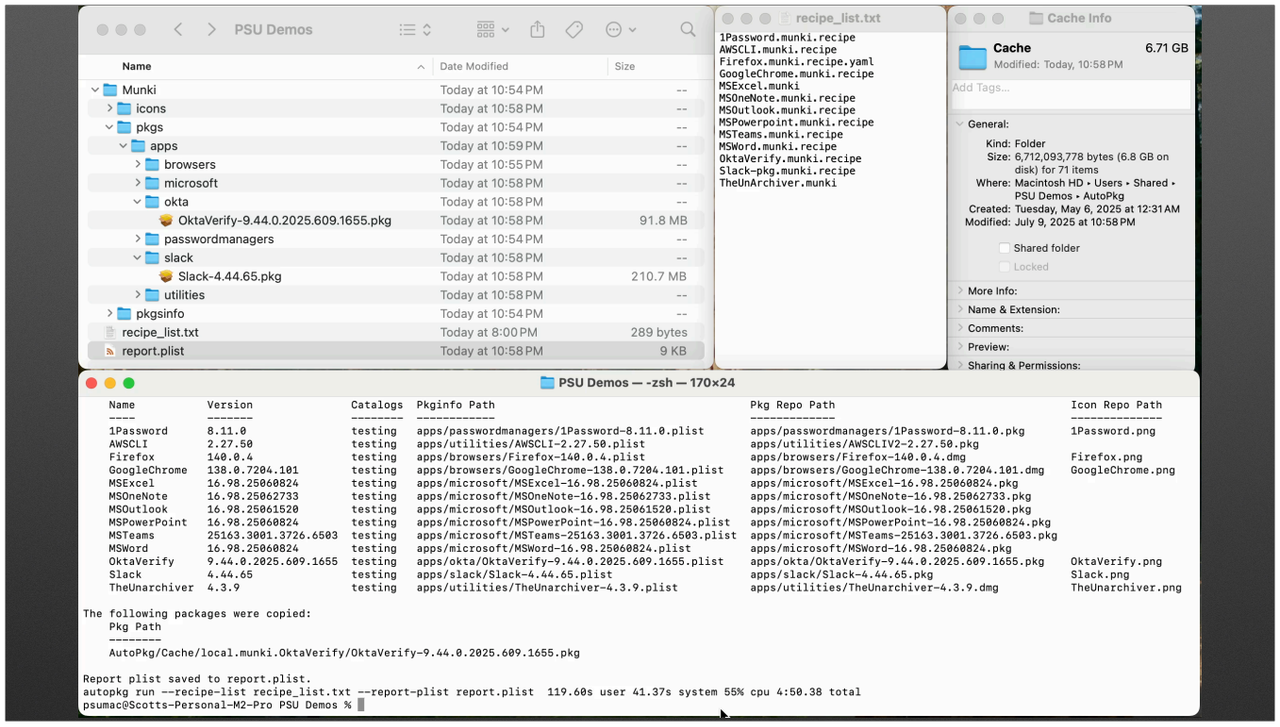


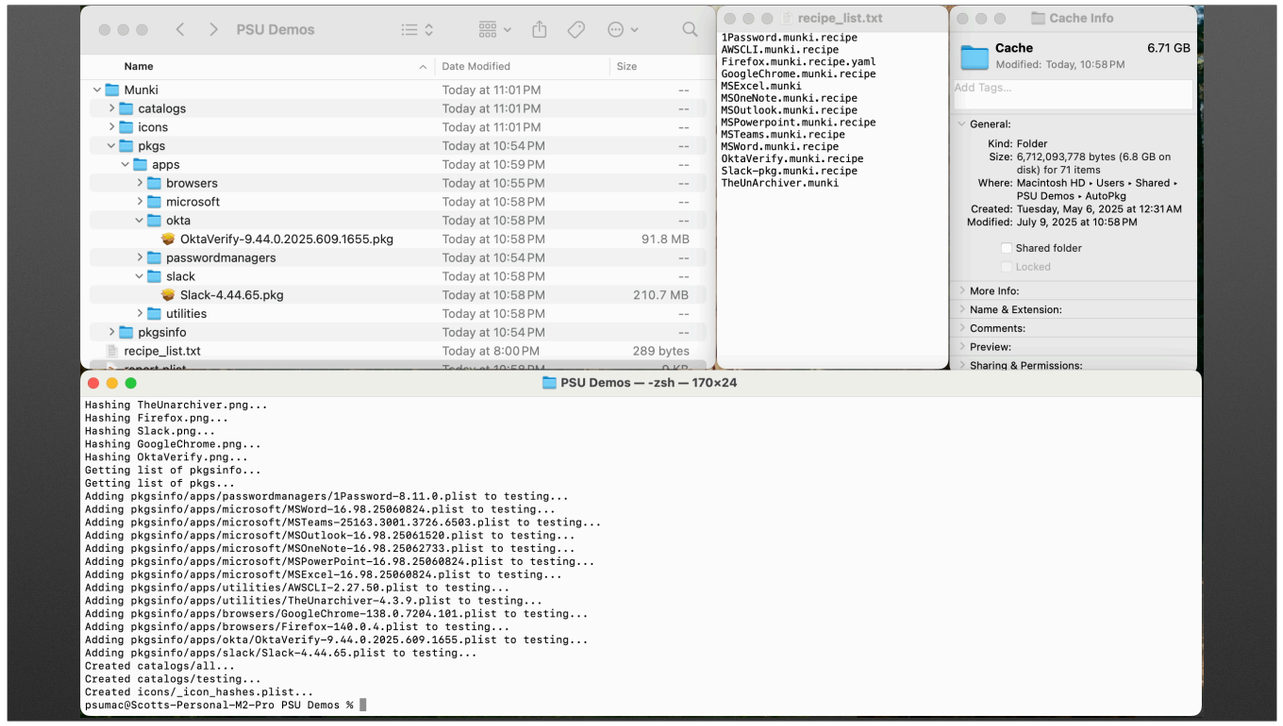
And we can open report.plist and see the structured output from that run.


PSU Demos
Name
Date Modified
Size
Munki
Today at 10:54 PM
icons
Today at 10:58 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
report.plist
Today at 10:58 PM
9 KB
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
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Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Name Version Catalogs Pkginfo Path Pkg Repo Path Icon Repo Path
1Password 8.11.0 testing apps/passwordmanagers/1Password-8.11.0.plist apps/passwordmanagers/1Password-8.11.0.pkg 1Password.png
AWSCLI 2.27.50 testing apps/utilities/AWSCLI-2.27.50.plist apps/utilities/AWSCLIV2-2.27.50.pkg
Firefox 140.0.4 testing apps/browsers/Firefox-140.0.4.plist apps/browsers/Firefox-140.0.4.dmg Firefox.png
GoogleChrome 138.0.7204.101 testing apps/browsers/GoogleChrome-138.0.7204.101.plist apps/browsers/GoogleChrome-138.0.7204.101.dmg GoogleChrome.png
MSExcel 16.98.25060824 testing apps/microsoft/MSExcel-16.98.25060824.plist apps/microsoft/MSExcel-16.98.25060824.pkg
MSOneNote 16.98.25062733 testing apps/microsoft/MSOneNote-16.98.25062733.plist apps/microsoft/MSOneNote-16.98.25062733.pkg
MSOutlook 16.98.25061520 testing apps/microsoft/MSOutlook-16.98.25061520.plist apps/microsoft/MSOutlook-16.98.25061520.pkg
MSPowerPoint 16.98.25060824 testing apps/microsoft/MSPowerPoint-16.98.25060824.plist apps/microsoft/MSPowerPoint-16.98.25060824.pkg
MSTeams 25163.3001.3726.6503 testing apps/microsoft/MSTeams-25163.3001.3726.6503.plist apps/microsoft/MSTeams-25163.3001.3726.6503.pkg
MSWord 16.98.25060824 testing apps/microsoft/MSWord-16.98.25060824.plist apps/microsoft/MSWord-16.98.25060824.pkg
OktaVerify 9.44.0.2025.609.1655 testing apps/okta/OktaVerify-9.44.0.2025.609.1655.plist apps/okta/OktaVerify-9.44.0.2025.609.1655.pkg OktaVerify.png
Slack 4.44.65 testing apps/slack/Slack-4.44.65.plist apps/slack/Slack-4.44.65.pkg Slack.png
TheUnarchiver 4.3.9 testing apps/utilities/TheUnarchiver-4.3.9.plist apps/utilities/TheUnarchiver-4.3.9.dmg TheUnarchiver.png
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
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autopkg run --recipe-list recipe_list.txt --report-plist report.plist 119.60s user 41.37s system 55% cpu 4:50.38 total
psumac@Scotts-Personal-M2-Pro PSU Demos %



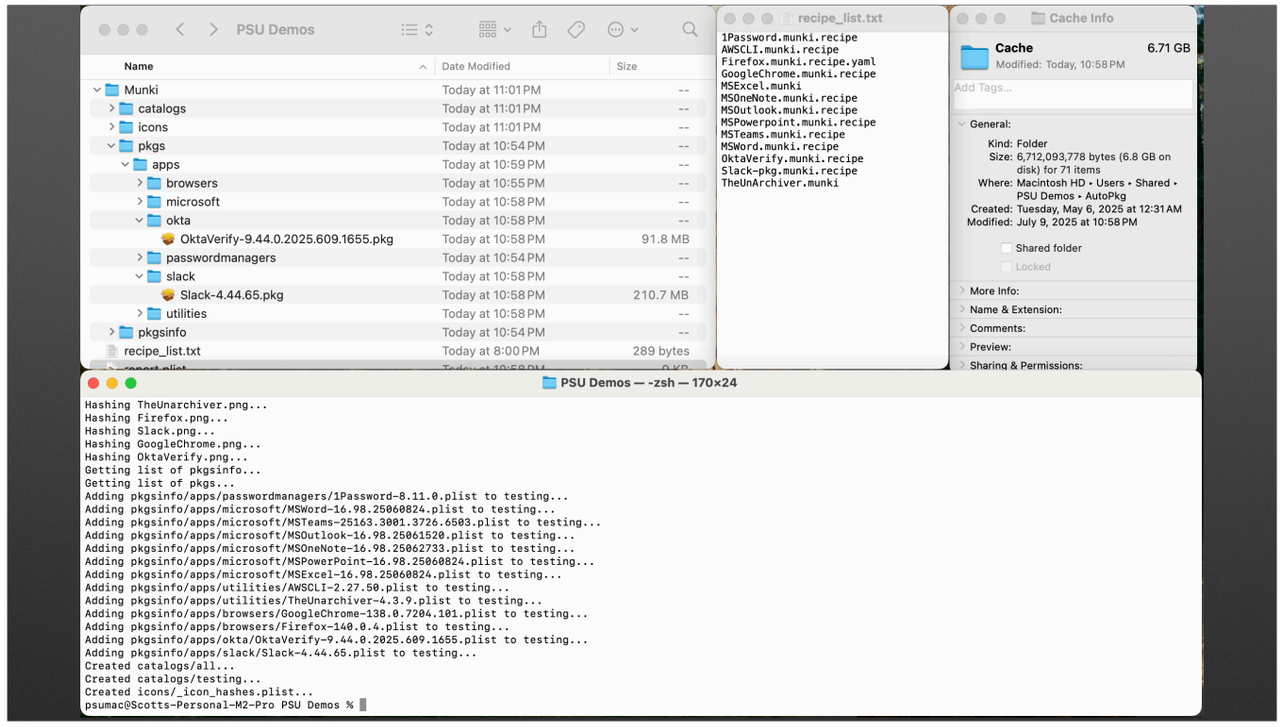
Now that AutoPkg is done, we can run `makecatalogs` on the Munki directory to prevent duplicate imports and keep our comparisons as clean as possible.

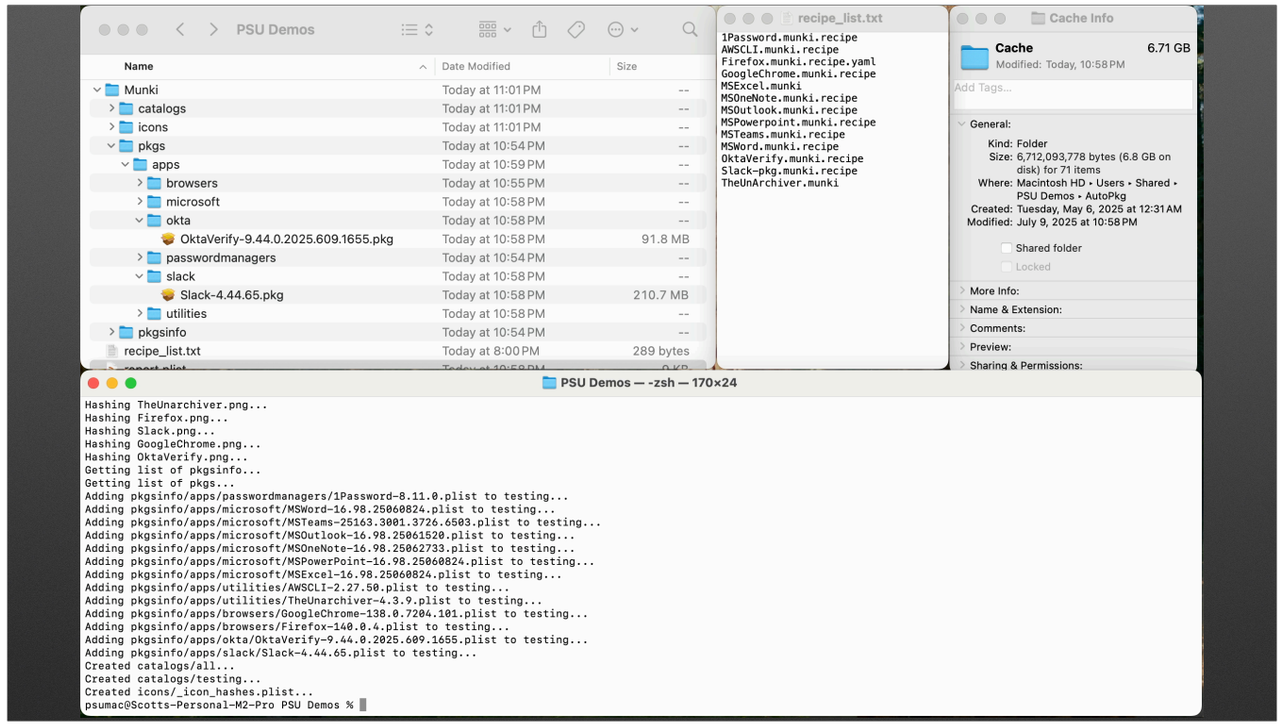

PSU Demos
Name
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Today at 10:54 PM
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Today at 10:58 PM
pkgs
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Today at 10:59 PM
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Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
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Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
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Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
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PSU Demos — -zsh — 170×24
Name Version Catalogs Pkginfo Path Pkg Repo Path Icon Repo Path
1Password 8.11.0 testing apps/passwordmanagers/1Password-8.11.0.plist apps/passwordmanagers/1Password-8.11.0.pkg 1Password.png
AWSCLI 2.27.50 testing apps/utilities/AWSCLI-2.27.50.plist apps/utilities/AWSCLIV2-2.27.50.pkg
Firefox 140.0.4 testing apps/browsers/Firefox-140.0.4.plist apps/browsers/Firefox-140.0.4.dmg Firefox.png
GoogleChrome 138.0.7204.101 testing apps/browsers/GoogleChrome-138.0.7204.101.plist apps/browsers/GoogleChrome-138.0.7204.101.dmg GoogleChrome.png
MSExcel 16.98.25060824 testing apps/microsoft/MSExcel-16.98.25060824.plist apps/microsoft/MSExcel-16.98.25060824.pkg
MSOneNote 16.98.25062733 testing apps/microsoft/MSOneNote-16.98.25062733.plist apps/microsoft/MSOneNote-16.98.25062733.pkg
MSOutlook 16.98.25061520 testing apps/microsoft/MSOutlook-16.98.25061520.plist apps/microsoft/MSOutlook-16.98.25061520.pkg
MSPowerPoint 16.98.25060824 testing apps/microsoft/MSPowerPoint-16.98.25060824.plist apps/microsoft/MSPowerPoint-16.98.25060824.pkg
MSTeams 25163.3001.3726.6503 testing apps/microsoft/MSTeams-25163.3001.3726.6503.plist apps/microsoft/MSTeams-25163.3001.3726.6503.pkg
MSWord 16.98.25060824 testing apps/microsoft/MSWord-16.98.25060824.plist apps/microsoft/MSWord-16.98.25060824.pkg
OktaVerify 9.44.0.2025.609.1655 testing apps/okta/OktaVerify-9.44.0.2025.609.1655.plist apps/okta/OktaVerify-9.44.0.2025.609.1655.pkg OktaVerify.png
Slack 4.44.65 testing apps/slack/Slack-4.44.65.plist apps/slack/Slack-4.44.65.pkg Slack.png
TheUnarchiver 4.3.9 testing apps/utilities/TheUnarchiver-4.3.9.plist apps/utilities/TheUnarchiver-4.3.9.dmg TheUnarchiver.png
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 119.60s user 41.37s system 55% cpu 4:50.38 total
psumac@Scotts-Personal-M2-Pro PSU Demos %



Let's clear the screen and run autopkg again.

Keep in mind that we just ran this. The cache is full, and it it *very* unlikely that any of these packages updated in the last 5 minutes or less.

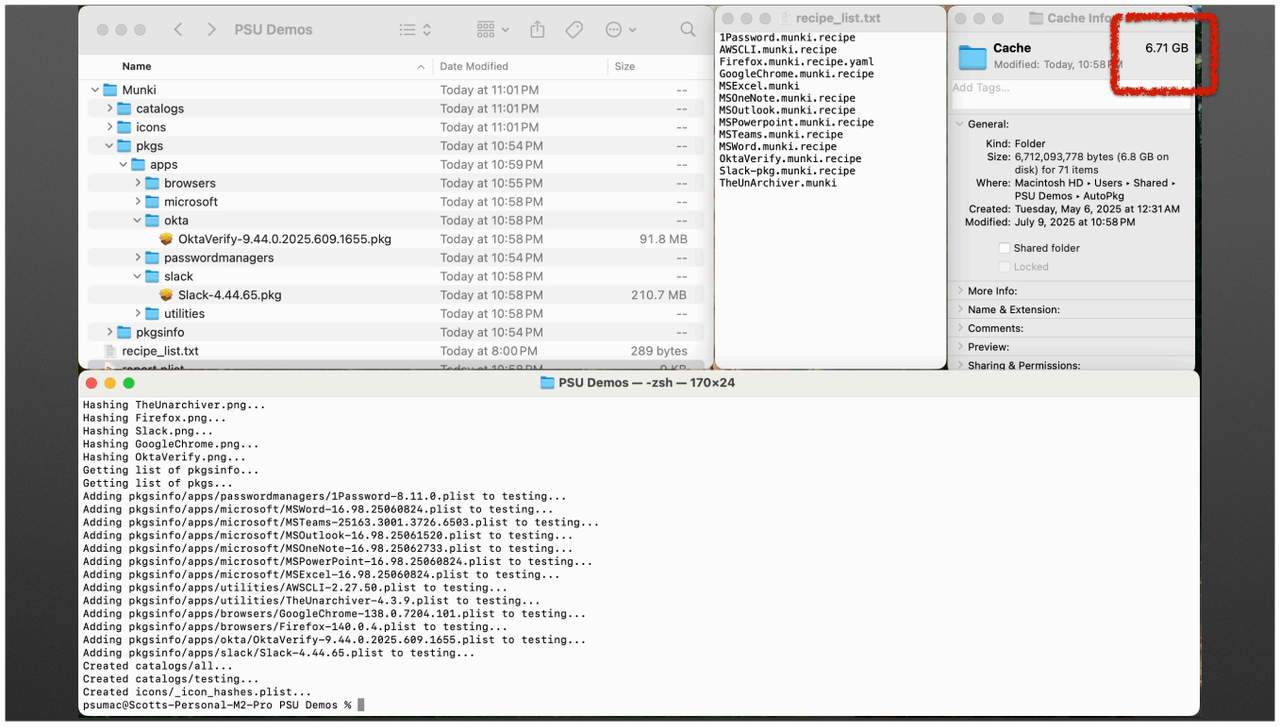

PSU Demos
Name
Date Modified
Size
Munki
Today at 11:01 PM
catalogs
Today at 11:01 PM
icons
Today at 11:01 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Hashing TheUnarchiver.png...
Hashing Firefox.png...
Hashing Slack.png...
Hashing GoogleChrome.png...
Hashing OktaVerify.png...
Getting list of pkgsinfo...
Getting list of pkgs...
Adding pkgsinfo/apps/passwordmanagers/1Password-8.11.0.plist to testing...
Adding pkgsinfo/apps/microsoft/MSWord-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSTeams-25163.3001.3726.6503.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOutlook-16.98.25061520.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOneNote-16.98.25062733.plist to testing...
Adding pkgsinfo/apps/microsoft/MSPowerPoint-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSExcel-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/utilities/AWSCLI-2.27.50.plist to testing...
Adding pkgsinfo/apps/utilities/TheUnarchiver-4.3.9.plist to testing...
Adding pkgsinfo/apps/browsers/GoogleChrome-138.0.7204.101.plist to testing...
Adding pkgsinfo/apps/browsers/Firefox-140.0.4.plist to testing...
Adding pkgsinfo/apps/okta/OktaVerify-9.44.0.2025.609.1655.plist to testing...
Adding pkgsinfo/apps/slack/Slack-4.44.65.plist to testing...
Created catalogs/all...
Created catalogs/testing...
Created icons/_icon_hashes.plist...
psumac@Scotts-Personal-M2-Pro PSU Demos %

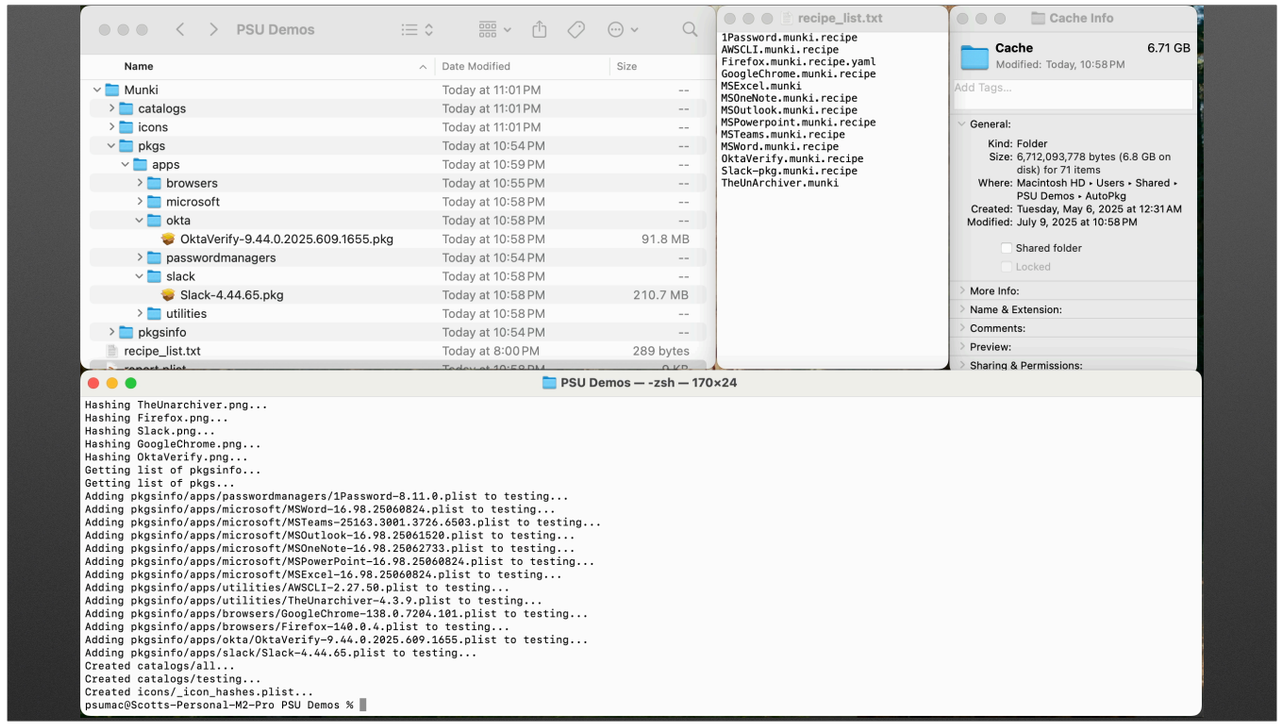


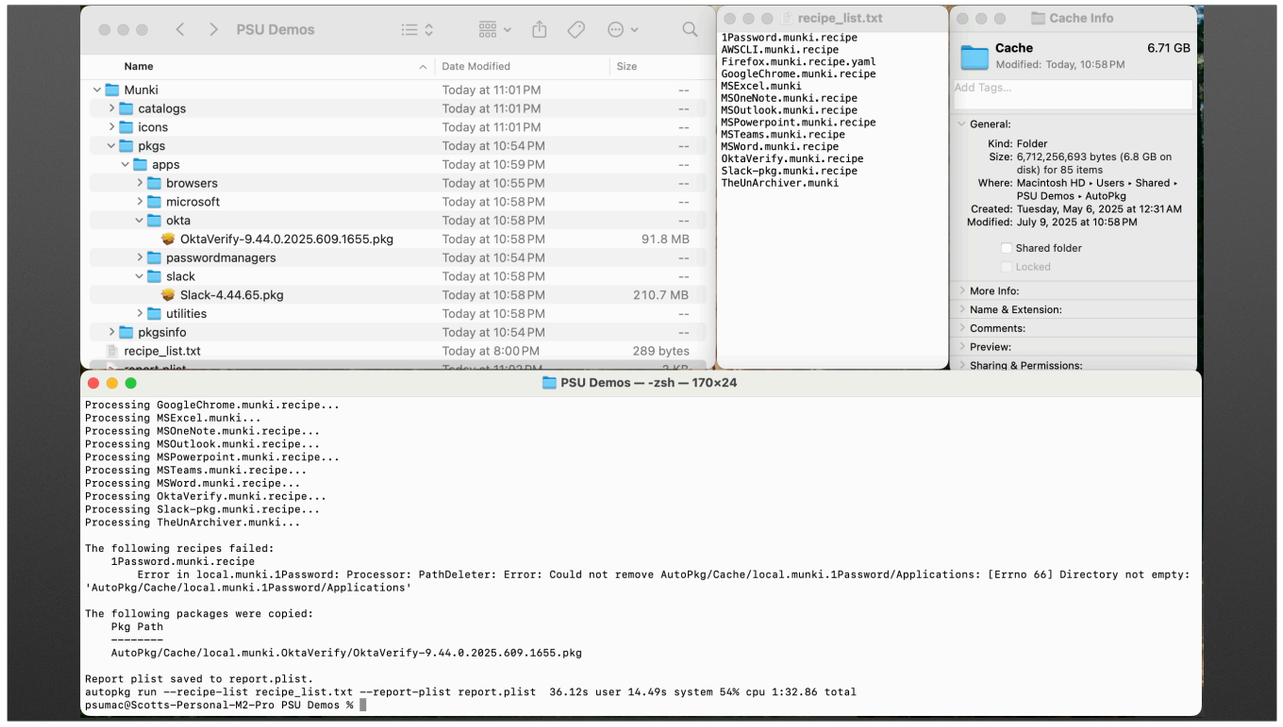
Note that the cache size is changing. Each recipe is different in how it determines if it needs to download a new version. For instance, some recipes need to unpack the cached files to determine existing versions. This is some of the behavior we are trying to improve.

Not only that, but the 1Password recipe failed for some reason. That wasn't even planned, but it helps to highlight that we want to avoid reprocessing files whenever possible.

[Let video finish]


PSU Demos
Name
Date Modified
Size
Munki
Today at 11:01 PM
catalogs
Today at 11:01 PM
icons
Today at 11:01 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
Modified: Today, 10:58 PM
6.71 GB
Add Tags...
General:
Kind: Folder
Size: 6,712,093,778 bytes (6.8 GB on disk) for 71 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Hashing TheUnarchiver.png...
Hashing Firefox.png...
Hashing Slack.png...
Hashing GoogleChrome.png...
Hashing OktaVerify.png...
Getting list of pkgsinfo...
Getting list of pkgs...
Adding pkgsinfo/apps/passwordmanagers/1Password-8.11.0.plist to testing...
Adding pkgsinfo/apps/microsoft/MSWord-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSTeams-25163.3001.3726.6503.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOutlook-16.98.25061520.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOneNote-16.98.25062733.plist to testing...
Adding pkgsinfo/apps/microsoft/MSPowerPoint-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSExcel-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/utilities/AWSCLI-2.27.50.plist to testing...
Adding pkgsinfo/apps/utilities/TheUnarchiver-4.3.9.plist to testing...
Adding pkgsinfo/apps/browsers/GoogleChrome-138.0.7204.101.plist to testing...
Adding pkgsinfo/apps/browsers/Firefox-140.0.4.plist to testing...
Adding pkgsinfo/apps/okta/OktaVerify-9.44.0.2025.609.1655.plist to testing...
Adding pkgsinfo/apps/slack/Slack-4.44.65.plist to testing...
Created catalogs/all...
Created catalogs/testing...
Created icons/_icon_hashes.plist...
psumac@Scotts-Personal-M2-Pro PSU Demos %

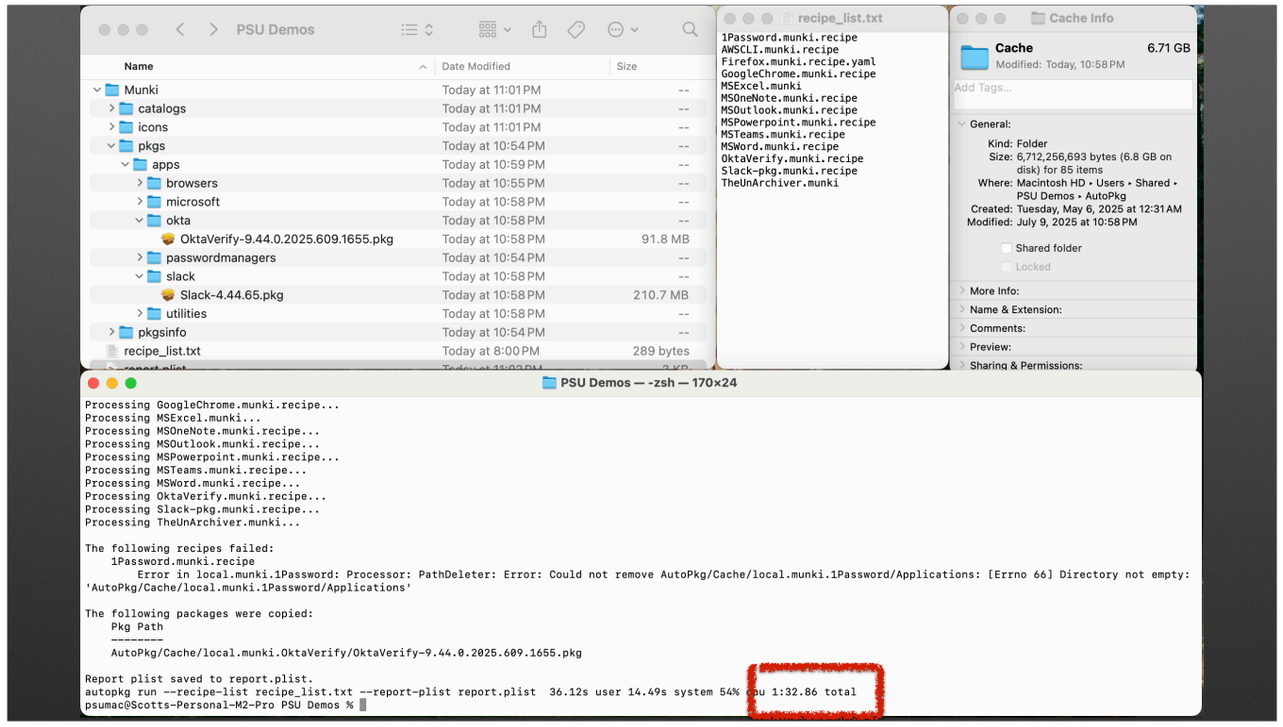

PSU Demos
Name
Date Modified
Size
Munki
Today at 11:01 PM
catalogs
Today at 11:01 PM
icons
Today at 11:01 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
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More Info:
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Sharing & Permissions:
PSU Demos — -zsh — 170×24
Hashing TheUnarchiver.png...
Hashing Firefox.png...
Hashing Slack.png...
Hashing GoogleChrome.png...
Hashing OktaVerify.png...
Getting list of pkgsinfo...
Getting list of pkgs...
Adding pkgsinfo/apps/passwordmanagers/1Password-8.11.0.plist to testing...
Adding pkgsinfo/apps/microsoft/MSWord-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSTeams-25163.3001.3726.6503.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOutlook-16.98.25061520.plist to testing...
Adding pkgsinfo/apps/microsoft/MSOneNote-16.98.25062733.plist to testing...
Adding pkgsinfo/apps/microsoft/MSPowerPoint-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/microsoft/MSExcel-16.98.25060824.plist to testing...
Adding pkgsinfo/apps/utilities/AWSCLI-2.27.50.plist to testing...
Adding pkgsinfo/apps/utilities/TheUnarchiver-4.3.9.plist to testing...
Adding pkgsinfo/apps/browsers/GoogleChrome-138.0.7204.101.plist to testing...
Adding pkgsinfo/apps/browsers/Firefox-140.0.4.plist to testing...
Adding pkgsinfo/apps/okta/OktaVerify-9.44.0.2025.609.1655.plist to testing...
Adding pkgsinfo/apps/slack/Slack-4.44.65.plist to testing...
Created catalogs/all...
Created catalogs/testing...
Created icons/_icon_hashes.plist...
psumac@Scotts-Personal-M2-Pro PSU Demos %

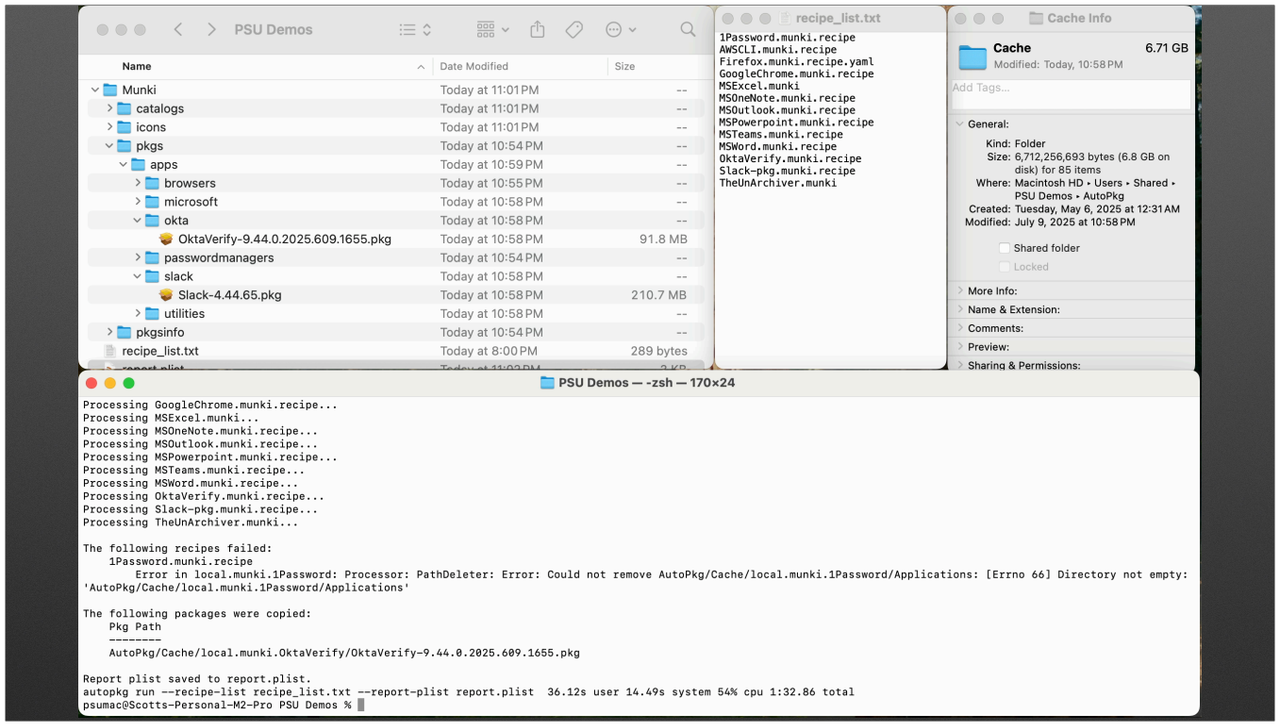


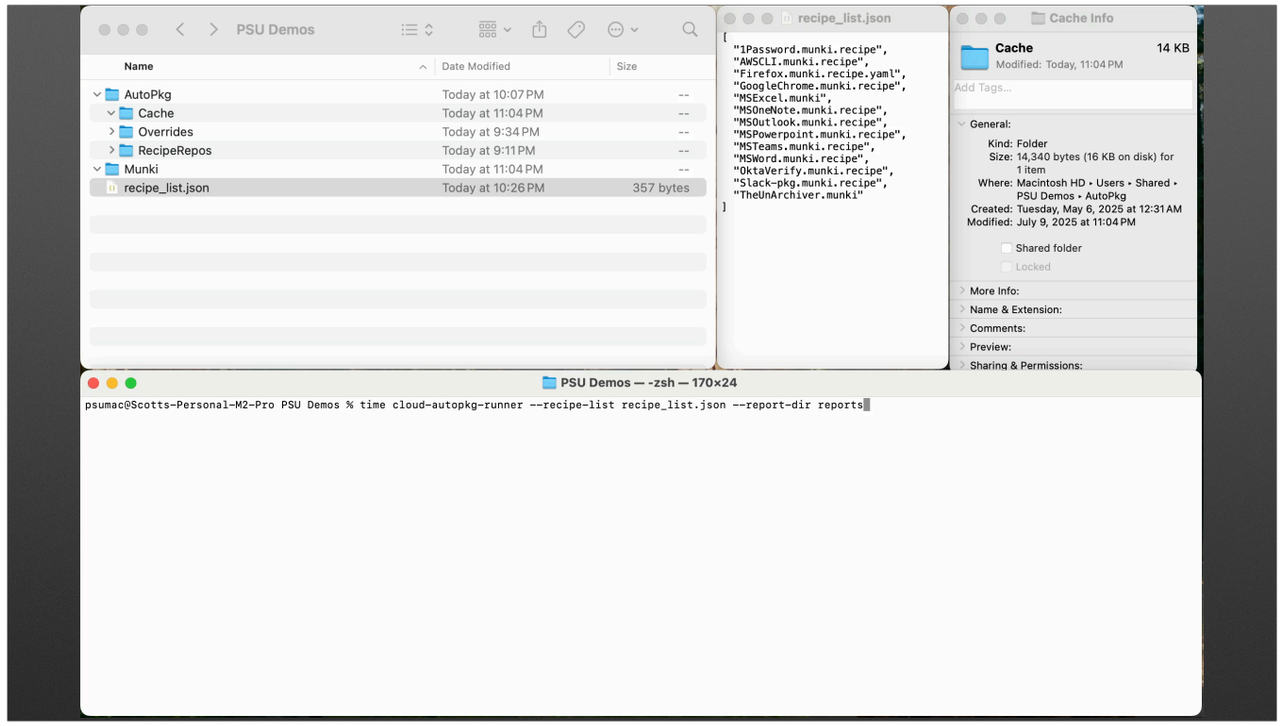
Running that second time took 1 minute and 33 seconds. That's certainly much faster than the original 5 minutes, and it highlights how good native AutoPkg can be about preventing redownloads. [to clear red circle]

Unfortunately, not everyone is running in an environment where that cache persists.

That's where Cloud AutoPkg Runner comes in. It excels where the cache is wiped between runs, but it also runs significantly faster even when the cache exists.


PSU Demos
Name
Date Modified
Size
Munki
Today at 11:01 PM
catalogs
Today at 11:01 PM
icons
Today at 11:01 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,256,693 bytes (6.8 GB on disk) for 85 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Processing GoogleChrome.munki.recipe...
Processing MSExcel.munki...
Processing MSOneNote.munki.recipe...
Processing MSOutlook.munki.recipe...
Processing MSPowerpoint.munki.recipe...
Processing MSTeams.munki.recipe...
Processing MSWord.munki.recipe...
Processing OktaVerify.munki.recipe...
Processing Slack-pkg.munki.recipe...
Processing TheUnArchiver.munki...
The following recipes failed:
1Password.munki.recipe
Error in local.munki.1Password: Processor: PathDeleter: Error: Could not remove AutoPkg/Cache/local.munki.1Password/Applications: [Errno 66] Directory not empty: 'AutoPkg/Cache/local.munki.1Password/Applications'
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 36.12s user 14.49s system 54% cpu 1:32.86 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

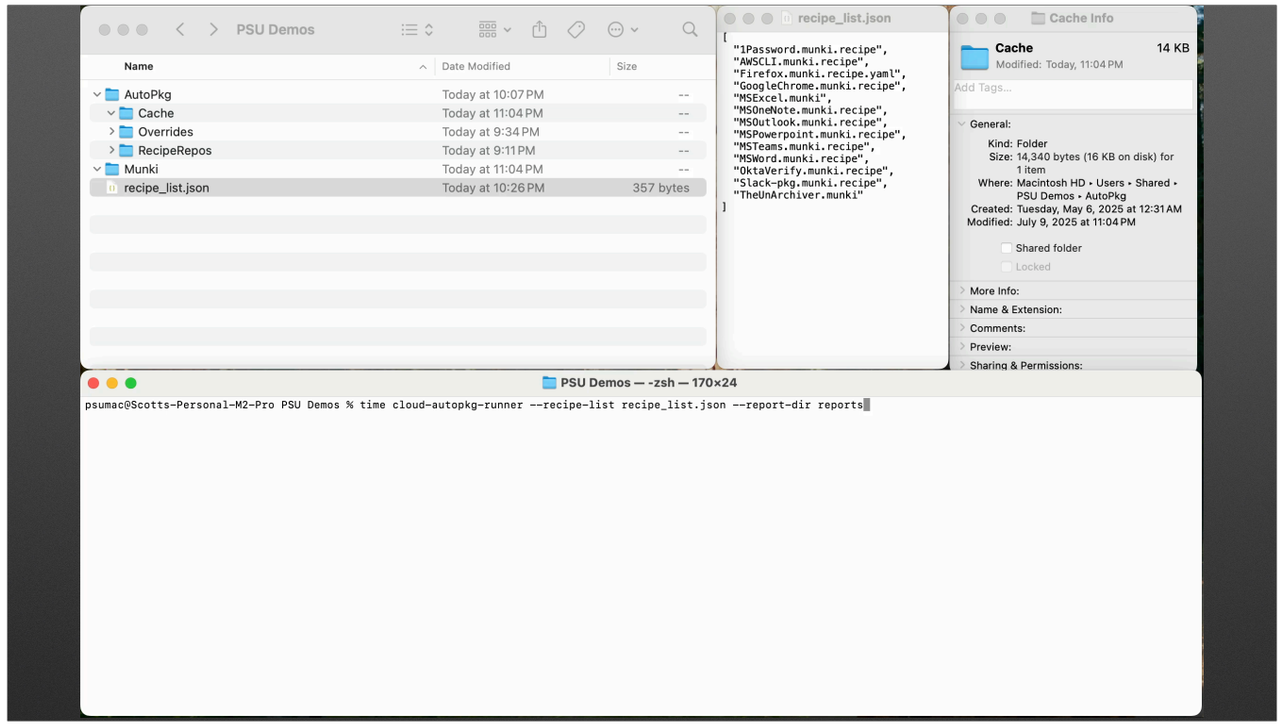

PSU Demos
Name
Date Modified
Size
Munki
Today at 11:01 PM
catalogs
Today at 11:01 PM
icons
Today at 11:01 PM
pkgs
Today at 10:54 PM
apps
Today at 10:59 PM
browsers
Today at 10:55 PM
microsoft
Today at 10:58 PM
okta
Today at 10:58 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 10:58 PM
91.8 MB
passwordmanagers
Today at 10:54 PM
slack
Today at 10:58 PM
Slack-4.44.65.pkg
Today at 10:58 PM
210.7 MB
utilities
Today at 10:58 PM
pkgsinfo
Today at 10:54 PM
recipe_list.txt
Today at 8:00 PM
289 bytes
recipe_list.txt
1Password.munki.recipe
AWSCLI.munki.recipe
Firefox.munki.recipe.yaml
GoogleChrome.munki.recipe
MSExcel.munki
MSOneNote.munki.recipe
MSOutlook.munki.recipe
MSPowerpoint.munki.recipe
MSTeams.munki.recipe
MSWord.munki.recipe
OktaVerify.munki.recipe
Slack-pkg.munki.recipe
TheUnArchiver.munki
Cache Info
Cache
6.71 GB
Modified: Today, 10:58 PM
Add Tags...
General:
Kind: Folder
Size: 6,712,256,693 bytes (6.8 GB on disk) for 85 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 10:58 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
Processing GoogleChrome.munki.recipe...
Processing MSExcel.munki...
Processing MSOneNote.munki.recipe...
Processing MSOutlook.munki.recipe...
Processing MSPowerpoint.munki.recipe...
Processing MSTeams.munki.recipe...
Processing MSWord.munki.recipe...
Processing OktaVerify.munki.recipe...
Processing Slack-pkg.munki.recipe...
Processing TheUnArchiver.munki...
The following recipes failed:
1Password.munki.recipe
Error in local.munki.1Password: Processor: PathDeleter: Error: Could not remove AutoPkg/Cache/local.munki.1Password/Applications: [Errno 66] Directory not empty: 'AutoPkg/Cache/local.munki.1Password/Applications'
The following packages were copied:
Pkg Path
AutoPkg/Cache/local.munki.OktaVerify/OktaVerify-9.44.0.2025.609.1655.pkg
Report plist saved to report.plist.
autopkg run --recipe-list recipe_list.txt --report-plist report.plist 36.12s user 14.49s system 54% cpu 1:32.86 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

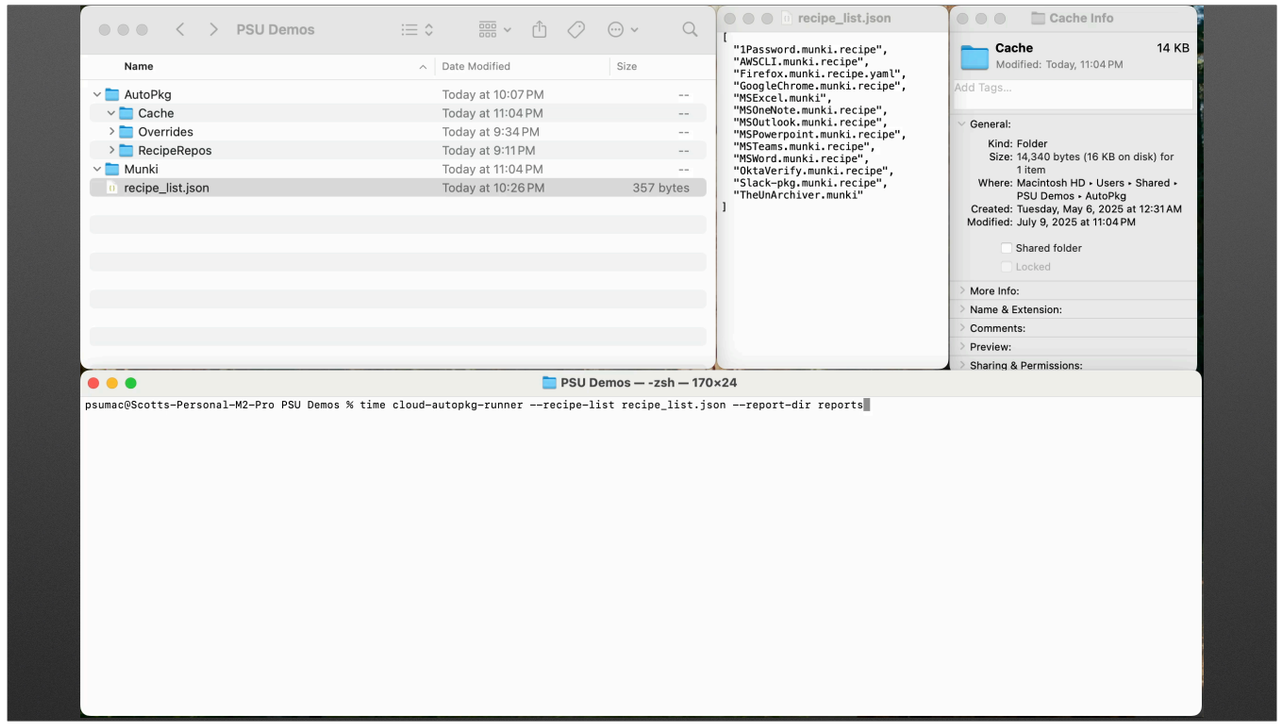


Let's reset the environment and I can show you both scenarios. You'll notice that everything is cleared out from the last run and we are running Cloud AutoPkg Runner in CLI mode.

This time, things start off a bit faster than before since the recipe list is running concurrently. Cache directories are created as each recipe is started.

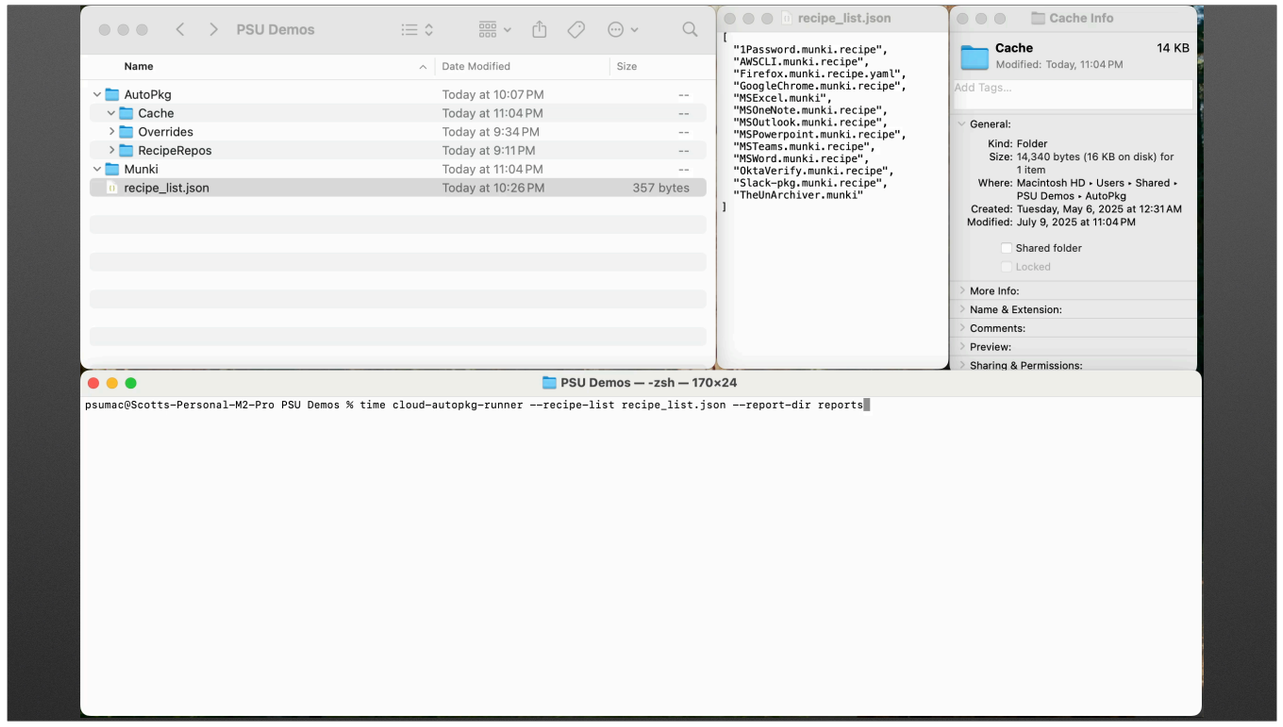
Scrolling down, you can see that there is now a separate report file for each recipe, and they are date stamped so they won't get overridden.

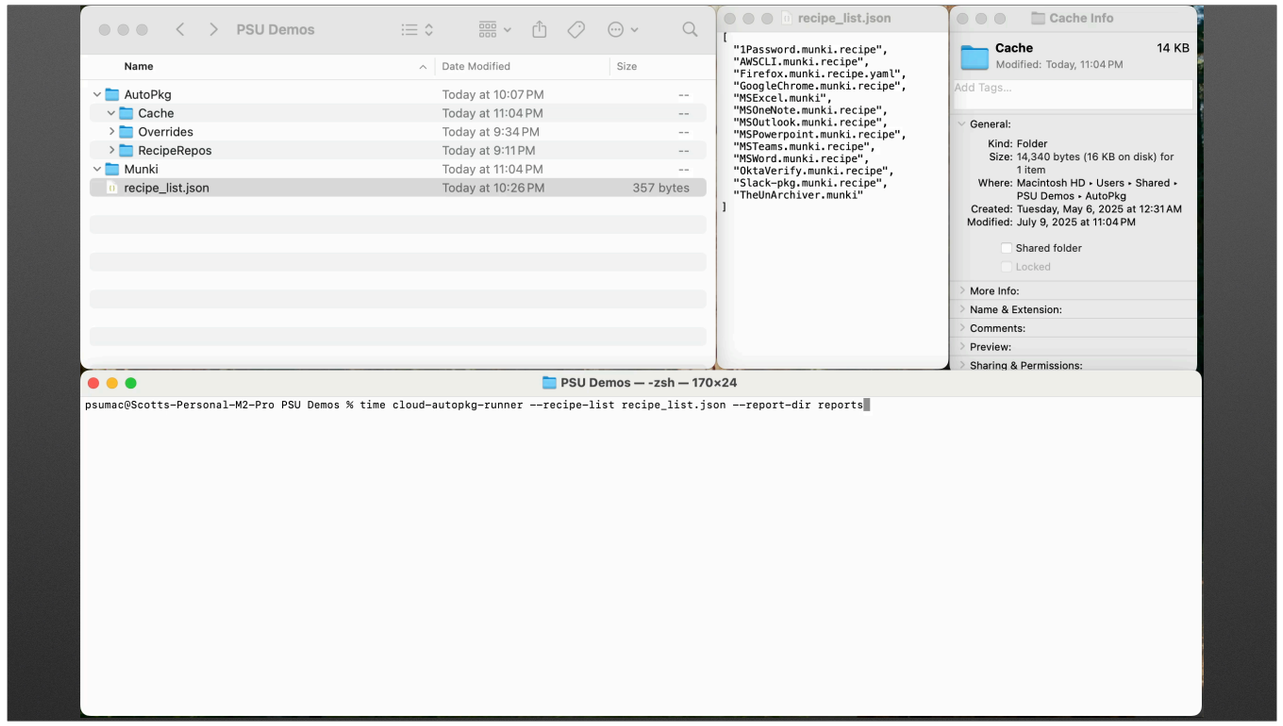

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:04 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:04 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
14 KB
Modified: Today, 11:04 PM
Add Tags...
General:
Kind: Folder
Size: 14,340 bytes (16 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:04 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports



I want to take this opportunity to point out that this recipe list is a JSON file instead of a list of recipes, but the list of recipes is the same.

The limiting factor here is my wifi bandwidth, but you'll note that the cache directory is trending up much faster this time. We don't have to wait for things like CodeSignVerifier or PkgCreator. The next recipe is already downloading while the first is working through those steps. This tool also creates secondary threads for blocking functions like reading and writing to disk. This means that the main thread isn't waiting on one file to complete to move on to the next.

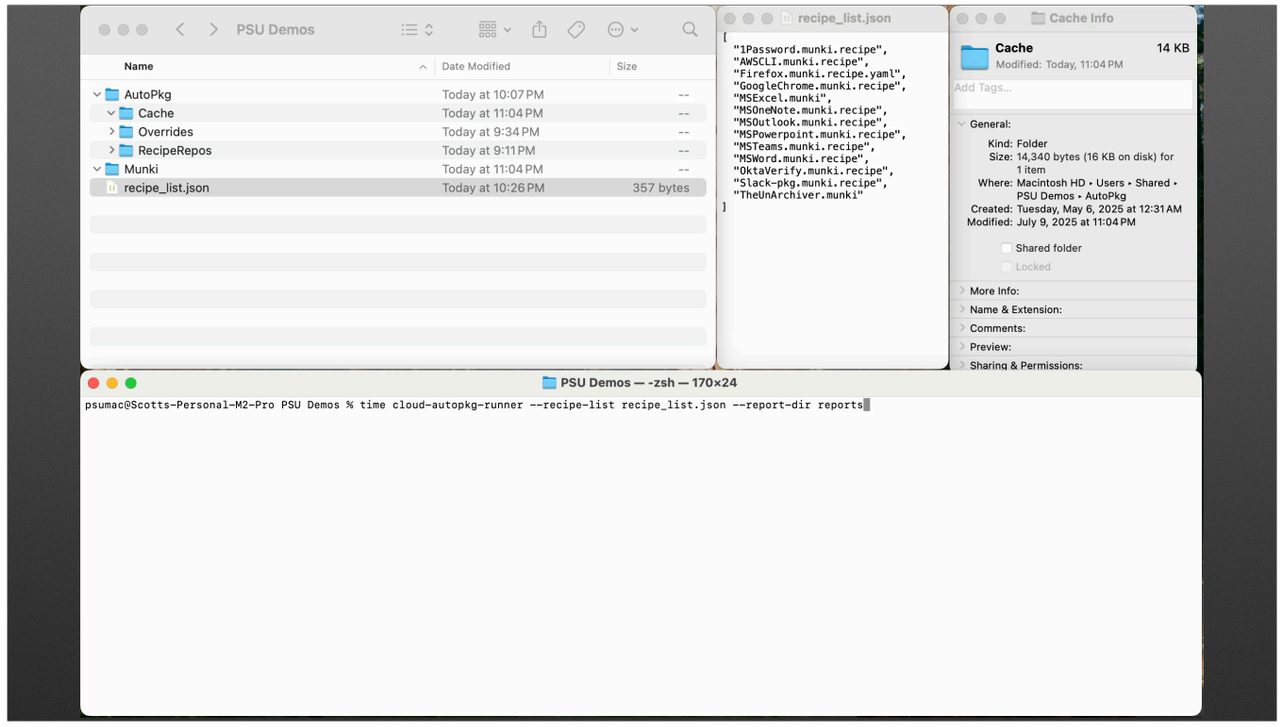

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:04 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:04 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
14 KB
Modified: Today, 11:04 PM
Add Tags...
General:
Kind: Folder
Size: 14,340 bytes (16 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:04 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports

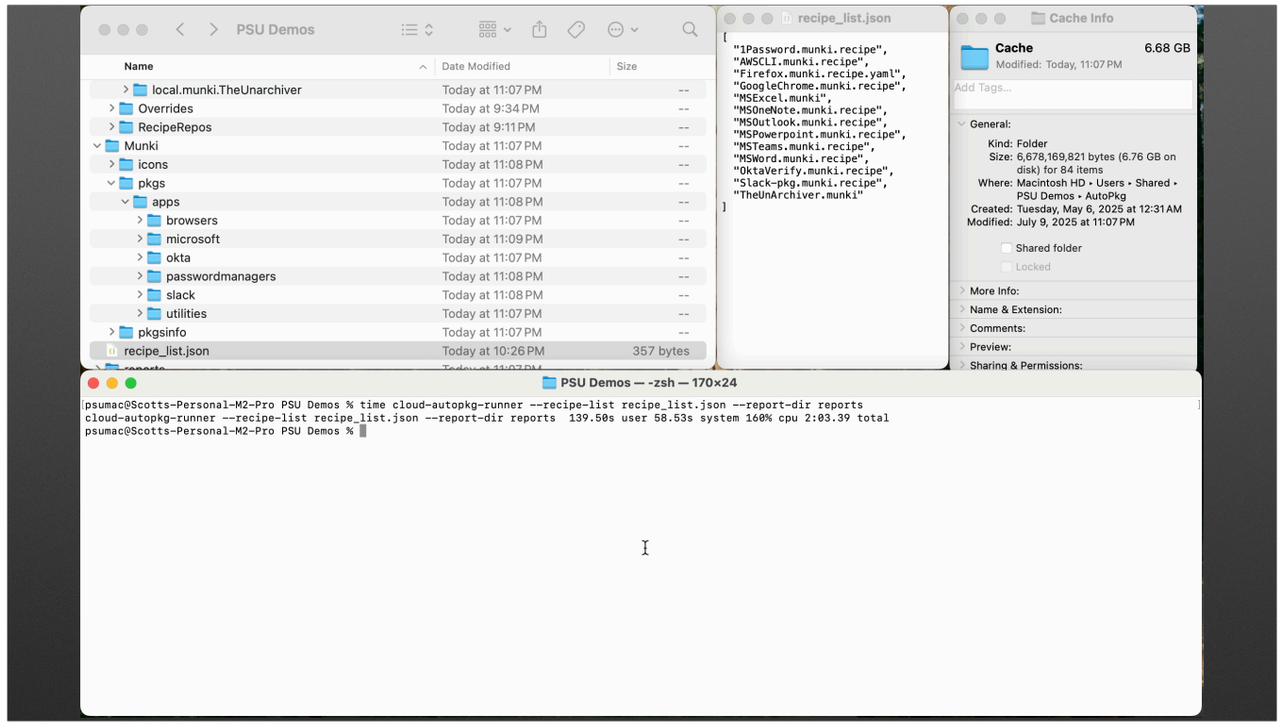


In my opinion, the coolest part of this project is that as each recipe completes, it stores metadata such as the last modified date, file size, and ETag for each file that gets downloaded. So if you have a recipe that downloads 2 files, the metadata for both are stored.

At the end of the run, all that metadata is written to disk since I'm using the JSON cache plugin. If you use a cloud plugin like S3, Azure, or Google, it is uploaded automatically. I'll show you this file in a minute.

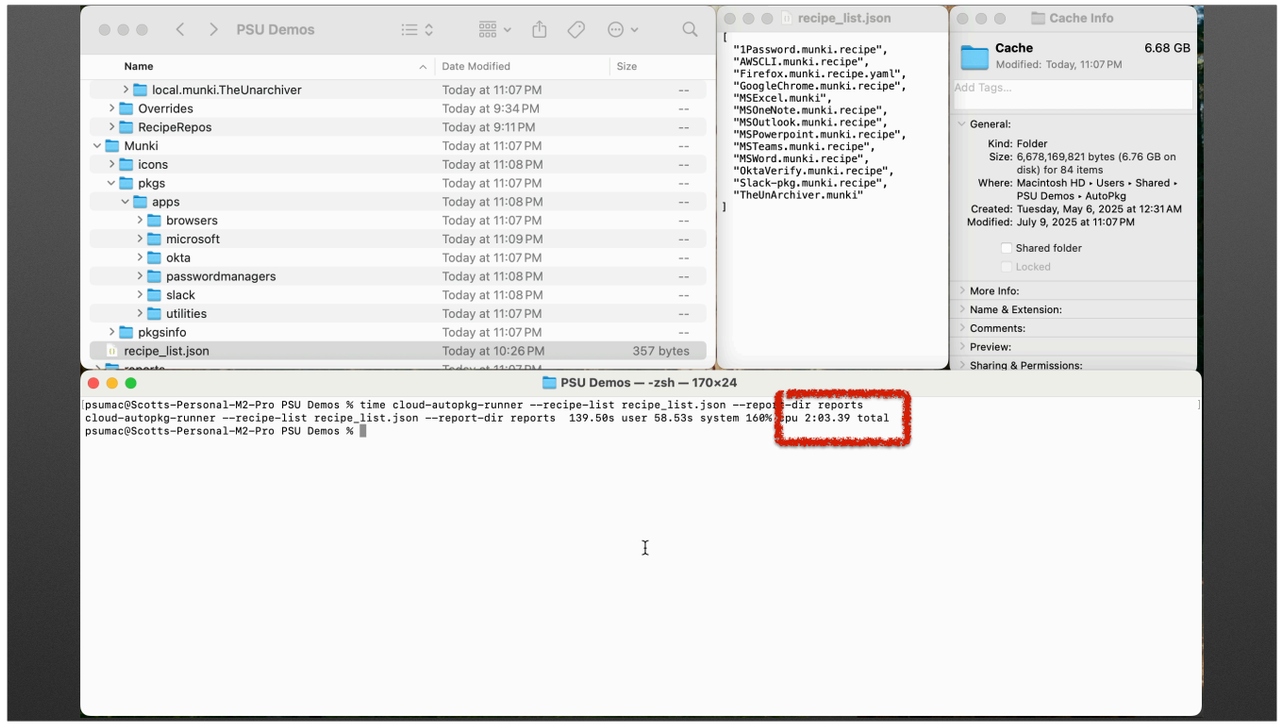
[Make sure video finishes.]

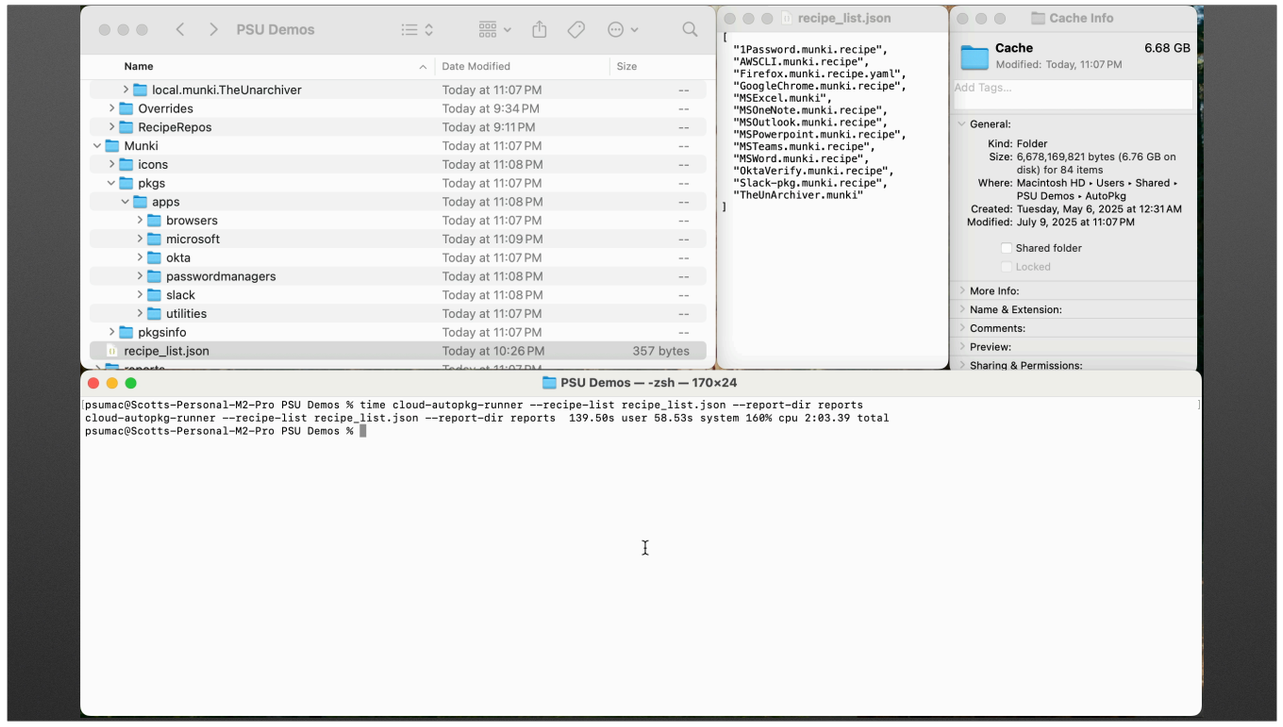

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:04 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:04 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
14 KB
Modified: Today, 11:04 PM
Add Tags...
General:
Kind: Folder
Size: 14,340 bytes (16 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:04 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports

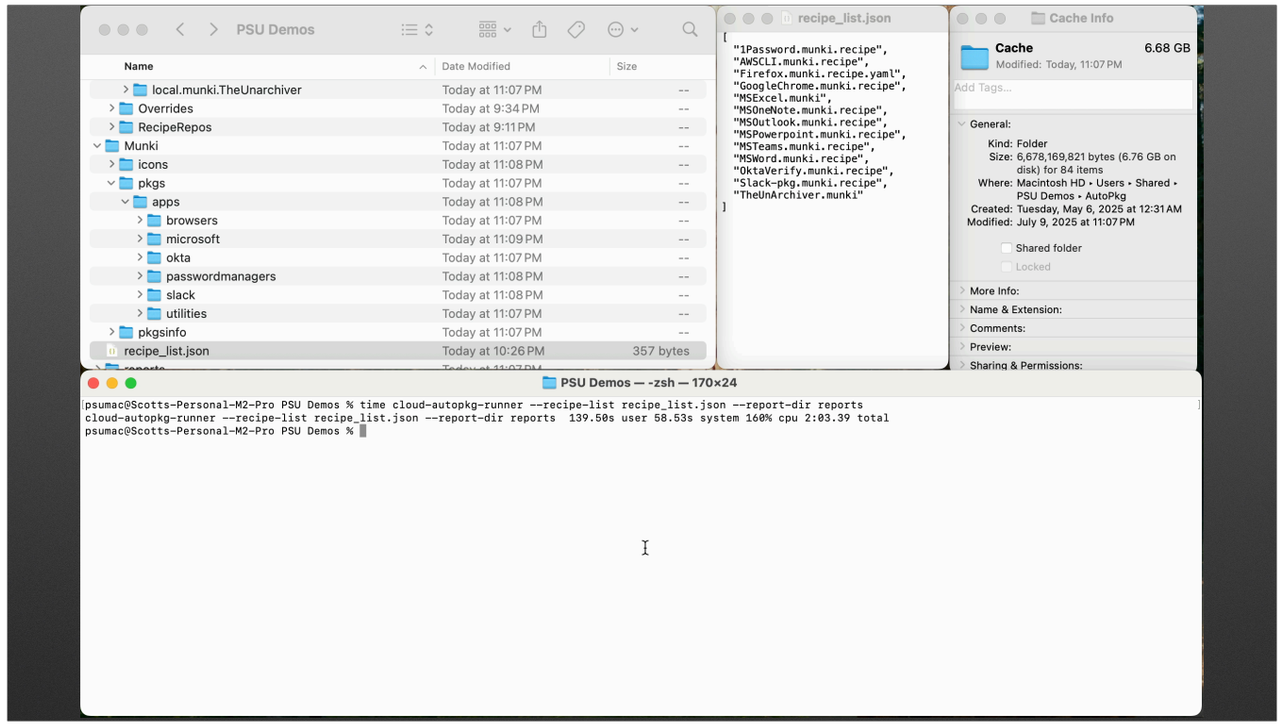


The total runtime is 2 minutes and 4 seconds. Remember that the native AutoPkg run was almost 5 minutes. So even without taking advantage of any of the caching functionalities, this is significantly faster.

I do want to apologize for this demo though, I forgot to increase the verbosity. I'm a firm believer that if your tool is designed to be used in automated workflows, it shouldn't output anything by default. If I would have added a -v to the command, we could see more of what's going on.

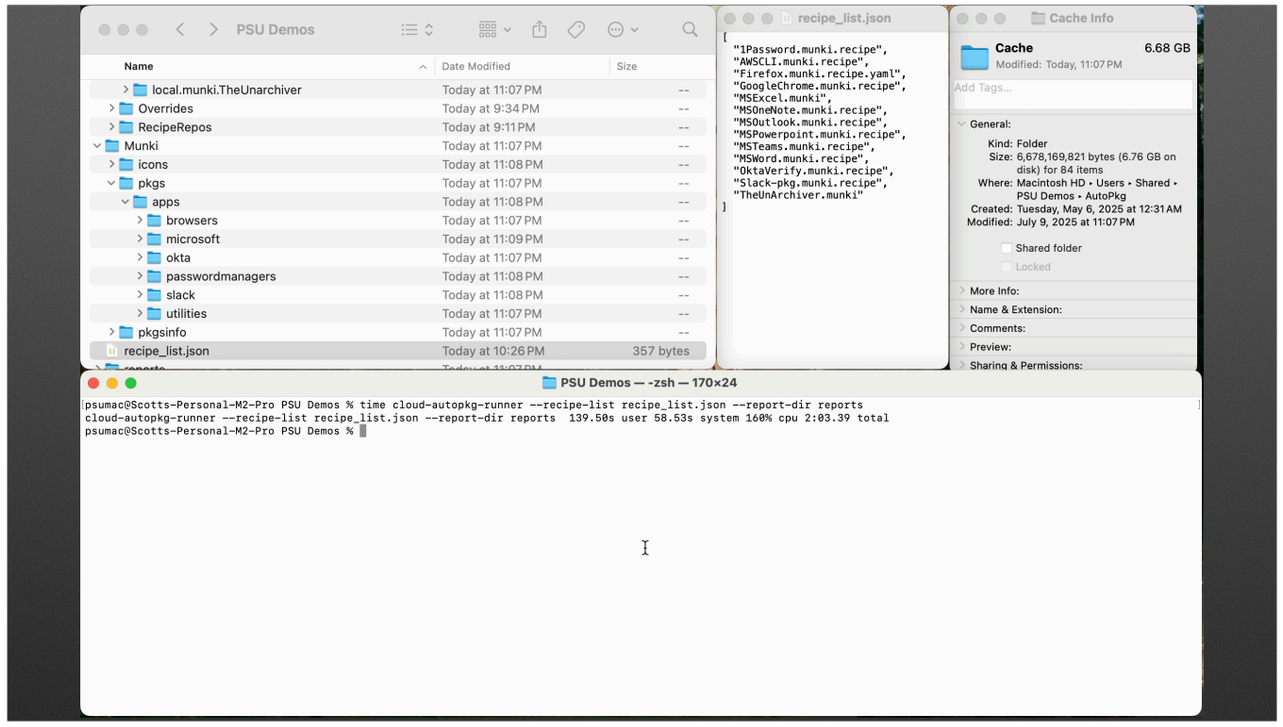

PSU Demos
Name
Date Modified
Size
local.munki.TheUnarchiver
Today at 11:07 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:07 PM
icons
Today at 11:08 PM
pkgs
Today at 11:07 PM
apps
Today at 11:08 PM
browsers
Today at 11:07 PM
microsoft
Today at 11:09 PM
okta
Today at 11:07 PM
passwordmanagers
Today at 11:08 PM
slack
Today at 11:08 PM
utilities
Today at 11:07 PM
pkgsinfo
Today at 11:07 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
Kind: Folder
Size: 6,678,169,821 bytes (6.76 GB on disk) for 84 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:07 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

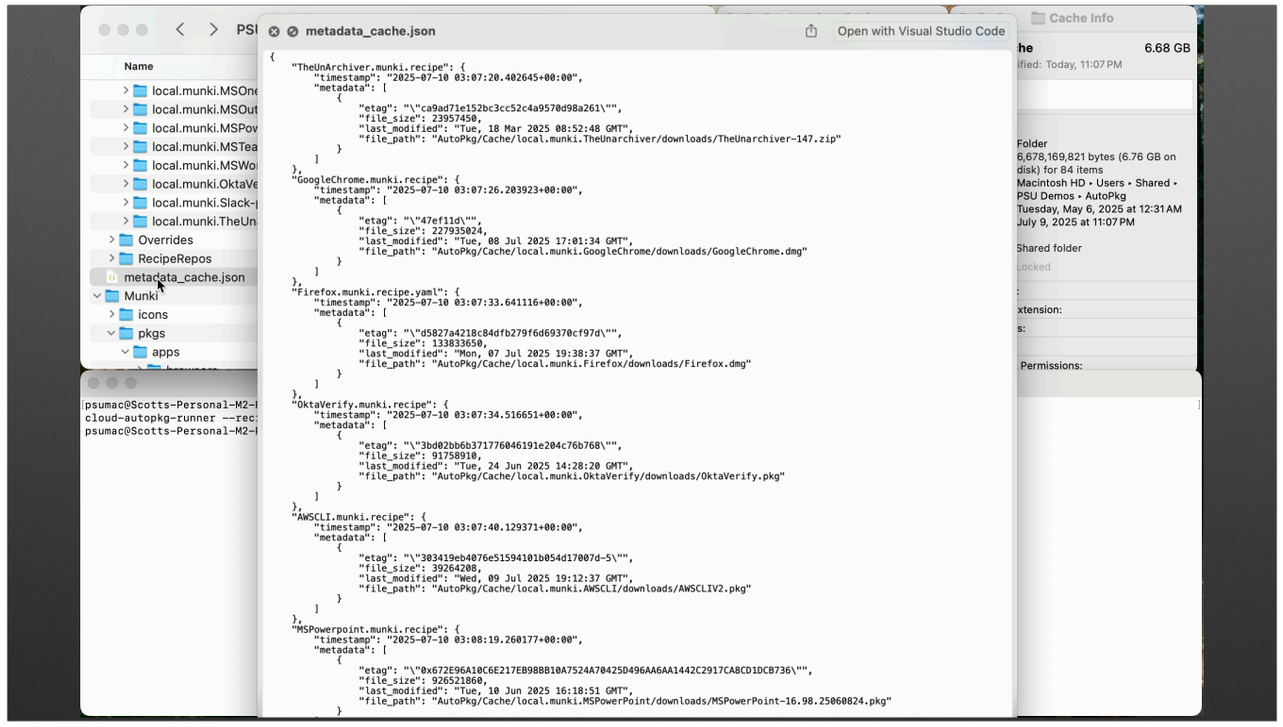

PSU Demos
Name
Date Modified
Size
local.munki.TheUnarchiver
Today at 11:07 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:07 PM
icons
Today at 11:08 PM
pkgs
Today at 11:07 PM
apps
Today at 11:08 PM
browsers
Today at 11:07 PM
microsoft
Today at 11:09 PM
okta
Today at 11:07 PM
passwordmanagers
Today at 11:08 PM
slack
Today at 11:08 PM
utilities
Today at 11:07 PM
pkgsinfo
Today at 11:07 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
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PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %



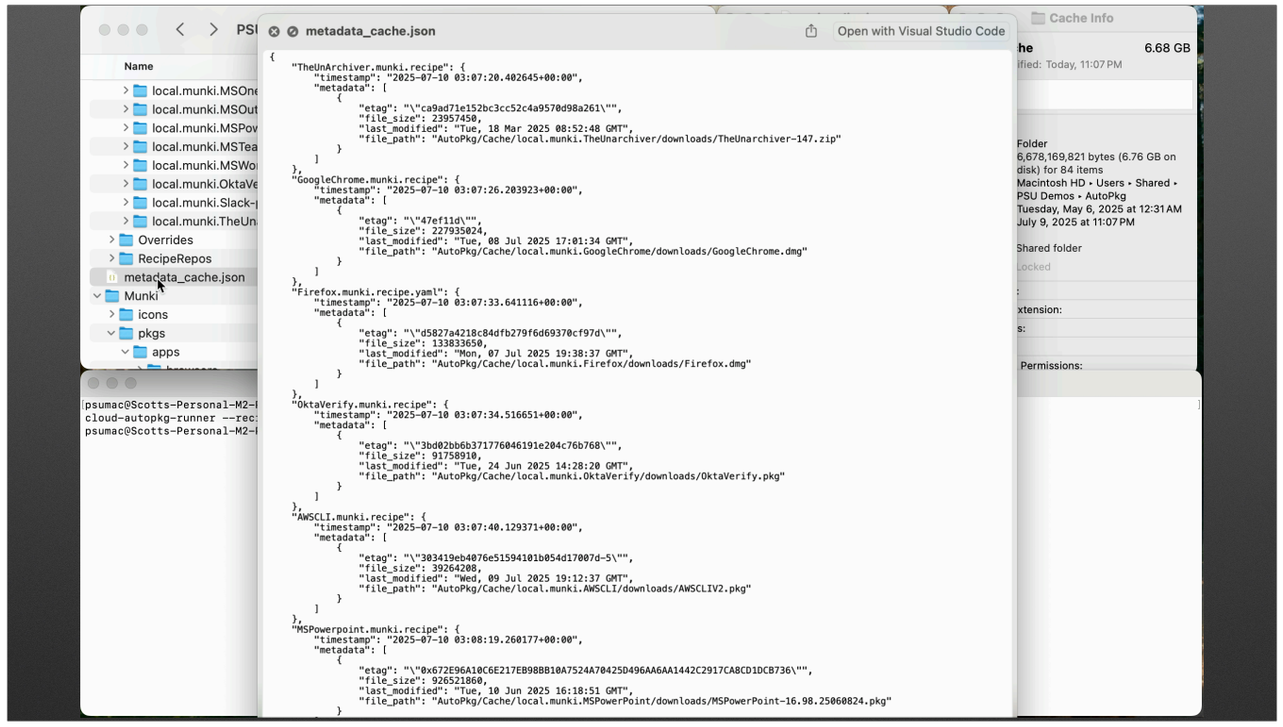
Just like before, all 13 recipes were successfully imported into Munki.

As I mentioned earlier, each report is separated into separate files for easier parsing. You can see the contents of one of them here.


PSU Demos
Name
Date Modified
Size
local.munki.TheUnarchiver
Today at 11:07 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
Munki
Today at 11:07 PM
icons
Today at 11:08 PM
pkgs
Today at 11:07 PM
apps
Today at 11:08 PM
browsers
Today at 11:07 PM
microsoft
Today at 11:09 PM
okta
Today at 11:07 PM
passwordmanagers
Today at 11:08 PM
slack
Today at 11:08 PM
utilities
Today at 11:07 PM
pkgsinfo
Today at 11:07 PM
recipe_list.json
Today at 10:26 PM
357 bytes
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
Kind: Folder
Size: 6,678,169,821 bytes (6.76 GB on disk) for 84 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:07 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %



And then here is the contents of the metadata cache file. I know it's hard to see, so here's a snippet of a single recipe.

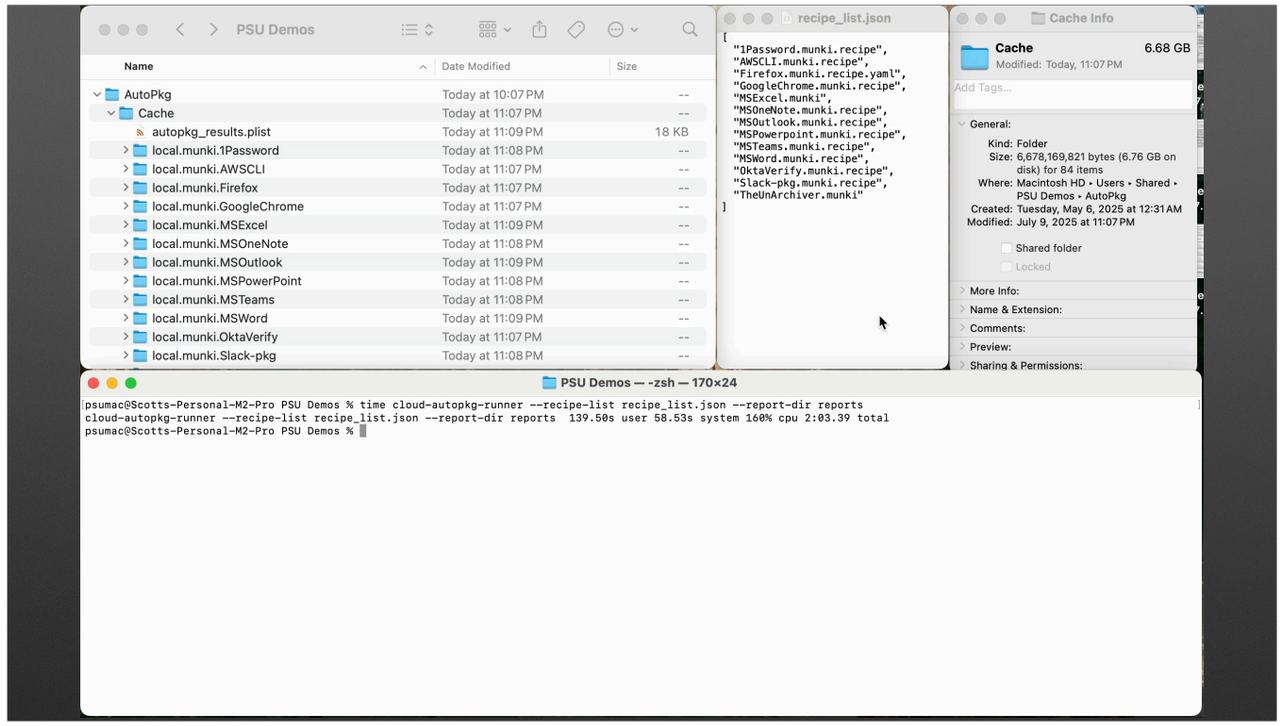

metadata_cache.json
Open with Visual Studio Code
Name
local.munki.MSOne
local.munki.MSOut
local.munki.MSPow
local.munki.MSTea
local.munki.MSWo
local.munki.OktaVe
local.munki.Slack-
local.munki.TheUn
Overrides
RecipeRepos
metadata_cache.json
Munki
icons
pkgs
apps
Cache Info
6.68 GB
Folder
6,678,169,821 bytes (6.76 GB on disk) for 84 items
Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Tuesday, May 6, 2025 at 12:31 AM
July 9, 2025 at 11:07 PM
Shared folder
Permissions:

```
{
    "GoogleChrome.munki.recipe": {
        "timestamp": "2025-07-10 03:07:26.203923+00:00",
        "metadata": [
            {
                "etag": "\"47ef11d\"",
                "file_size": 227935024,
                "last_modified": "Tue, 08 Jul 2025 17:01:34 GMT",
                "file_path": "AutoPkg/Cache/local.munki.GoogleChrome/downloads/GoogleChrome.dmg"
            }
        ]
    }
}
```


It's keyed by the recipe name, has a timestamp from when this entry was created, and a metadata key that contains a list of each file downloaded.

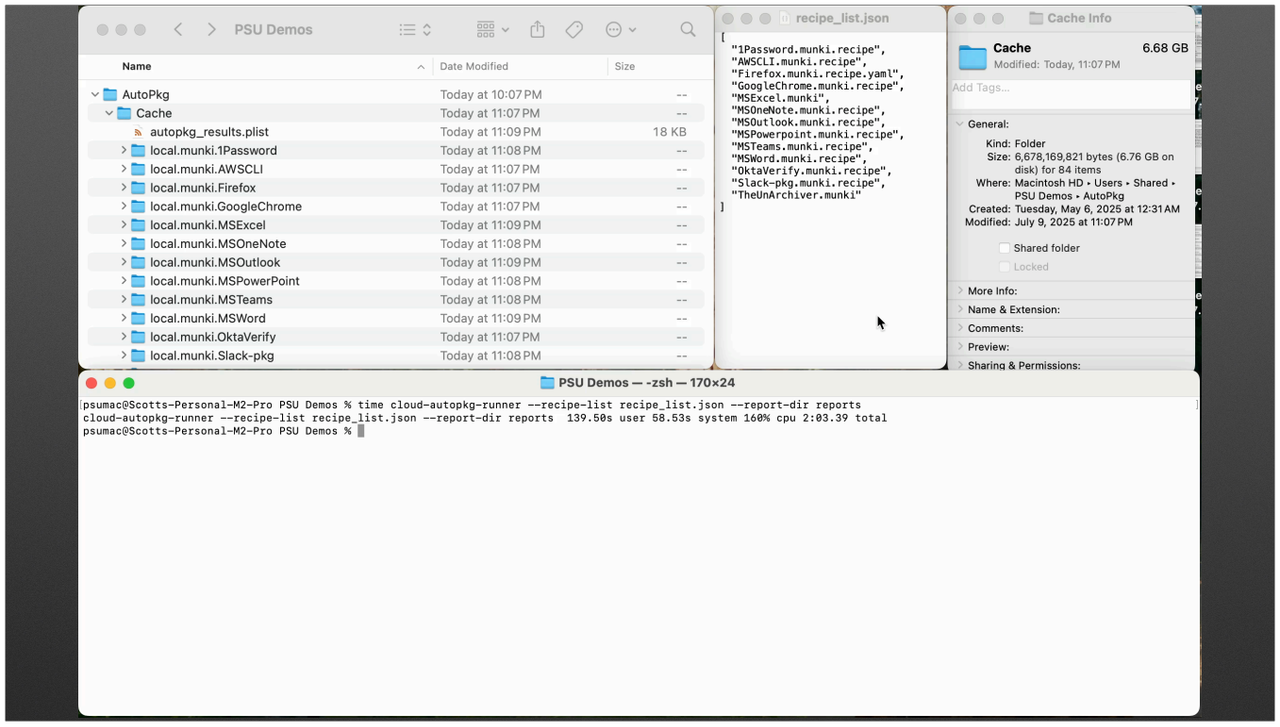
If the cache doesn't already exist, this script will create it. Otherwise, it will modify and append the changes.

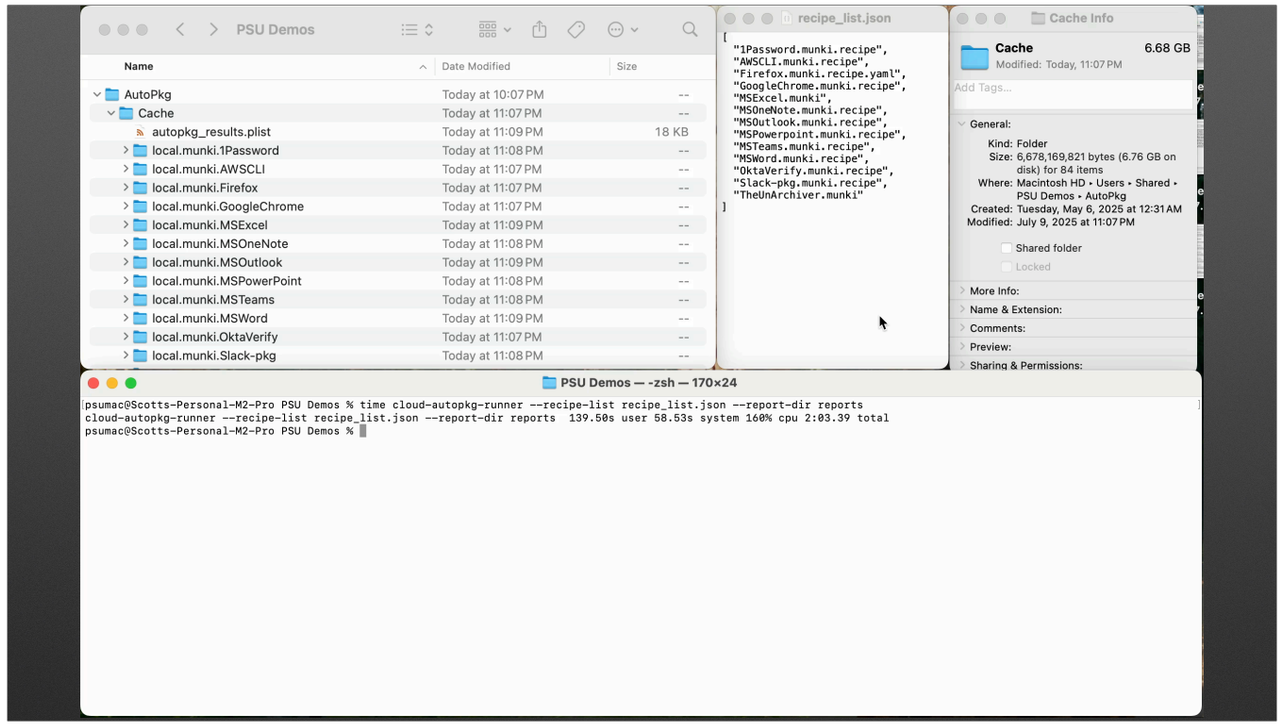
On the next run, this cache file is used to populate the AutoPkg cache directory with fake files if the real files are not present.



Back to the demo... just like the last time, we run `makecatalogs` on the Munki directory to prevent duplicate imports.

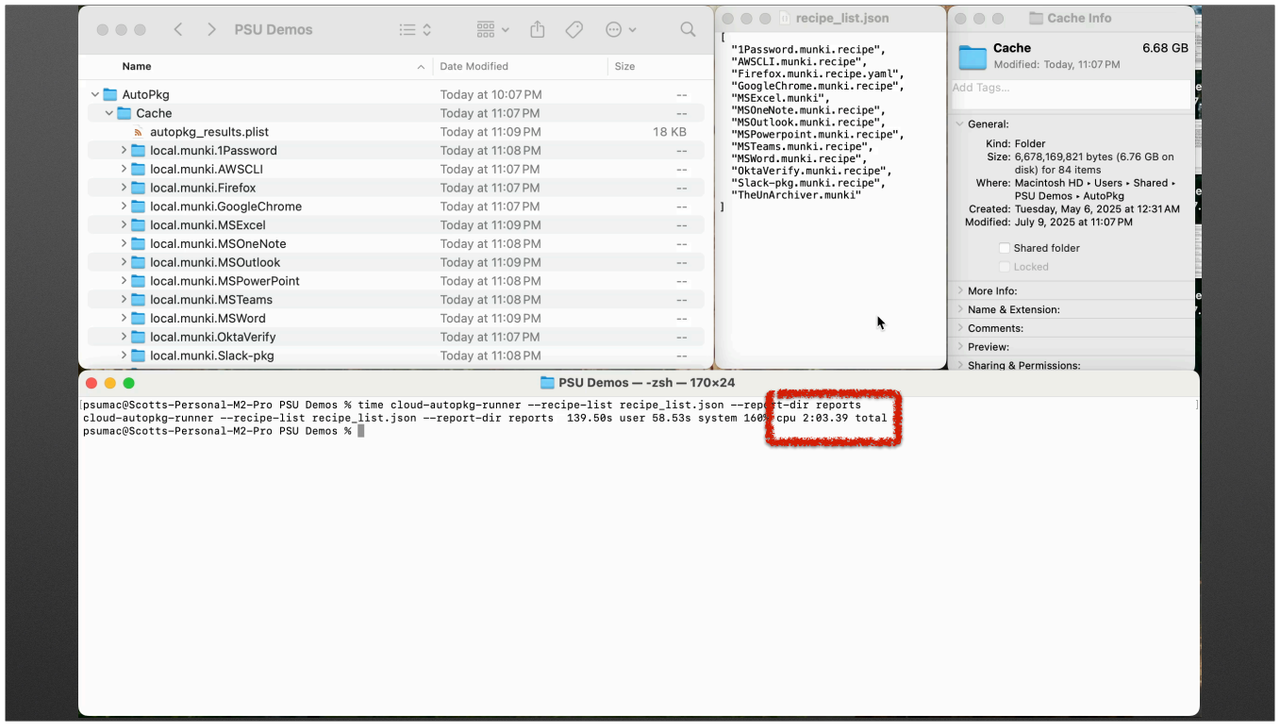
Clear the screen... and then run the same command again.

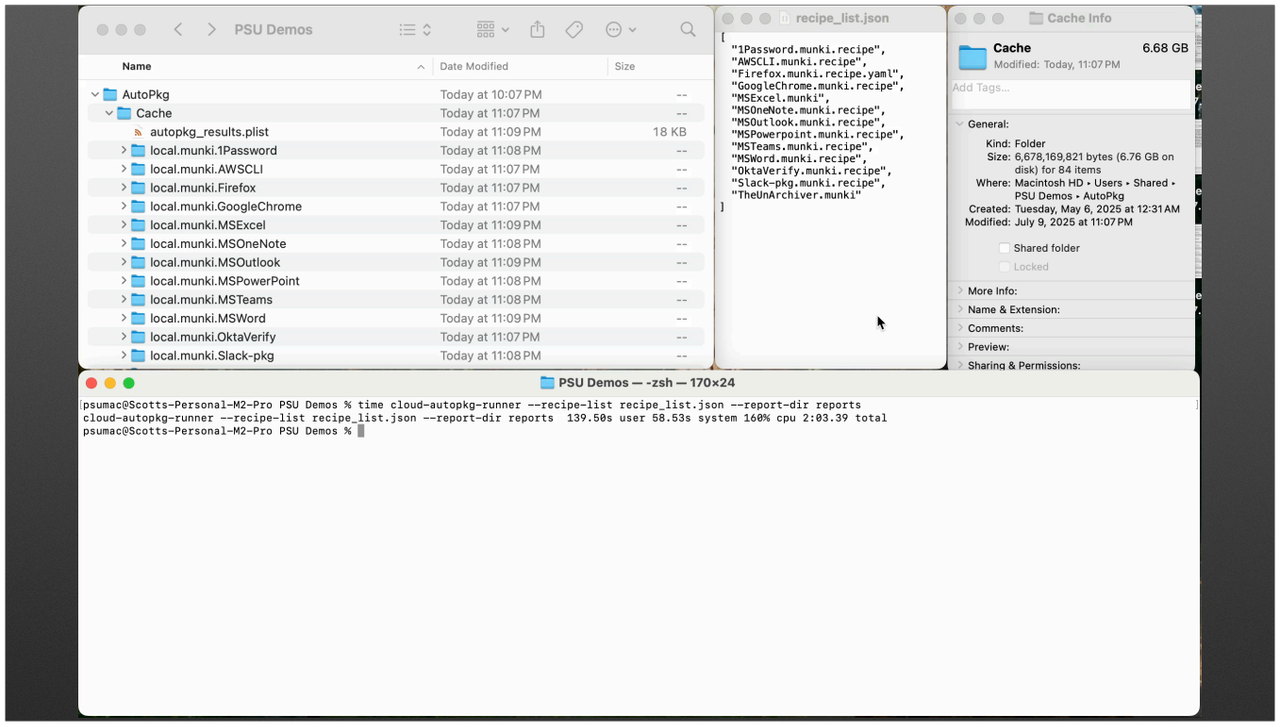

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:07 PM
autopkg_results.plist
Today at 11:09 PM
18 KB
local.munki.1Password
Today at 11:08 PM
local.munki.AWSCLI
Today at 11:07 PM
local.munki.Firefox
Today at 11:07 PM
local.munki.GoogleChrome
Today at 11:07 PM
local.munki.MSExcel
Today at 11:09 PM
local.munki.MSOneNote
Today at 11:08 PM
local.munki.MSOutlook
Today at 11:09 PM
local.munki.MSPowerPoint
Today at 11:08 PM
local.munki.MSTeams
Today at 11:08 PM
local.munki.MSWord
Today at 11:09 PM
local.munki.OktaVerify
Today at 11:07 PM
local.munki.Slack-pkg
Today at 11:08 PM
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
Kind: Folder
Size: 6,678,169,821 bytes (6.76 GB on disk) for 84 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:07 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

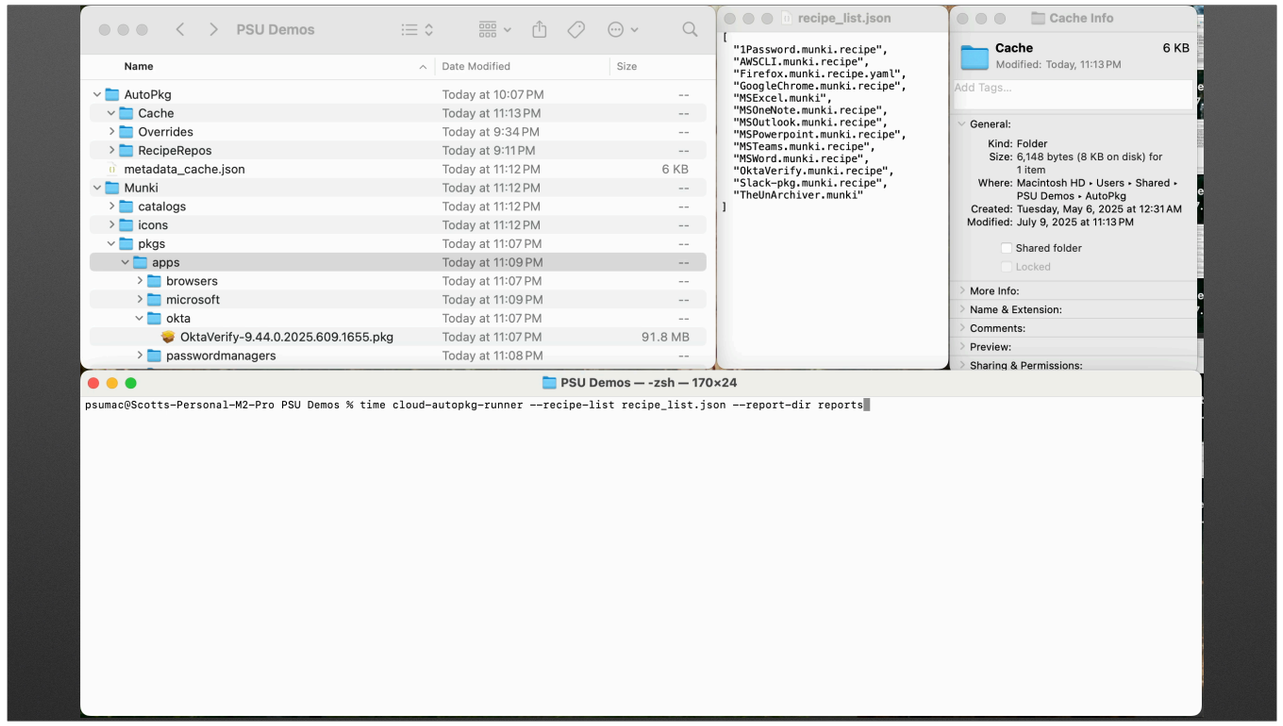


The cache is still intact, so it takes 8.4 seconds to complete. If you remember, this took 93 seconds.. a minute and a half using the native AutoPkg command.

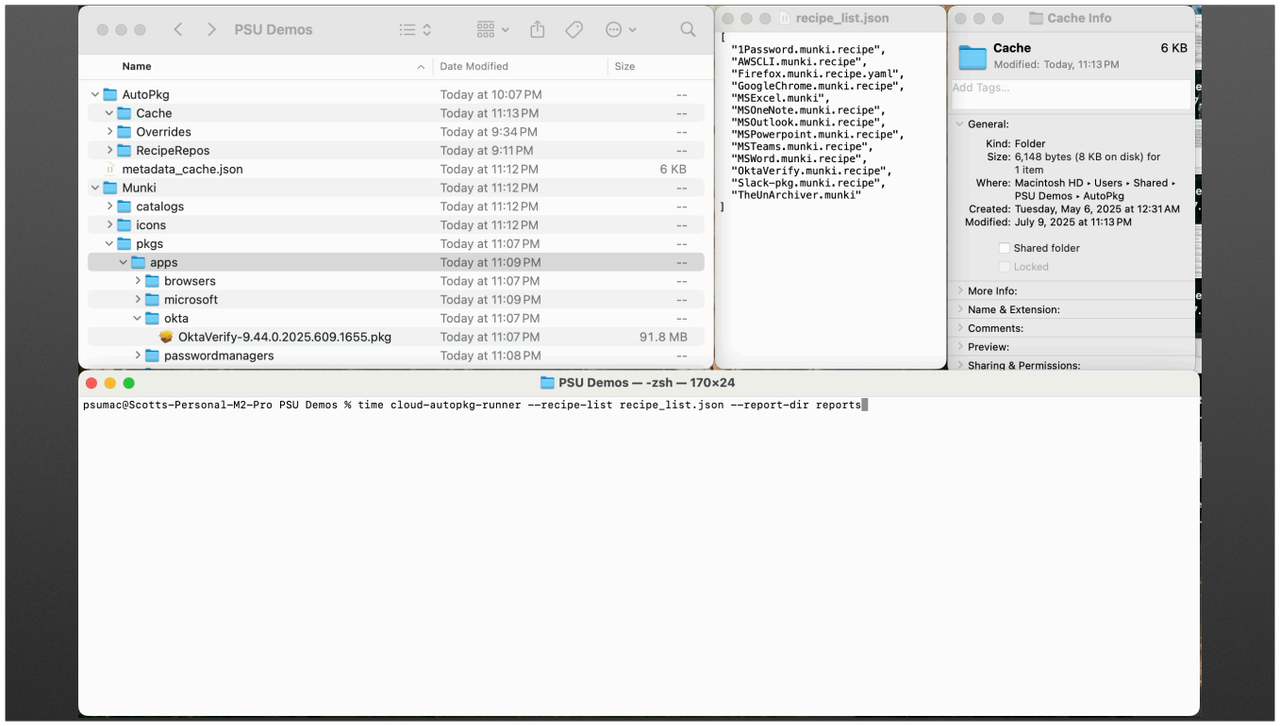
That's blazing fast, but like I said, the cache is still intact. Let's see what happens when the cache doesn't already exist.

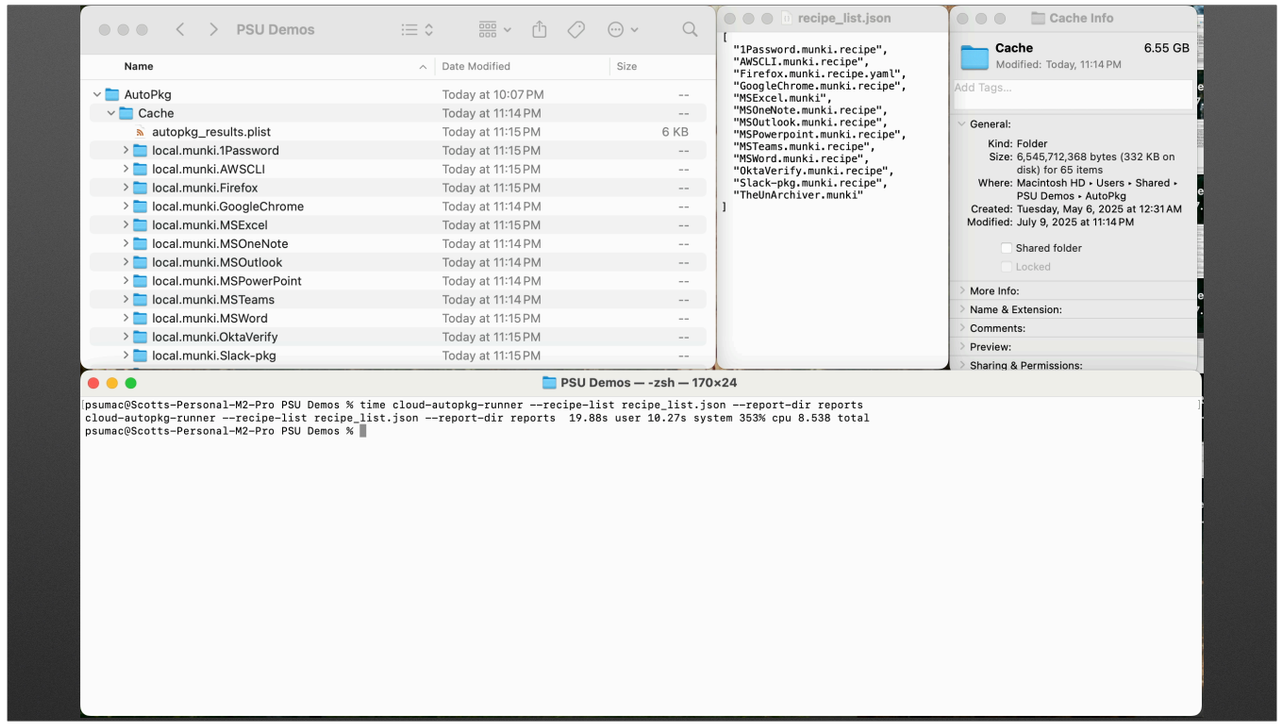

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:07 PM
autopkg_results.plist
Today at 11:09 PM
18 KB
local.munki.1Password
local.munki.AWSCLI
local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
local.munki.MSWord
local.munki.OktaVerify
local.munki.Slack-pkg
recipe_list.json
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
Kind: Folder
Size: 6,678,169,821 bytes (6.76 GB on disk) for 84 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:07 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %


PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:07 PM
autopkg_results.plist
Today at 11:09 PM
18 KB
local.munki.1Password
Today at 11:08 PM
local.munki.AWSCLI
Today at 11:07 PM
local.munki.Firefox
Today at 11:07 PM
local.munki.GoogleChrome
Today at 11:07 PM
local.munki.MSExcel
Today at 11:09 PM
local.munki.MSOneNote
Today at 11:08 PM
local.munki.MSOutlook
Today at 11:09 PM
local.munki.MSPowerPoint
Today at 11:08 PM
local.munki.MSTeams
Today at 11:08 PM
local.munki.MSWord
Today at 11:09 PM
local.munki.OktaVerify
Today at 11:07 PM
local.munki.Slack-pkg
Today at 11:08 PM
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.68 GB
Modified: Today, 11:07 PM
Add Tags...
General:
Kind: Folder
Size: 6,678,169,821 bytes (6.76 GB on disk) for 84 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:07 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 139.50s user 58.53s system 160% cpu 2:03.39 total
psumac@Scotts-Personal-M2-Pro PSU Demos %

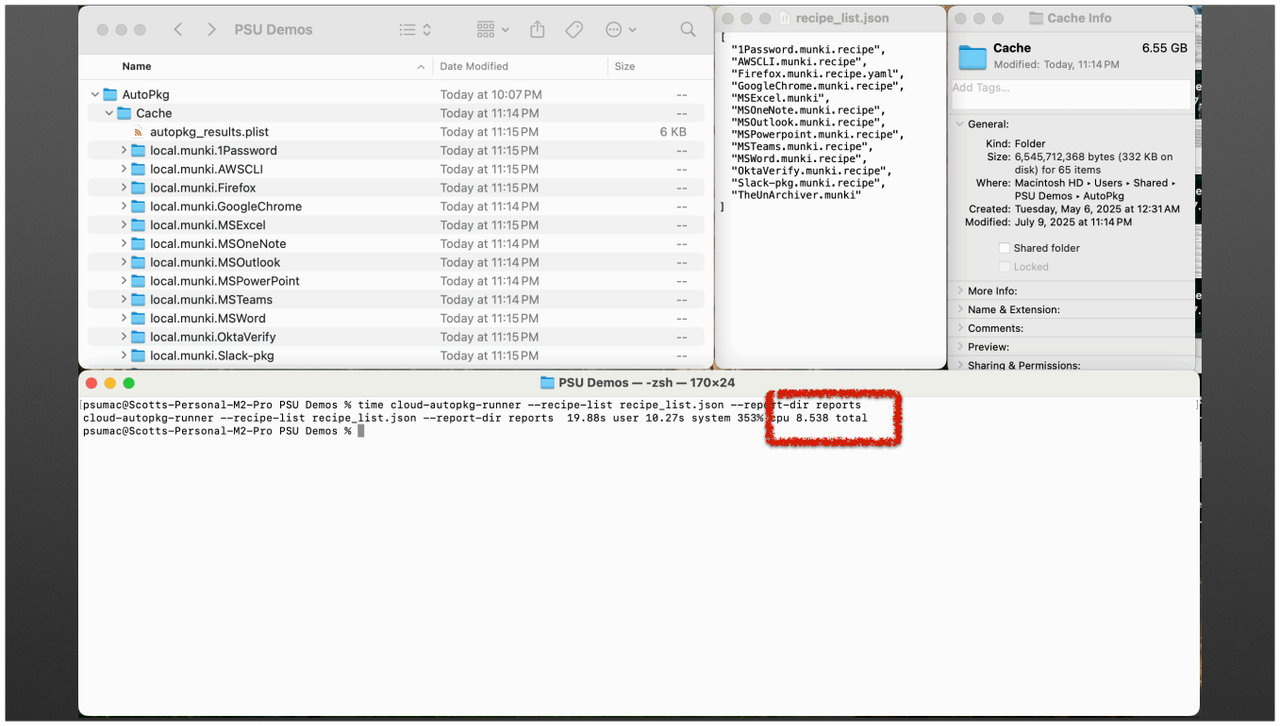


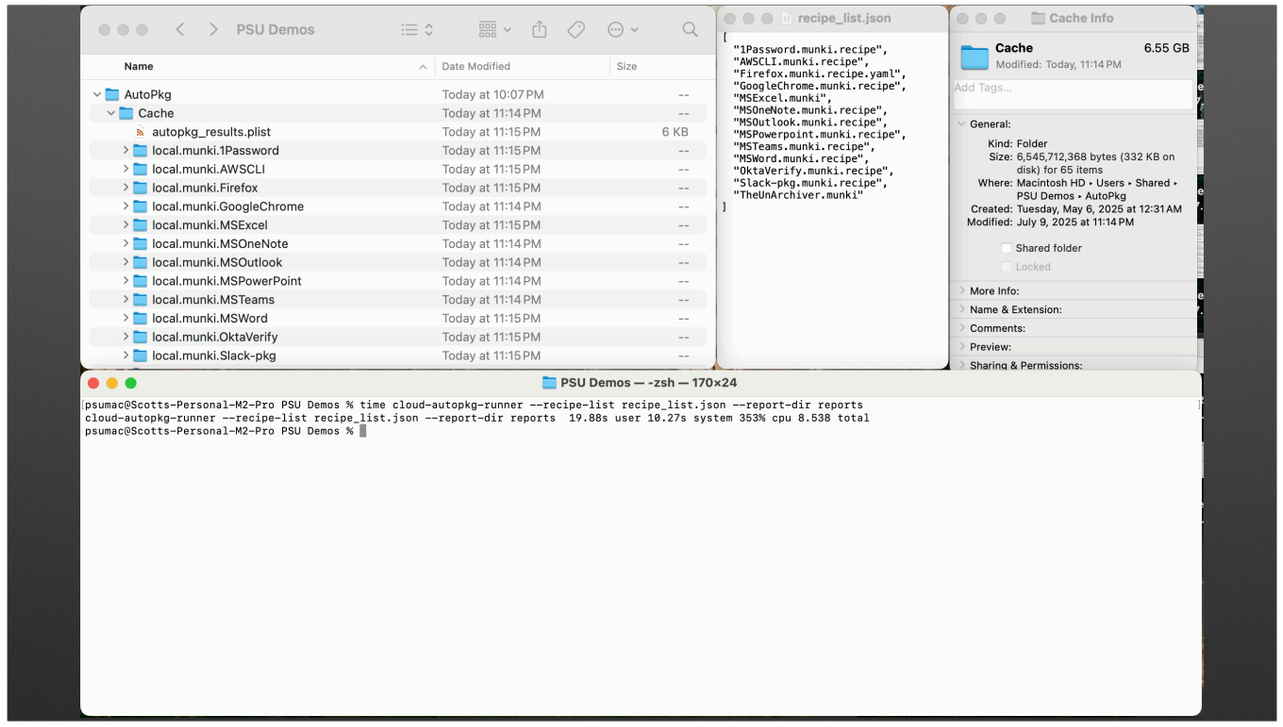
Okay, so we've emptied out the cache directory. Let's run this again using the metadata_cache.json file.

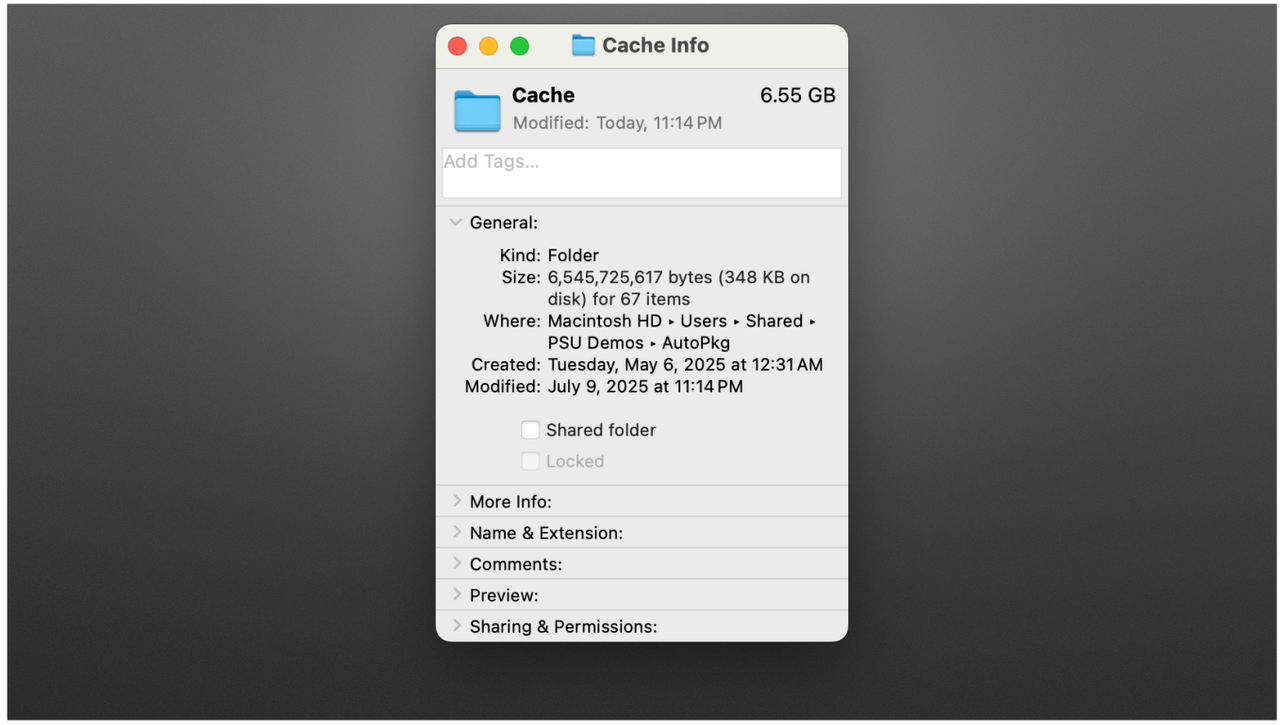

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:13 PM
Overrides
Today at 9:34 PM
RecipeRepos
Today at 9:11 PM
metadata_cache.json
Today at 11:12 PM
6 KB
Munki
Today at 11:12 PM
catalogs
Today at 11:12 PM
icons
Today at 11:12 PM
pkgs
Today at 11:07 PM
apps
Today at 11:09 PM
browsers
Today at 11:07 PM
microsoft
Today at 11:09 PM
okta
Today at 11:07 PM
OktaVerify-9.44.0.2025.609.1655.pkg
Today at 11:07 PM
91.8 MB
passwordmanagers
Today at 11:08 PM
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6 KB
Modified: Today, 11:13 PM
Add Tags...
General:
Kind: Folder
Size: 6,148 bytes (8 KB on disk) for 1 item
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:13 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports



And it's done after 8.5 seconds this time.

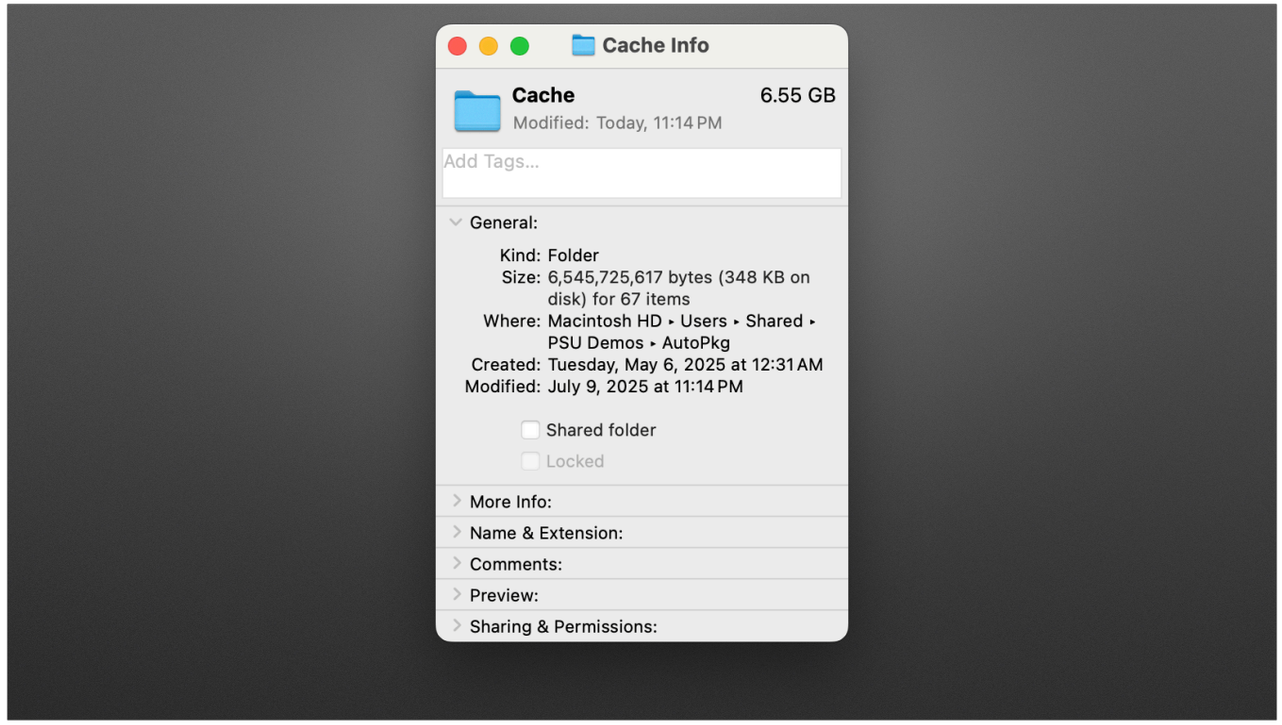

PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:14 PM
autopkg_results.plist
Today at 11:15 PM
6 KB
local.munki.1Password
local.munki.AWSCLI
local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
local.munki.MSWord
local.munki.OktaVerify
local.munki.Slack-pkg
recipe_list.json
[
"1Password.munki.recipe",
"AWSCLI.munki.recipe",
"Firefox.munki.recipe.yaml",
"GoogleChrome.munki.recipe",
"MSExcel.munki",
"MSOneNote.munki.recipe",
"MSOutlook.munki.recipe",
"MSPowerpoint.munki.recipe",
"MSTeams.munki.recipe",
"MSWord.munki.recipe",
"OktaVerify.munki.recipe",
"Slack-pkg.munki.recipe",
"TheUnArchiver.munki"
]
Cache Info
Cache
6.55 GB
Modified: Today, 11:14 PM
Add Tags...
General:
Kind: Folder
Size: 6,545,712,368 bytes (332 KB on disk) for 65 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:14 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:
PSU Demos — -zsh — 170×24
psumac@Scotts-Personal-M2-Pro PSU Demos % time cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports
cloud-autopkg-runner --recipe-list recipe_list.json --report-dir reports 19.88s user 10.27s system 353% cpu 8.538 total
psumac@Scotts-Personal-M2-Pro PSU Demos %


PSU Demos
Name
Date Modified
Size
AutoPkg
Today at 10:07 PM
Cache
Today at 11:14 PM
autopkg_results.plist
Today at 11:15 PM
6 KB
local.munki.1Password
local.munki.AWSCLI
local.munki.Firefox
local.munki.GoogleChrome
local.munki.MSExcel
local.munki.MSOneNote
local.munki.MSOutlook
local.munki.MSPowerPoint
local.munki.MSTeams
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recipe_list.json
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"MSExcel.munki",
"MSOneNote.munki.recipe",
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"MSWord.munki.recipe",
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PSU Demos — -zsh — 170×24
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psumac@Scotts-Personal-M2-Pro PSU Demos %



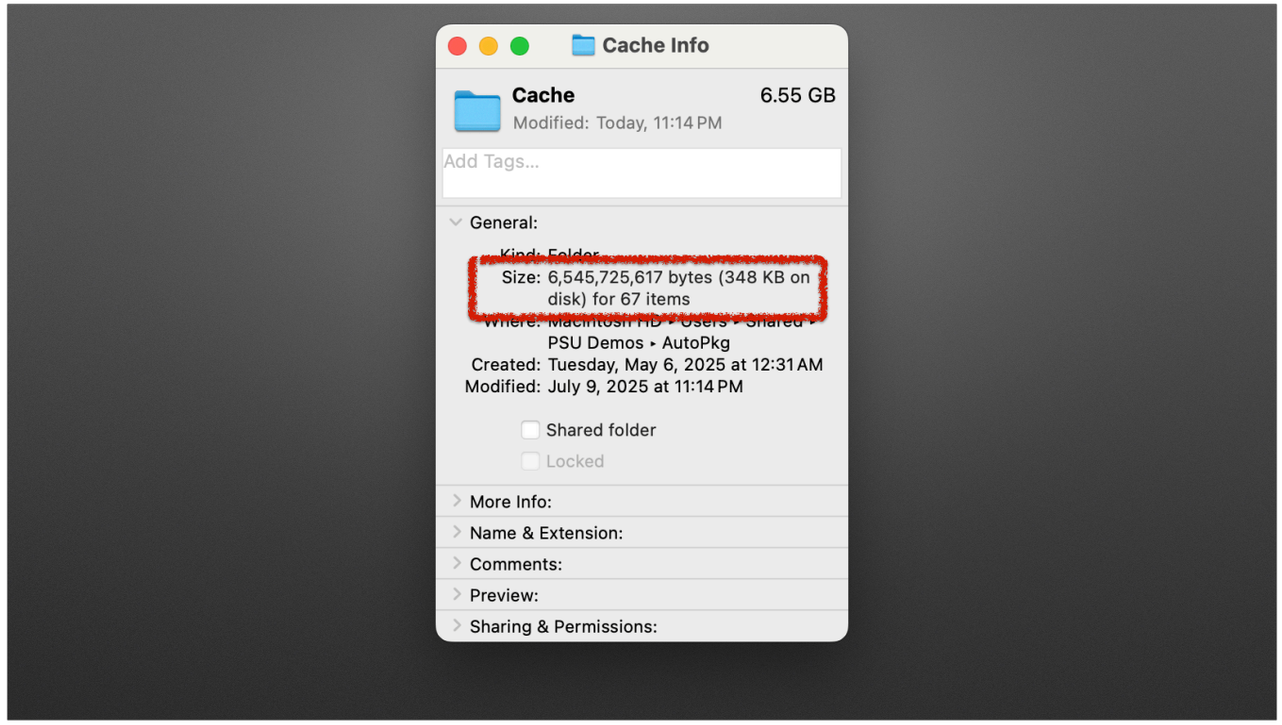
Let's take a closer look at that cache directory. It says it's using 6.55 GB of data, but if you look closely, it says it's using only 348 KB on disk.

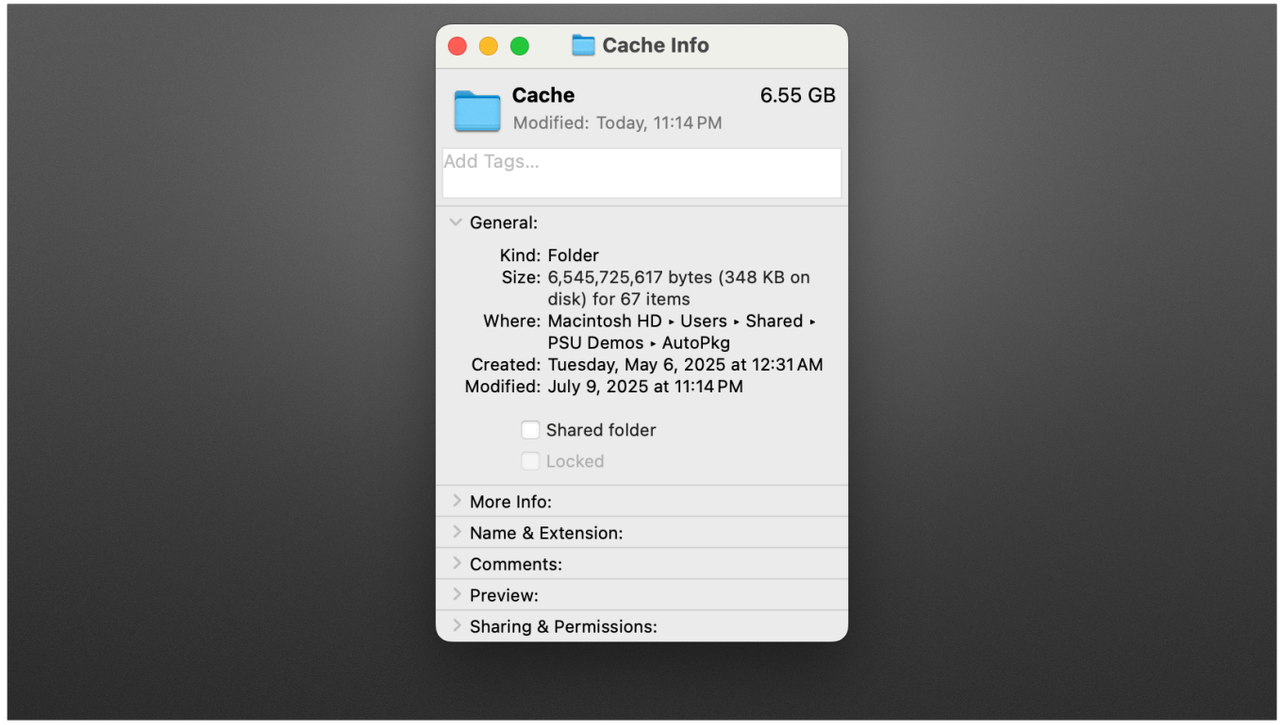
Since the cache files didn't exist, file stubs were created in their place. This is why I say this is the tool to beat if you are running in a CI/CD or similar environment where the cache doesn't persist between runs.

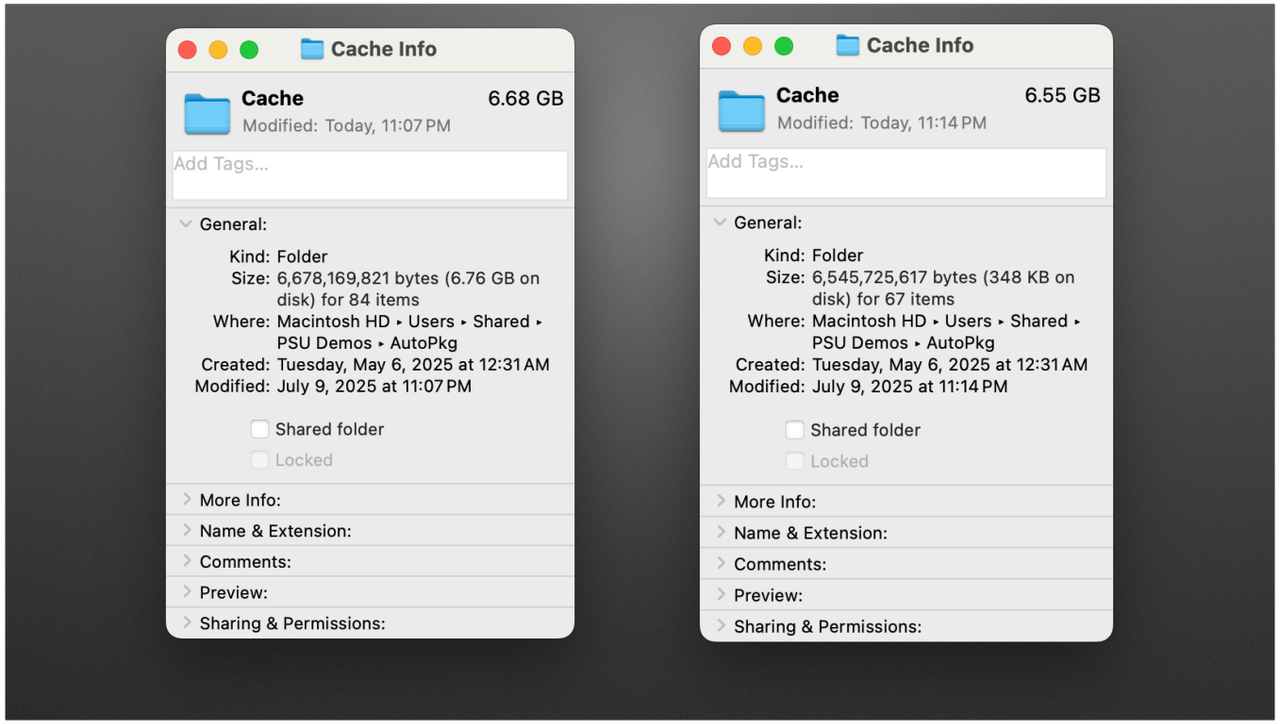


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Since the cache files didn't exist, file stubs were created in their place. This is why I say this is the tool to beat if you are running in a CI/CD or similar environment where the cache doesn't persist between runs.


Cache Info
Cache
6.55 GB
Modified: Today, 11:14 PM
Add Tags...
General:
Kind: Folder
Size: 6,545,725,617 bytes (348 KB on disk) for 67 items
Where: Macintosh HD ▸ Users ▸ Shared ▸ PSU Demos ▸ AutoPkg
Created: Tuesday, May 6, 2025 at 12:31 AM
Modified: July 9, 2025 at 11:14 PM
Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:


Cache Info
Cache
6.55 GB
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General:
Kind: Folder
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Shared folder
Locked
More Info:
Name & Extension:
Comments:
Preview:
Sharing & Permissions:



The earlier run where the cache was already there is a full 6.76 GB on disk.

And don't ask me why the values are slightly different after each of these runs. I'm chalking that up to "just APFS things."



While we're comparing things... I mentioned it earlier, but I wanted to highlight another difference. Cloud AutoPkg Runner requires the recipe-list to be a valid JSON file while AutoPkg is expecting a text file.

Another thing worth noting is that partial matching works just like with AutoPkg, as you can see with Excel and The Unarchiver. Also, the Firefox recipe is in YAML and that intermixing just works.

What?? No CI/CD Demo?

I can hear your thoughts... No CI/CD demo?

I figured that I would tempt the Demo Gods and do this one live since I'm not downloading anything to this laptop directly.

https://github.com/MScottBlake/cloud-autopkg-runner-examples/

Takeaways

I hear a lot of people say that speed is nice-to-have, but it's no big deal if it takes a while. I'm here to say that it matters more than you think. If you have to wait 30, 60, 90 minutes for your script to complete, that can take you out of your flow. It can make debugging tedious. It can limit how often you check for updates.

The faster your scripts run, the less likely you are to hit transient network issues.

Takeaways

There are financial reasons as well. For instance, GitHub Actions are billed by the minute, and the freedom to throw away your AutoPkg cache after each run can lower storage fees.

It might not seem like much, but it can add up.

What's Next?

Support for the .info.json files that are created by URLDownloaderPython.

Integration tests - For instance, we need to test that files are properly stored in cloud containers.

Configuration file support - I want to be able to specify all settings in a TOML file or something similar.

Resources

- MacAutoPkg https://code.google.com/archive/p/macautopkg/
- AutoPkg https://autopkg.github.io/autopkg
- AutoPkgr https://www.lindegroup.com/resources/autopkgr
- Recipe Robot https://github.com/homebysix/recipe-robot
- Facebook autopkg_tools.py https://github.com/facebook/IT-CPE/tree/main/legacy/autopkg_tools
- AutoPkg on GitLab CI https://rickheil.com/autopkg-on-gitlab-ci
- Jenkins & AutoPKG – Love At First Sight https://geekygordo.com/2013/11/01/jenkins-autopkg-love-at-first-sight
- Gusto autopkg_tools.py https://github.com/Gusto/it-cpe-opensource/tree/main/autopkg
- AutoPkg Conductor https://github.com/rtrouton/autopkg-conductor
- Cloud AutoPkg Runner https://github.com/MScottBlake/cloud-autopkg-runner
- Cloud AutoPkg Runner Example Workflows https://github.com/MScottBlake/cloud-autopkg-runner-examples

To finish things out... If you're interested in any of the projects that I've mentioned in this talk, I think I have them all linked here.

Snap a picture if you want, but remember that the slides are all available online.

Questions?

I intentionally left a lot of time for questions, so please, fire away.

I promised I would show the link to this presentation again at the end, so here it is.



https://github.com/MScottBlake/Presentations

I'm just going to leave this up while we do questions.

Other AutoPkg Sessions

- Create AutoPkg recipes for Mac/Win/Linux from scratch (James Stewart)
  3:15 PM today in 207
- From Mac to Windows: Expanding AutoPkg's Reach (Shoolin S., Michael Kearns, and Albert Koch)
  9:00 AM tomorrow in 208
- AutoAutoPkg - Manage your AutoPkg with GitHub Actions (Adam Anklewicz)
  1:15 PM tomorrow in 206
- Automate all the things - fully automated OS and App patching (Jacob Burley)
  10:45 AM Friday in Deans Hall 1

There are 4 other AutoPkg-related talks this week. Check them out.



https://bit.ly/psumac25-894765

And finally.... Thank you to Penn State for letting me get up here, and a big Thanks to you all.

Please make sure that you submit feedback. Good and bad are both welcome and appreciated.



https://bit.ly/psumac25-894765